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(12) **United States Patent**
Bennett et al.(10) **Patent No.: US 12,403,204 B2**(45) **Date of Patent: Sep. 2, 2025**(54) **GENE THERAPY FOR OCULAR DISORDERS**(71) Applicant: **The Trustees of the University of Pennsylvania**, Philadelphia, PA (US)(72) Inventors: **Jean Bennett**, Bryn Mawr, PA (US);
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(51) **Int. Cl.****A61K 48/00** (2006.01)**A61K 9/00** (2006.01)**C07K 14/47** (2006.01)**C12N 15/86** (2006.01)(52) **U.S. Cl.**CPC **A61K 48/0058** (2013.01); **A61K 9/0048** (2013.01); **C07K 14/47** (2013.01); **C12N 15/86** (2013.01); **C12N 2750/14143** (2013.01); **C12N 2799/022** (2013.01); **C12N 2799/04** (2013.01); **C12N 2800/107** (2013.01); **C12N 2800/22** (2013.01); **C12N 2840/002** (2013.01); **C12N 2840/007** (2013.01)(58) **Field of Classification Search**CPC **A61K 48/0058**
See application file for complete search history.(56) **References Cited****U.S. PATENT DOCUMENTS**5,139,941 A 8/1992 Muzyczka et al.
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Primary Examiner — Anoop K Singh*Assistant Examiner* — David A Montanari(74) *Attorney, Agent, or Firm* — Howson & Howson LLP; Colleen M. Schaller(57) **ABSTRACT**

Compositions and methods are provided for treating ocular disorders in a subject are provided. In one aspect, an adeno-associated viral vector is provided which includes a nucleic acid molecule comprising a sequence encoding CNGA3. In another aspect, an adeno-associated viral vector is provided which includes a nucleic acid molecule comprising a sequence encoding CNGB3. In another aspect, an adeno-associated viral vector is provided which includes a nucleic acid molecule comprising a sequence encoding REP-1. In desired embodiments, the subject is human, cat, dog, sheep, or non-human primate.

10 Claims, 27 Drawing Sheets**Specification includes a Sequence Listing.**

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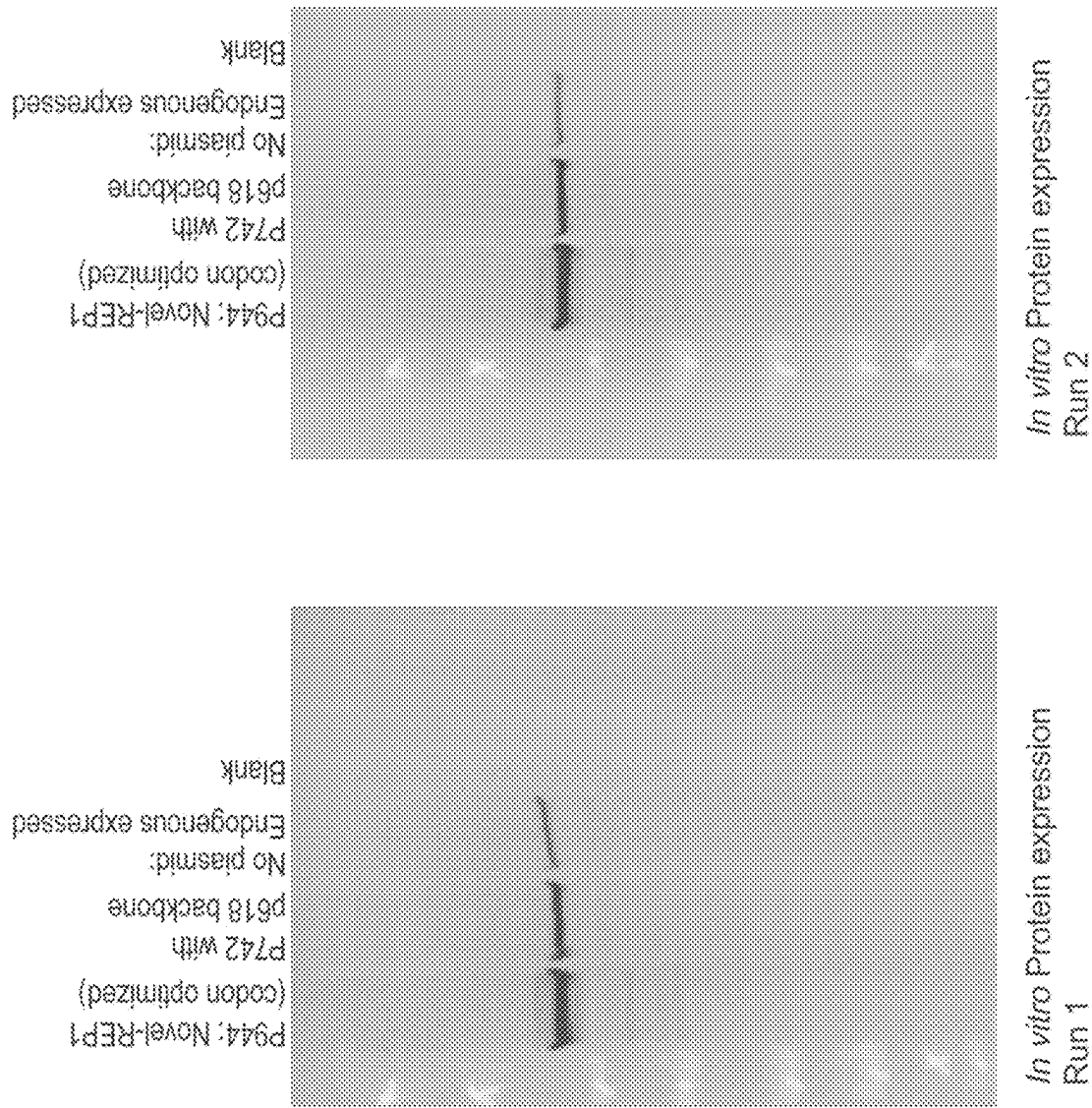
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QUERY = Native REP-1 (SEQ ID NO: 3)
SUBJECT = Codon-optimized REP-1 (SEQ ID NO: 1)

Score	Expect	Identities	Gaps	Strand
1267 bits(1404)	0.0	1455/1955(74%)	4/1955(0%)	Plus/Plus
Query 1	ATGGCGGATACTCTCCCTTCGGAGTTTGATGTGATCGTAATAGGGACGGTATTGCCCTGAA	60		
Sbjct 1	ATGGCTGATACCCCTGCCCTCTGAATTCGACGTGATTGTGATTGGAACCGGACTCCCCTGAA	60		
Query 61	TCCATCATTCAGCTGCATGTTCAAGAAGTGGCCGGAGAGTTCTGCATGTTGATTCAAGA	120		
Sbjct 61	TCGATCATCGCGCGGCTGTTCOCGGTCCGGTCCGGCGCGTGTGACAGTCGATTTCGAGA	120		
Query 121	AGCTACTATGGAGGAAAACCTGGGCCAGTTTTTAGCTTTTCAGGACTATTGTCCCTGGCTAAAG	180		
Sbjct 121	AGCTACTACGGAGGGAATTGGGCCCTCATTCTCCTTCTCCGGACTGCTCTCCCTGGCTGAAG	180		
Query 181	GAATACCAGGAAAAACAGTGACATTGTAAGTGACAGTCCAGTGTGGCAAGACCAGATCCTT	240		
Sbjct 181	GAGTATCAGGAGAACCTCCGACATTGTCTCCGACTCACCTGTGTGGCAGGACCAGATCCTG	240		
Query 241	GAAAATGAAGAAGCCATTGCTCTTAGCAGGAAGGACA AAAA C T A T T C A A C A T G T G G A A G T A	300		
Sbjct 241	GAAAACGAGGAAGCAATAGCCCTGAGCCGGGAGGACAAGACCATCCAGCACGTGGAGGTG	300		

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FIG. 2
(continued)

Sbjct	661	ATCAAGGAGGGGGGAGTTTAACTGATCGATCTGGTGTGGAAGCTGCTGTACAGCCGCGGT	720
Query	721	TTACTAATTGATCTTCTTAATCAAAATCTAATGTTAGTCGATAT--GCAGAGTTTAAAAATA	778
Sbjct	721	CTGCTGATCGATCTGCTCATTAAGTCGAACGT--GTGAGATACGCCGAGTTCAAGAACA	778
Query	779	TTACCAGGATTCCTTGCAATTCGAGAAGGACGAGTGGAAACAGGTTCCGTTGCCAGAGCAG	838
Sbjct	779	TCACAAAGGATTCTCGCCTTCGGGAAGGAAGAGTGGAAACAAGTGCCGTGCTCCGGGCGCG	838
Query	839	ATGTCCTTAATAGCAAAACACTTACTATGGTAGAAAAGCGAATGCTAATGAAATTTCTTA	898
Sbjct	839	ACGTGTTCACTCAAAAGCAACTTACCATGGTGGAAAAGCGCATGCTGATGAAATTCCTGA	898
Query	899	CAATTTGTATGGAAATATGAGAAATATCCTGATGAATATAAAGGATATGAAGAGATCACAT	958
Sbjct	899	CCTTCGTGATGGAGTACGAAAAGTACCCCTGATGAGTACAAGGGTTACGAAGAATAACTT	958
Query	959	TTTATGAATATTAAAGACTCAAAAATTAACCCCCCAACCTCCAATATATTGTCATGCATT	1018
Sbjct	959	TCTACGAGTACCTCAAGACCCAGAGCTGACCCCGGAATCTGCAGTACATTGTGATGCAC	1018
Query	1019	CAATTGCAATGACATCAGAGACAGCCAGCAGCACCATAGATGGTCTCAAGAGCTACCAAAA	1078
Sbjct	1019	CAATCGCAATGACCTCCGAAACCGCCTCTCTGACCATCGACGGGCTCAAGGCCACCAGA	1078
Query	1079	ACTTTCTTCACGTCTTTGGGCGGTATGGCAACACTCCATTTTGTTCCTTTATATGGCC	1138

FIG. 2
(continued)

FIG. 2
(continued)

Query	1499	TGAAAGGCACCTATTGTTGATTTGACCTTGACATCTTCTAAACAGCAAGAGAATT	1558
Sbjct	1499	TGAAAGGCACCTTACCTGGTGCACCTGACGNGCACCTCATCGAAAACCGCTAGAGAGGACC	1558
Query	1559	TAGAATCAGTTGTGCAGAAAATTGTTGTTCCTATATACTGAAATGAGAGATAGAAAATGAAC	1618
Sbjct	1559	TGGAATCCGTCGTCCAAAAGCTGTGTCGCTTACCCGAGATGGAATTGAAAACGAAC	1618
Query	1619	AAGTAGAAAAGCCAAAGAAATCTGTGGGCTCTTTACTTCAATATAGAGATTCGTCAGACA	1678
Sbjct	1619	AAGTGGAGAAAGCCCGCATCCTTTGGGCCCTGTACTTTAACAANGCGGATTCCTCCGATA	1678
Query	1679	TCAGCAGGAGCTGTATAATGATTAACCATCCAAACGTTTATGTCTGTCTGGCCACAGATT	1738
Sbjct	1679	TCTCGCGTCCCTGCTATAACGACTTGCCCTTCGAACGTCTACGTCTGCTCCGGGCCAGACT	1738
Query	1739	GTGGTTTAGGAAATGATAATGCAGTCAAAACAGGCTGAACACACTTTTCCAGGAAATCTGCC	1798
Sbjct	1739	GCGGTCTTGGCAACGACAAATGCCGIGAGAGCGGCGGAAAACACTGTTCCAAGAGATCTGCC	1798
Query	1799	CCAAATGAAGATTTCTGTCCCCCTCCACCAAAATCCTGAAGACATTATCCTTGATGGAGACA	1858
Sbjct	1799	CTAACGAGGATTTTGGCCCGGCCCCCAACCCCGAGGATATCATCTTTGGACGGAGACA	1858
Query	1859	GTTTACAGCCAGAGGCTTCAGAAATCCAGTGCCATACCAGAGGCTAACTCGGAGACTTTCA	1918
Sbjct	1859	GCCTGCAGCCAGAACATCCGAGTCCAGGCCCATCCCGGAGGCCCAACAGCGAAACCTTCA	1918

FIG. 2
(continued)

FIG. 2
(continued)

Query	1919	AGGAAAGGCACAAAACCTTGGAAACCTAGAGGAGTCC	1953
Sbjct	1919	AGGAGAGCACAATAACCTGGGCAACCTGGAACAGTCC	1953

Codon optimized (Query) SEQ ID NO: 9 vs. Native (subject) SEQ ID NO: 13
CNGA3

Score	Expect	Identities	Gaps
1856 bits(2058)	0.0	1678/2108(80%)	2/2108(0%)
Query 1	GCGGCCGCCACCATGGCTAAGATTAAACACCCAGTACTCACATCCCGGCACTCACCTC 60		
Sbjct 1	GCGGCCGCCACCATGGCCCAAGATCAACACCCCAATACTCCCACCCCCTCCAGGACCCACCTC 60		
Query 61	AAAGTCAAGACCTCCGATCGGGATCTGAACCGGGCTGAGAATGGGCTGTGGGGGCCAC 120		
Sbjct 61	AAGTAAAGACCTCAGACCGGGATCTCAATCGCGCTGAAAATGSCCTCAGCAGAGCCCAC 120		
Query 121	TCGTCGTCGGAGGAAACCAAGCAGCAGC-GTGTCTCCAGCCGGGCATCGCCATGGAAACTAGGGG 179		
Sbjct 121	TCGTCAAGTGAGGAGAC-ATCGTCAGTGTGTGAGCCGGGGATCGCCATGGAGACCAAGAG 179		
Query 180	GCTGGCGGACTCCGGACAGGGATCCTTCACTGGACAGGGTATTGCCCGGCTGAGCAGACT 239		
Sbjct 180	ACTGGCTGACTCCGGGCAGGGCTCCTTCACCGGCCAGGGGATCGCCAGGGCTGTGGCGCCT 239		
Query 240	GATCTTCCTGCTTCGCCGCTGGGGGCCAGACACAGTGCACCATCAGGACCAAGGACCTTGA 299		
Sbjct 240	CATCTTCTTGTGCGCAGGTGGGCTGCCAGGTCATGTGCACCACCAAGGACCAAGGACCGGA 299		

FIG. 3

FIG. 3
(continued)

FIG. 3
(continued)

Query	1140	AGAGTACCTGTTTCGTGGTGGTGGACTTCCTGGTCCGAGTGTGATTTTCGCCACCATGT	1199
Sbjct	1140	GGAGTATCTCTTTTGTGGTCGTAGACTTCTTGGTGGGTGTTCTGATTTTGGCCACCATGT	1199
Query	1200	GGGAAACGTGGGCTCCATGATCTCCAACATGAACGGTCGAGAGCTGAGTTCACCAAGCCAA	1259
Sbjct	1200	GGGCAATGTGGGCTCCATGATCTCGAATATGAATGCTCACGGGCAGAGTTCCAGGGCCAA	1259
Query	1260	GATCGACTCCATTAAAGCAGTACATGCAGTTTCAGAAAGGTCACCAAGGACCTTGGAAACCCAG	1319
Sbjct	1260	GATTGATTCCATCAAGCAGTACATGCAGTTCCTCGCAAGGTCACCAAGGACTTGGAGACCGG	1319
Query	1320	GGTCATCCGCTGGTTCGACTACCTGTGGGCCCAACAAGACTGTGGACGAAAGGAAGT	1379
Sbjct	1320	GGTTATCCGGTGGTTTGACTACCTGTGGGCCCAACAAGAACGGTGGATGAGAAGGAGGT	1379
Query	1380	GCTGAAGTCGCTGCCGGATAAGCTGAAGGCCGAAATCGCCATTAAACGTGCACCTTGACAC	1439
Sbjct	1380	GCTCAAGAGCCTCCAGACAAAGCTGAAGGCTGAGATCGCCATCAACGTGCACCTGGACAC	1439
Query	1440	CCTGAAGAAAGTCCGGATCTTCCAAGACTGTGAAGCCGGCCTCCTGGTGGAGCTCGTGTCT	1499
Sbjct	1440	GCTGAAGAAGGTTCCGATCTTCCAGGACTGTGAGGCAGGGCTGCTGGTGGAGCTGTGTGCT	1499
Query	1500	CAAGCTGCGGCCCAACCGTGTTCAGCCCGGAGATTACATTTCGAAGAAGGGCGATATCGG	1559
Sbjct	1500	GAAGCTGCGACCCACTGTGTTCAGCCCTGGCGATTATATCTGCAAGAAGGGAGATATTGG	1559

FIG. 3
(continued)

Query	1560	CAAAGAGATGTACATCATCAACGAGGAAAGCTGGCCGTGGTCGGGACGACGGCGTGAC	1619
Sbjct	1560	GAAGGAGATGTACATCATCAACGAGGCAAGCTGGCCGTGGTGGCTGATGATGGGTAC	1619
Query	1620	CCAGTTCGTGGTGTCTCCGACGGATCCTACTTCGGTGAAATCTCAATCCTCAACATCAA	1679
Sbjct	1620	CCAGTTCGTGGTCTCAGCGATGBCAGCTACTTCGGGGAGATCAGCATCTGAACATCAA	1679
Query	1680	GGGGTCCAAGTCCGGCAACCGGAGAACTGCCAACATTCCGCTCCATCGGATACAGCGACCT	1739
Sbjct	1680	GGGAGCAAGTCGGGGAACCGGAGCGGCCAACATCCGAGCATTTGGCTACTCAGACCT	1739
Query	1740	GTTTTCCTGTCCAAGGATGACTGTATGGAGGCTCTGACTGAGTACCTGAAGCGAAGAA	1799
Sbjct	1740	GTTCTCCTCTCAAAGGACGATCTCATGGAGGCCCTCACCGAGTACCCCGAAGCCAAGAA	1799
Query	1800	GGCTTTGGAGGAAAAGGGCGGCGAGATTCTGATGAAGGACAATTTGATCGACGAGGAGCT	1859
Sbjct	1800	GGCCCTGGAGGAGAAAGGACGGCAGATCCTGATGAAAGACAACCTGATCGATGAGGAGCT	1859
Query	1860	CGCACGGGCGGCGCGACCCCAAGGATCTCGAAGAGAGAGGTGCGAACACAGCTGGGTCTTC	1919
Sbjct	1860	GGCCAGGGCGGCGGCGACCCCAAGACCTTGAGGAGAAAAGTGGAGACAGCTGGGTCTTC	1919
Query	1920	GCTTGATACCTGCAACCCGATTCGCGCGGCTGCTCGCCGAGTACAACGCGACCCAGAT	1979
Sbjct	1920	CCTGGACACCCTGCAGACCAGGTTTGACGCCCTCCTGGCTGAGTACAGCCACCCAGAT	1979

FIG. 3
(continued)

Query	1980	GAAGATGAAGCAGAGAGACTGTACAGTTGGAAATCCCAAGTCAAGGGCGGAGGCGACAAGCC	2039
Sbjct	1980	GAAGATGAAGCAGAGCGTCTCAGCCCACTGGAAAGCCAGGTGAAGGTTGGTGGGGACAAGCC	2039
Query	2040	GCTGGCGGACGGGGAAGTGCCCGGGGACGCCACCAAGACTGAGGACAAAGCAGCAGTGATC	2099
Sbjct	2040	CCTGGCTGATGGGGAAGTTCCCGGGGATGCTACAAAACAGAGGACAAACAACAGTGATC	2099
Query	2100	ATAGATCT	2107
Sbjct	2100	ATAGATCT	2107

FIG. 3
(continued)

Locations of silent mutations are highlighted

CLUSTAL W (1.83) multiple sequence alignment

CNGB3_with_modified_ends - SEQ ID NO: 23

CNGB3_modified_ORF - SEQ ID NO: 21

native_CNGB3_ORF - SEQ ID NO: 19

CNGB3_with_modified
CNGB3_modified_ORF
native_CNGB3_ORF
GCGCGCGCCACCATGTTTAAATCGCTGACAAAAGTCAACAAGGTGAAGCCTATAGGAGAG
-----ATGTTTAAATCGCTGACAAAAGTCAACAAGGTGAAGCCTATAGGAGAG
-----ATGTTTAAATCGCTGACAAAAGTCAACAAGGTGAAGCCTATAGGAGAG

CNGB3_with_modified
CNGB3_modified_ORF
native_CNGB3_ORF
AACAAATGAGAATGAACAAAGTTCTCGTCGGAATGAAGAAGGCTCTCACCCCAAGTAATCAG
AACAAATGAGAATGAACAAAGTTCTCGTCGGAATGAAGAAGGCTCTCACCCCAAGTAATCAG
AACAAATGAGAATGAACAAAGTTCTCGTCGGAATGAAGAAGGCTCTCACCCCAAGTAATCAG

CNGB3_with_modified
CNGB3_modified_ORF
native_CNGB3_ORF
TCTCAGCAAAACCACAGCACAGGAGAAAACAAGGTGAAGAGAAAATCTCTCAAAACCAAG
TCTCAGCAAAACCACAGCACAGGAGAAAACAAGGTGAAGAGAAAATCTCTCAAAACCAAG
TCTCAGCAAAACCACAGCACAGGAGAAAACAAGGTGAAGAGAAAATCTCTCAAAACCAAG

CNGB3_with_modified
CNGB3_modified_ORF
native_CNGB3_ORF
TCAACTCCAGTCACGTCTGAAGGCCACACACCAACATACAAGACAAAATCTCCAAAGAAA
TCAACTCCAGTCACGTCTGAAGGCCACACACCAACATACAAGACAAAATCTCCAAAGAAA
TCAACTCCAGTCACGTCTGAAGGCCACACACCAACATACAAGACAAAATCTCCAAAGAAA

CNGB3_with_modified
CNGB3_modified_ORF
native_CNGB3_ORF
AATTCTCTGGAGATCTGACCACAAACCCCTGACCCCTCAAATGCAGCAGAACCAACTGGA
AATTCTCTGGAGATCTGACCACAAACCCCTGACCCCTCAAATGCAGCAGAACCAACTGGA
AATTCTCTGGAGATCTGACCACAAACCCCTGACCCCTCAAATGCAGCAGAACCAACTGGA

CNGB3_with_modified
CNGB3_modified_ORF
native_CNGB3_ORF
ACAGTGCCAGAGCAGAAGGAAATGGACCCCGGGAAGAGGTTCCAAACAGCCCAAAAAC
ACAGTGCCAGAGCAGAAGGAAATGGACCCCGGGAAGAGGTTCCAAACAGCCCAAAAAC
ACAGTGCCAGAGCAGAAGGAAATGGACCCCGGGAAGAGGTTCCAAACAGCCCAAAAAC

CNGB3_with_modified
AAACCGCCAGCAGCTCTGTTATAAATGAGTATGCCGATGCCAGCTACACACCTGGTG

FIG. 4

CNGB3__modified_ORF
native__CNGB3__ORF

CNGB3__with__modified
CNGB3__modified_ORF
native__CNGB3__ORF

CNGB3__with__modified
CNGB3__modified_ORF
native__CNGB3__ORF

CNGB3__with__modified
CNGB3__modified_ORF
native__CNGB3__ORF

CNGB3__with__modified
CNGB3__modified_ORF
native__CNGB3__ORF

CNGB3__with__modified
CNGB3__modified_ORF
native__CNGB3__ORF

CNGB3__with__modified
CNGB3__modified_ORF
native__CNGB3__ORF

CNGB3__with__modified
CNGB3__modified_ORF
native__CNGB3__ORF

AAACCGCCAGCAGCTCCTGTTATATAAATGAGTATGCCGATGCCAGCTACACAACCTGGTG
AAACCGCC[]GCAGCTCCTGTTATATAAATGAGTATGCCGATGCCAGCTACACAACCTGGTG

AAAGAATGCGTC[]AAAGAACAGCCCTCTACAAGAAAAAGTTGGTAGAGGGAGATCTCTCC
AAAGAATGCGTC[]AAAGAACAGCCCTCTACAAGAAAAAGTTGGTAGAGGGAGATCTCTCC
AAAGAATGCGTC[]AAAGAACAGCCCTCTACAAGAAAAAGTTGGTAGAGGGAGATCTCTCC

TCACCCGAAGCCAGCCCAAACTGCAAGCCACGGCTGTACCACCAAGTAAAAAGAAAGC
TCACCCGAAGCCAGCCCAAACTGCAAGCCACGGCTGTACCACCAAGTAAAAAGAAAGC
TCACCCGAAGCCAGCCCAAACTGCAAGCCACGGCTGTACCACCAAGTAAAAAGAAAGC

GATGATAAGCCCAACAGAACATTACTACAGGCTGTTGGTTCAAAGTCAAAAAGATGCCCT
GATGATAAGCCCAACAGAACATTACTACAGGCTGTTGGTTCAAAGTCAAAAAGATGCCCT
GATGATAAGCCCAACAGAACATTACTACAGGCTGTTGGTTCAAAGTCAAAAAGATGCCCT

TTAACAGAGTACTTAAAGCGAATTAAACTTCCAAACAGCATAGATTTCATACACAGATCGA
TTAACAGAGTACTTAAAGCGAATTAAACTTCCAAACAGCATAGATTTCATACACAGATCGA
TTAACAGAGTACTTAAAGCGAATTAAACTTCCAAACAGCATAGATTTCATACACAGATCGA

CTCTATCTCCTGTGGCTCTTGGCTTGTCACTCTTGCCCTATAACTGGAACCTGCTTTTATA
CTCTATCTCCTGTGGCTCTTGGCTTGTCACTCTTGCCCTATAACTGGAACCTGCTTTTATA
CTCTATCTCCTGTGGCTCTTGGCTTGTCACTCTTGCCCTATAACTGGAACCTGCTTTTATA

CCACTGGCCCTCGCTTCCCAATATCAAAACCGCAGACAACATACACTACTGGCTTATTTGGC
CCACTGGCCCTCGCTTCCCAATATCAAAACCGCAGACAACATACACTACTGGCTTATTTGGC
CCACTGGCCCTCGCTTCCCAATATCAAAACCGCAGACAACATACACTACTGGCTTATTTGGC

GACATCATCTGTGATATCATCTACCTTTATGATATGCTATTTATCCAGCCCACTCCAG
GACATCATCTGTGATATCATCTACCTTTATGATATGCTATTTATCCAGCCCACTCCAG
GACATCATCTGTGATATCATCTACCTTTATGATATGCTATTTATCCAGCCCACTCCAG

TTTGTAAAGAGGAGGAGACATAATAGTGGATTCAAATGAGCTAAGGAAACACTACAGGACT
TTTGTAAAGAGGAGGAGACATAATAGTGGATTCAAATGAGCTAAGGAAACACTACAGGACT

FIG. 4
(continued)

native__CNGB3__ORF	TTTGTAAAGAGGAGACATAAATAGTGGATTCAATGAGCTAAGGAAACACTACAGGACT

CNGB3__with__modified	TTACAAAAATTCAGTTGGATGTCGCATCAATAAACCATTGGATATTGCTACCTCTTC
CNGB3__modified__ORF	TCTACAAAAATTCAGTTGGATGTCGCATCAATAAACCATTGGATATTGCTACCTCTTC
native__CNGB3__ORF	TCTACAAAAATTCAGTTGGATGTCGCATCAATAAACCATTGGATATTGCTACCTCTTC

CNGB3__with__modified	TTTGGGTTAATCCAAATGTTTAGAGCAAAATAGGATGTTAAAGTACACTTCATTTTGGAA
CNGB3__modified__ORF	TTTGGGTTAATCCAAATGTTTAGAGCAAAATAGGATGTTAAAGTACACTTCATTTTGGAA
native__CNGB3__ORF	TTTGGGTTAATCCAAATGTTTAGAGCAAAATAGGATGTTAAAGTACACTTCATTTTGGAA

CNGB3__with__modified	TTTAATCATCACCTAGAGTCTATAATGGACAAAGCATATATCTACAGAGTTATTCGAACA
CNGB3__modified__ORF	TTTAATCATCACCTAGAGTCTATAATGGACAAAGCATATATCTACAGAGTTATTCGAACA
native__CNGB3__ORF	TTTAATCATCACCTAGAGTCTATAATGGACAAAGCATATATCTACAGAGTTATTCGAACA

CNGB3__with__modified	ACTGGATACTTGCTGTTTATTCTGCACATTAATGCCCTGTGTTTATTACTGGGCTTCAAAC
CNGB3__modified__ORF	ACTGGATACTTGCTGTTTATTCTGCACATTAATGCCCTGTGTTTATTACTGGGCTTCAAAC
native__CNGB3__ORF	ACTGGATACTTGCTGTTTATTCTGCACATTAATGCCCTGTGTTTATTACTGGGCTTCAAAC

CNGB3__with__modified	TATGAAGGAATTGGCACTACTAGATGGGTGTATGATGGGGAAGGAAACGAGTATCTGAGA
CNGB3__modified__ORF	TATGAAGGAATTGGCACTACTAGATGGGTGTATGATGGGGAAGGAAACGAGTATCTGAGA
native__CNGB3__ORF	TATGAAGGAATTGGCACTACTAGATGGGTGTATGATGGGGAAGGAAACGAGTATCTGAGA

CNGB3__with__modified	TGTTATTATTGGGCAGTTCGAACTTTAATTACCATTTGGTGGCCTTCCAGAACCCACAAACT
CNGB3__modified__ORF	TGTTATTATTGGGCAGTTCGAACTTTAATTACCATTTGGTGGCCTTCCAGAACCCACAAACT
native__CNGB3__ORF	TGTTATTATTGGGCAGTTCGAACTTTAATTACCATTTGGTGGCCTTCCAGAACCCACAAACT

CNGB3__with__modified	TTATTGAAATTTGTTTTTCAACTCTTGAATTTTTTCTGGAGTTTTTGTGTTCTCCAGT
CNGB3__modified__ORF	TTATTGAAATTTGTTTTTCAACTCTTGAATTTTTTCTGGAGTTTTTGTGTTCTCCAGT
native__CNGB3__ORF	TTATTGAAATTTGTTTTTCAACTCTTGAATTTTTTCTGGAGTTTTTGTGTTCTCCAGT

CNGB3__with__modified	TTAATTGGTCAGATGAGAGATGTGATTTGGAGCAGCTACAGCCCAATCAGAACTACTTCCGC
CNGB3__modified__ORF	TTAATTGGTCAGATGAGAGATGTGATTTGGAGCAGCTACAGCCCAATCAGAACTACTTCCGC
native__CNGB3__ORF	TTAATTGGTCAGATGAGAGATGTGATTTGGAGCAGCTACAGCCCAATCAGAACTACTTCCGC

FIG. 4
(continued)

CNGB3__with_modified CNGB3__modified_ORF native__CNGB3__ORF	GCCTGCATGGATGACACACCATTCCTACATGAACAATTACTCCATTCCCTAAACTTGTGCAA GCCTGCATGGATGACACACCATTCCTACATGAACAATTACTCCATTCCCTAAACTTGTGCAA GCCTGCATGGATGACACACCATTCCTACATGAACAATTACTCCATTCCCTAAACTTGTGCAA *****
CNGB3__with_modified CNGB3__modified_ORF native__CNGB3__ORF	AAGCGAGTTCGGACTTGGTATGAATATACATGGGACTCTCAAAGAATGCTAGATGAGTCT AAGCGAGTTCGGACTTGGTATGAATATACATGGGACTCTCAAAGAATGCTAGATGAGTCT AAGCGAGTTCGGACTTGGTATGAATATACATGGGACTCTCAAAGAATGCTAGATGAGTCT *****
CNGB3__with_modified CNGB3__modified_ORF native__CNGB3__ORF	GATTTGCTTAAGACCCCTACCAACTACGGTCCAGTTAGCCCTCGCCATTGATGTGAACCTTC GATTTGCTTAAGACCCCTACCAACTACGGTCCAGTTAGCCCTCGCCATTGATGTGAACCTTC GATTTGCTTAAGACCCCTACCAACTACGGTCCAGTTAGCCCTCGCCATTGATGTGAACCTTC *****
CNGB3__with_modified CNGB3__modified_ORF native__CNGB3__ORF	AGCATCATCAGCAAAAGTTGACTTGTTCAAGGGTTGTGATACACAGATGATTATGACATG AGCATCATCAGCAAAAGTTGACTTGTTCAAGGGTTGTGATACACAGATGATTATGACATG AGCATCATCAGCAAAAGTTGACTTGTTCAAGGGTTGTGATACACAGATGATTATGACATG *****
CNGB3__with_modified CNGB3__modified_ORF native__CNGB3__ORF	TTGCTAAGATTGAATCCGTTCTCTATTTGCCTGGTGACTTTGCTGCAAAAAGGGAGAA TTGCTAAGATTGAATCCGTTCTCTATTTGCCTGGTGACTTTGCTGCAAAAAGGGAGAA TTGCTAAGATTGAATCCGTTCTCTATTTGCCTGGTGACTTTGCTGCAAAAAGGGAGAA *****
CNGB3__with_modified CNGB3__modified_ORF native__CNGB3__ORF	ATTGGCAAGGAAATGTATATCATCAAGCATGGAGAAAGTCCAAAGTCTTTGGAGGCCCTGAT ATTGGCAAGGAAATGTATATCATCAAGCATGGAGAAAGTCCAAAGTCTTTGGAGGCCCTGAT ATTGGCAAGGAAATGTATATCATCAAGCATGGAGAAAGTCCAAAGTCTTTGGAGGCCCTGAT *****
CNGB3__with_modified CNGB3__modified_ORF native__CNGB3__ORF	GCTACTAAAGTTCTGTTACTCTGAAAGCTGGTGGTGGTGGTGGAGAAATCAGCCTTCTA GCTACTAAAGTTCTGTTACTCTGAAAGCTGGTGGTGGTGGTGGAGAAATCAGCCTTCTA GCTACTAAAGTTCTGTTACTCTGAAAGCTGGTGGTGGTGGTGGAGAAATCAGCCTTCTA *****
CNGB3__with_modified CNGB3__modified_ORF native__CNGB3__ORF	GCAGCAGGAGGAGGAAACCGTCGAACCTGCCAATGTGGTGGCCACGGGTTTGCCCAATCTT GCAGCAGGAGGAGGAAACCGTCGAACCTGCCAATGTGGTGGCCACGGGTTTGCCCAATCTT GCAGCAGGAGGAGGAAACCGTCGAACCTGCCAATGTGGTGGCCACGGGTTTGCCCAATCTT *****

FIG. 4
(continued)

CNGB3__with_modified
CNGB3__modified_ORF
native_CNGB3__ORF

TTAACTCTAGACAAAAGACCCCTCAAGAAATTTCTAGTGCATTATCCAGATTCTGAAAGA
TTAACTCTAGACAAAAGACCCCTCAAGAAATTTCTAGTGCATTATCCAGATTCTGAAAGA
TTAACTCTAGACAAAAGACCCCTCAAGAAATTTCTAGTGCATTATCCAGATTCTGAAAG

CNGB3__with_modified
CNGB3__modified_ORF
native_CNGB3__ORF

ATCCTCATGAAGAAAGCCAGAGTGCTTTTAAAGCAGAAAGGCTAAGACCGCAGAAAGCAACC
ATCCTCATGAAGAAAGCCAGAGTGCTTTTAAAGCAGAAAGGCTAAGACCGCAGAAAGCAACC
ATCCTCATGAAGAAAGCCAGAGTGCTTTTAAAGCAGAAAGGCTAAGACCGCAGAAAGCAACC

CNGB3__with_modified
CNGB3__modified_ORF
native_CNGB3__ORF

CCTCCAAGAAAAGATCTTGCCCTCCTCTCCCAAGAAAGAGACACCCAAACTGTTT
CCTCCAAGAAAAGATCTTGCCCTCCTCTCCCAAGAAAGAGACACCCAAACTGTTT
CCTCCAAGAAAAGATCTTGCCCTCCTCTCCCAAGAAAGAGACACCCAAACTGTTT

CNGB3__with_modified
CNGB3__modified_ORF
native_CNGB3__ORF

AAAACTCTCTAGGAGGCACAGGAAAAGCAAGTCTTCCAAGACTACTCAAAATTGAAGCGA
AAAACTCTCTAGGAGGCACAGGAAAAGCAAGTCTTCCAAGACTACTCAAAATTGAAGCGA
AAAACTCTCTAGGAGGCACAGGAAAAGCAAGTCTTCCAAGACTACTCAAAATTGAAGCGA

CNGB3__with_modified
CNGB3__modified_ORF
native_CNGB3__ORF

GACCAAGCAGCTCAGAAGAAAAGAAAATTTCTGAAGGAGGAGAGGAAAAGAAAAT
GACCAAGCAGCTCAGAAGAAAAGAAAATTTCTGAAGGAGGAGAGGAAAAGAAAAT
GACCAAGCAGCTCAGAAGAAAAGAAAATTTCTGAAGGAGGAGAGGAAAAGAAAAT

CNGB3__with_modified
CNGB3__modified_ORF
native_CNGB3__ORF

GAAGATAACAAAAAGAAAATGAAGATAAACAAAAAGAAAATGAAGATAAGGAAAAGAA
GAAGATAACAAAAAGAAAATGAAGATAAACAAAAAGAAAATGAAGATAAGGAAAAGAA
GAAGATAACAAAAAGAAAATGAAGATAAACAAAAAGAAAATGAAGATAAGGAAAAGAA

CNGB3__with_modified
CNGB3__modified_ORF
native_CNGB3__ORF

AATGAAGATAAAGATAAAGGAAGACAGAGCCAGAAAGCCTGACAGACCTGAATGT
AATGAAGATAAAGATAAAGGAAGACAGAGCCAGAAAGCCTGACAGACCTGAATGT
AATGAAGATAAAGATAAAGGAAGACAGAGCCAGAAAGCCTGACAGACCTGAATGT

CNGB3__with_modified
CNGB3__modified_ORF
native_CNGB3__ORF

ACAGCAAGTCCTATTGTCAGTGGAGGAAGAACCCCACTCAGTTAGAAGGACAGTTTACCC
ACAGCAAGTCCTATTGTCAGTGGAGGAAGAACCCCACTCAGTTAGAAGGACAGTTTACCC
ACAGCAAGTCCTATTGTCAGTGGAGGAAGAACCCCACTCAGTTAGAAGGACAGTTTACCC

FIG. 4
(continued)

FIG. 4
(continued)

CNGB3__with__modified	AGAGGGACTTCTCGTCAATCACTCATTATCAGCATGGCTCCTTCTGCTGAGGGCGGAGAA
CNGB3__modified__ORF	AGAGGGACTTCTCGTCAATCACTCATTATCAGCATGGCTCCTTCTGCTGAGGGCGGAGAA
native__CNGB3__ORF	AGAGGGACTTCTCGTCAATCACTCATTATCAGCATGGCTCCTTCTGCTGAGGGCGGAGAA

CNGB3__with__modified	GAGGTTCTTACTATTGAAGTCAAAGAAAGGCTAAGCAATGATCATAACTGCAG-MDIFI
CNGB3__modified__ORF	GAGGTTCTTACTATTGAAGTCAAAGAAAGGCTAAGCAATGA-----
native__CNGB3__ORF	GAGGTTCTTACTATTGAAGTCAAAGAAAGGCTAAGCAATGA-----

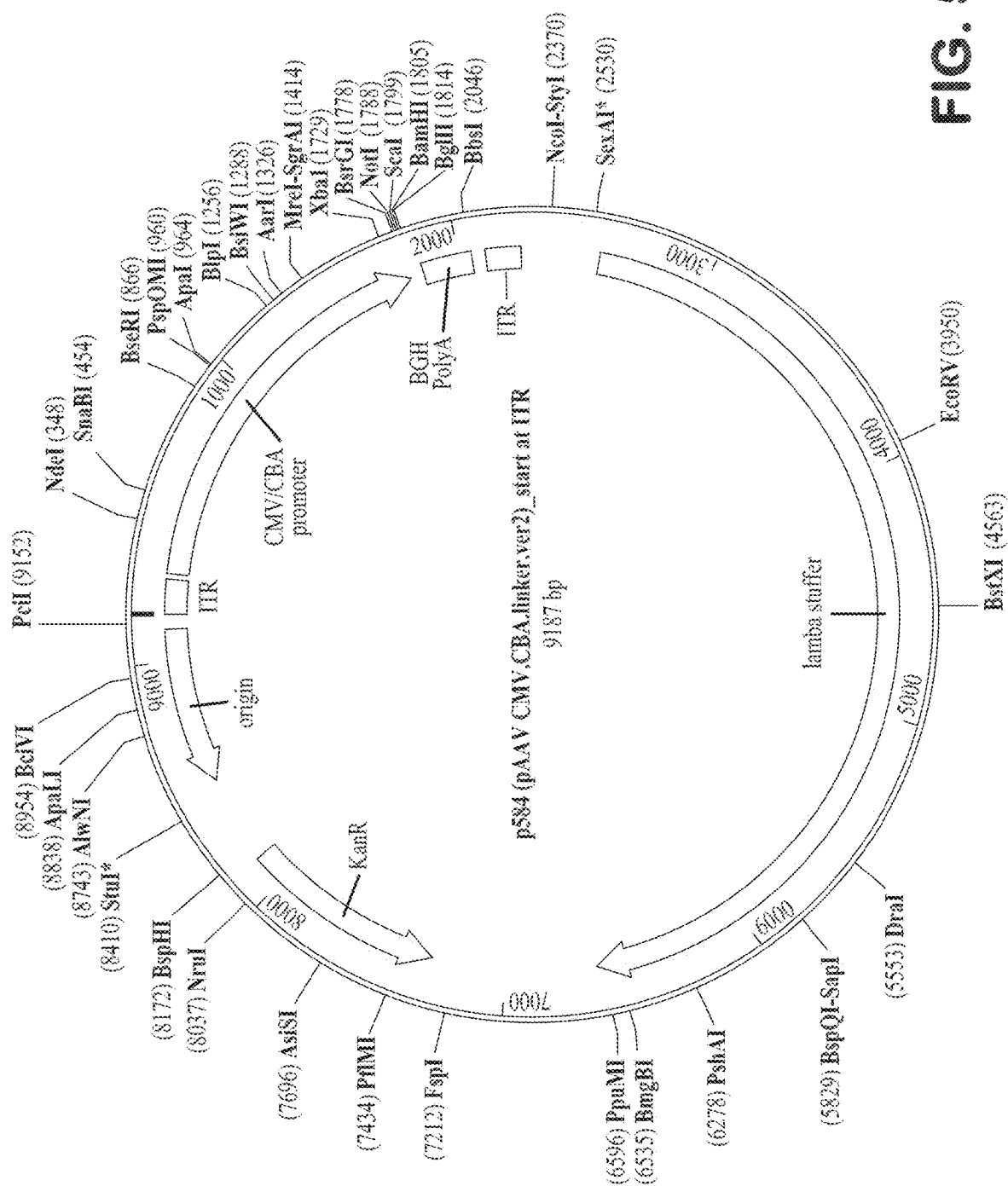
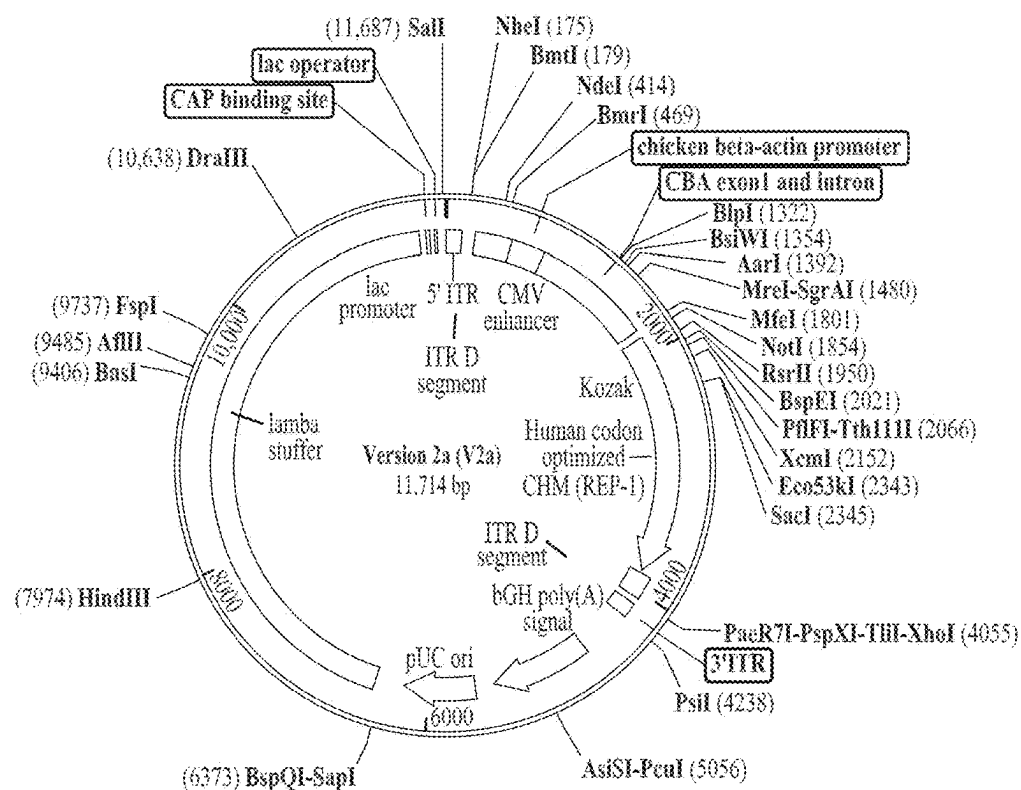


FIG. 5

Version 2a (V2a)

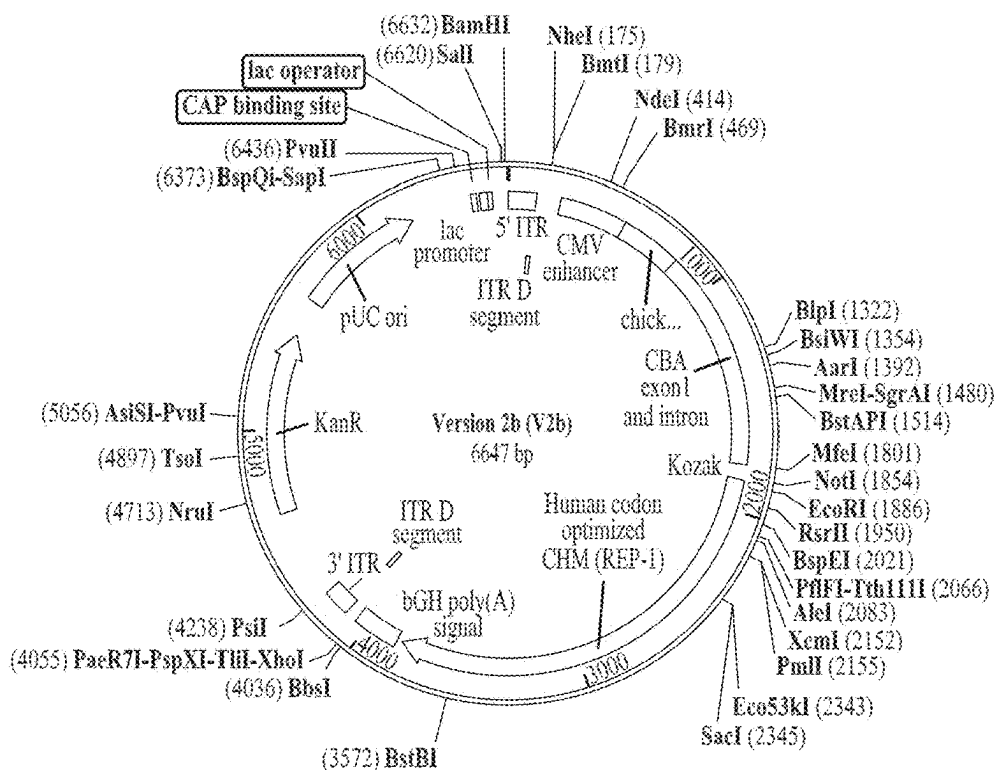


Features

Feature	Location	Size	Directionality
5' ITR	1...130	130	==
ITR D segment	113...130	18	==
CMV enhancer	241...544	304	==
Chicken beta-actin promoter	546...823	278	=>
CBA exon1 and intron	824...1795	972	=>
Kozak	1859...1864	6	=>
Human codon optimized CHM (REP-1)	1865...3826	1962	=>
bGH poly(A) signal	3847...4054	208	==
3' ITR	4104...4233	130	==
ITR D segment	4104...4121	18	==
KanR	4631...5440	810	=>
pUC ori	5612...6200	589	=>
lamba stuffer	6437...11,503	5067	==
CAP binding site	11,555...11,576	22	==
lac promoter	11,591...11,621	31	==
lac operator	11,629...11,645	17	==

FIG. 6

Version 2b (V2b)

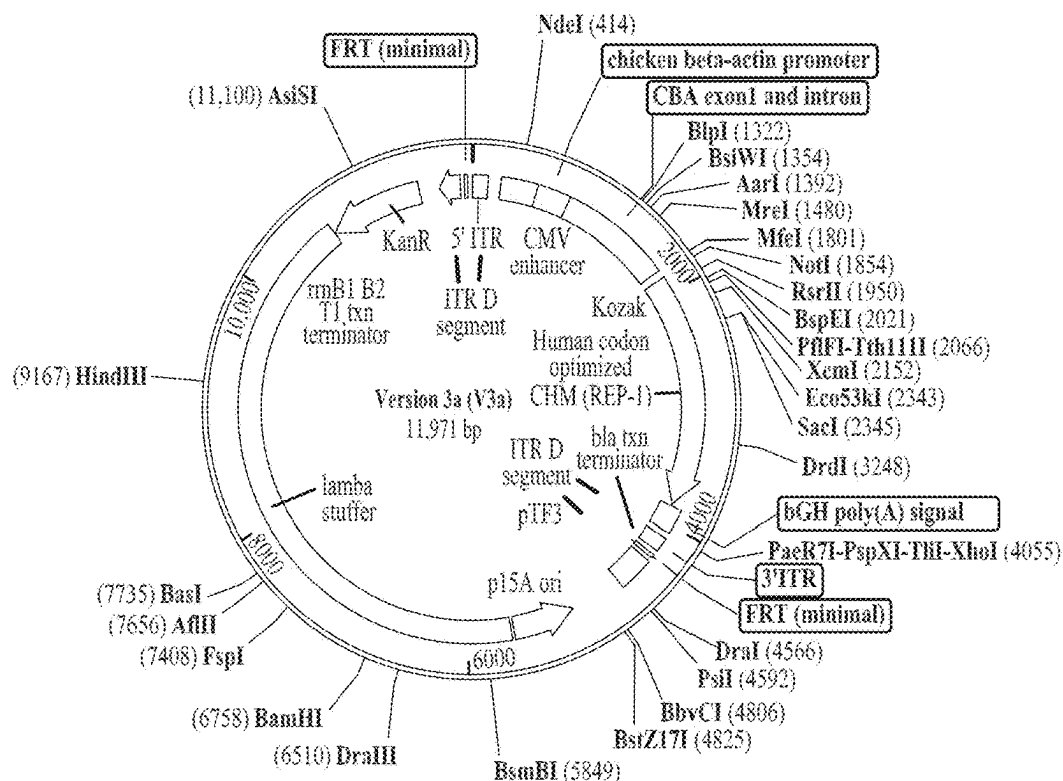


Features

Feature	Location	Size	Directionality
5' ITR	1...130	130	==
ITR D segment	113...130	18	==
CMV enhancer	241...544	304	==
Chicken beta-actin promoter	546...823	278	=>
CBA exon1 and intron	824...1795	972	=>
Kozak	1859...1864	6	=>
Human codon optimized CHM (REP-1)	1865...3826	1962	=>
bGH poly(A) signal	3847...4054	208	==
3' ITR	4104...4233	130	==
ITR D segment	4104...4121	18	==
KanR	4631...5440	810	=>
pUC ori	5612...6200	589	=>
CAP binding site	6488...6509	22	==
lac promoter	6524...6554	31	==
lac operator	6562...6578	17	==

FIG. 7

Version 3a (V3a)

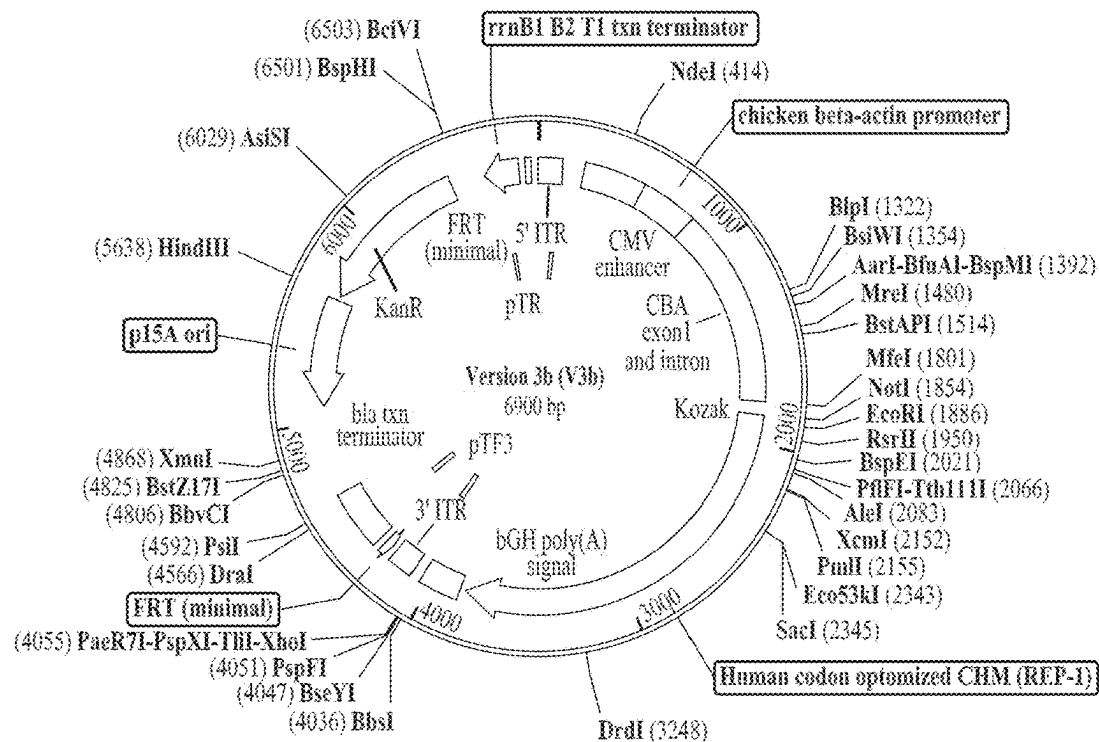


Features

Feature	Location	Size	Directionality
5' ITR	1...130	130	==
ITR D segment	113...130	18	==
CMV enhancer	241...544	304	==
Chicken beta-actin promoter	546...823	278	=>
CBA exon1 and intron	824...1795	972	=>
Kozak	1859...1864	6	=>
Human codon optimized CHM (REP-1)	1865...3826	1962	=>
bGH poly(A) signal	3847...4054	208	==
3' ITR	4104...4233	130	==
ITR D segment	4104...4121	18	==
FRT (minimal)	4264...4297	34	<=
bla txn terminator	4330...4630	301	==
pTF3	4421...4446	26	==
p15A ori	5077...5622	546	<=
lamba stuffer	5643...10,709	5067	==
KanR	10,715...11,524	810	<=
rmB1 B2 T1 txn terminator	11,703...11,877	175	<=
pTR	11,778...11,794	17	==
FRT (minimal)	11,909...11,942	34	=>

FIG. 8

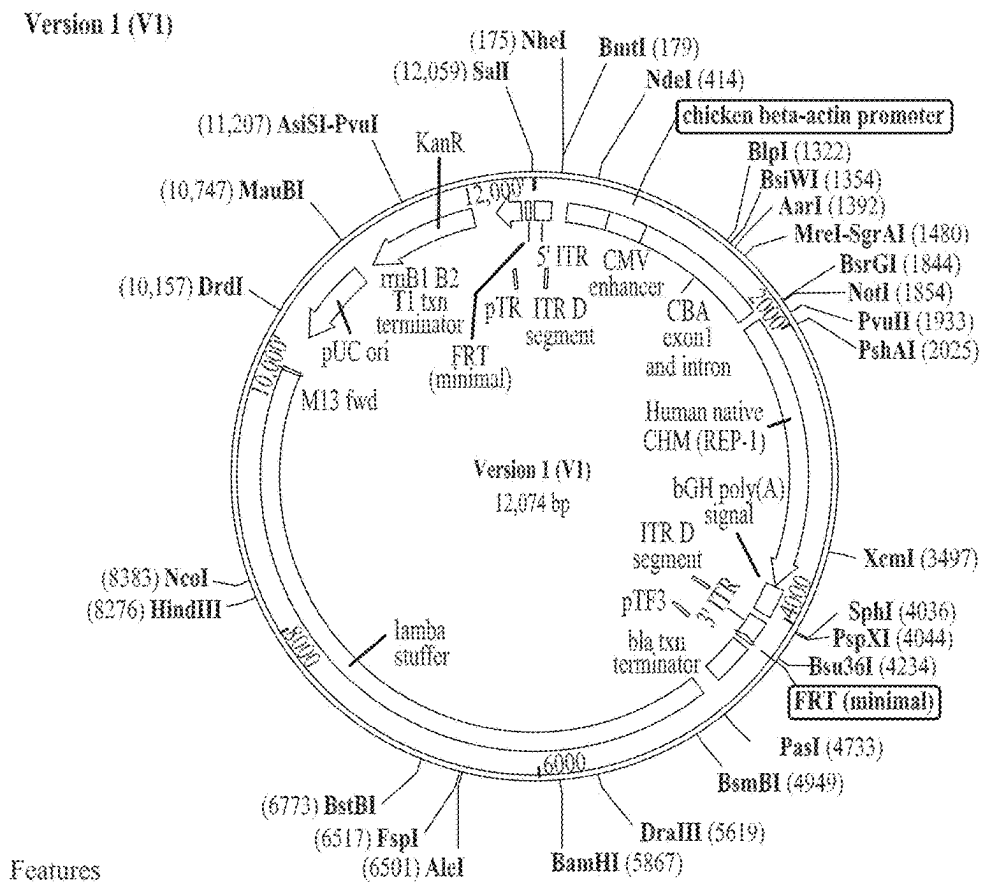
Version 3a (V3a)



Features

Feature	Location	Size	Directionality
5' ITR	1...130	130	==
ITR D segment	113...130	18	==
CMV enhancer	241...544	304	==
Chicken beta-actin promoter	546...823	278	=>
CBA exon1 and intron	824...1795	972	=>
Kozak	1859...1864	6	=>
Human codon optimized CHM (REP-1)	1865...3826	1962	=>
bGH poly(A) signal	3847...4054	208	==
3' ITR	4104...4233	130	==
ITR D segment	4104...4121	18	==
FRT (minimal)	4264...4297	34	<=
bla txn terminator	4330...4630	301	==
pTF3	4421...4446	26	==
p15A ori	5077...5622	546	<=
KanR	5644...6453	810	<=
rrnB1 B2 T1 txn terminator	6632...6806	175	<=
pTR	6707...6723	17	==
FRT (minimal)	6838...6871	34	=>

FIG. 9



Feature	Location	Size	Directionality
5' ITR	1...130	130	==
ITR D segment	113...130	18	==
CMV enhancer	241...544	304	==
Chicken beta-actin promoter	546...823	278	=>
CBA exon1 and intron	824...1795	972	=>
Human native CHM (REP-1)	1861...3822	1962	=>
bGH poly(A) signal	3836...4043	208	==
3' ITR	4093...4222	130	==
ITR D segment	4093...4110	18	==
FRT (minimal)	4250...4283	34	<=
bla txn terminator	4316...4616	301	==
pTF3	4407...4432	26	==
lambda stuffer	4752...9818	5067	==
M13 fwd	9824...9840	17	<=
pUC ori	10,110...10,698	589	<=
KanR	10,822...11,631	810	<=
rmB1 B2 T1 txn terminator	11,810...11,984	175	<=
pTR	11,885...11,901	17	==
FRT (minimal)	12,016...12,049	34	==

FIG. 10

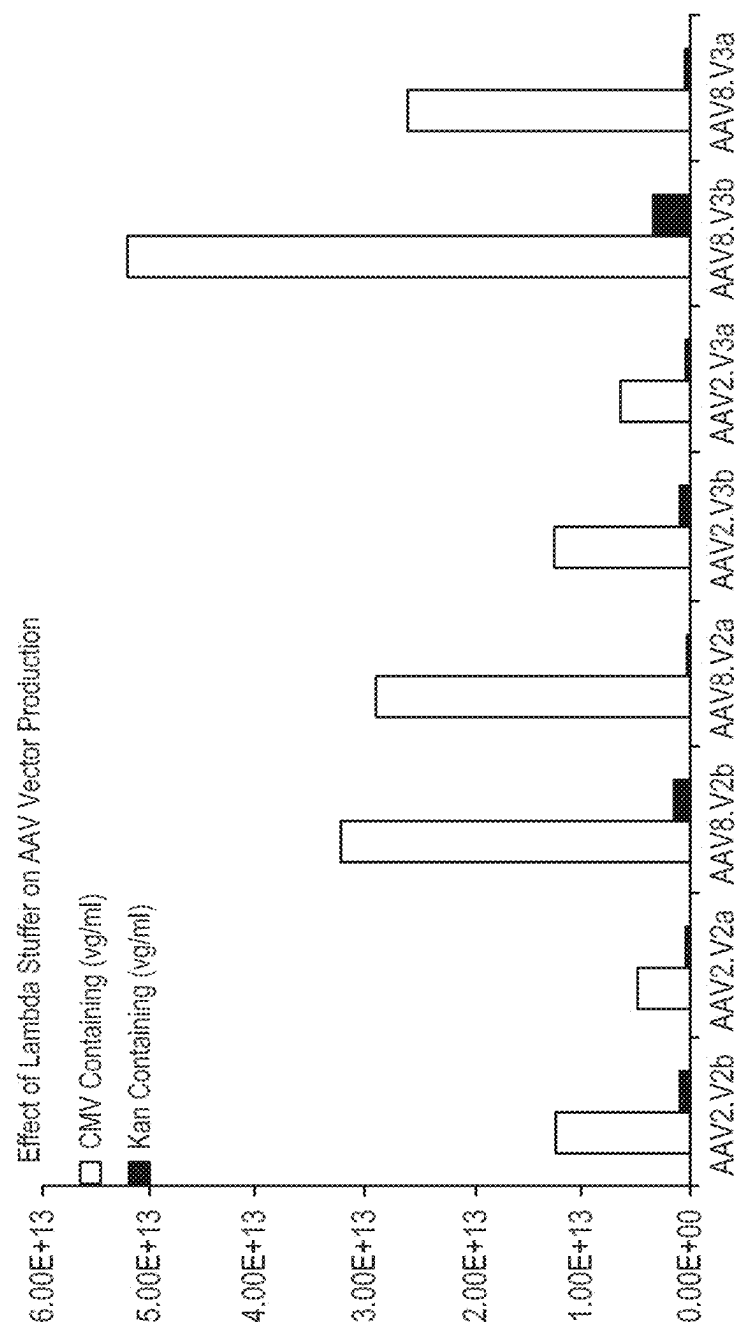


FIG. 11

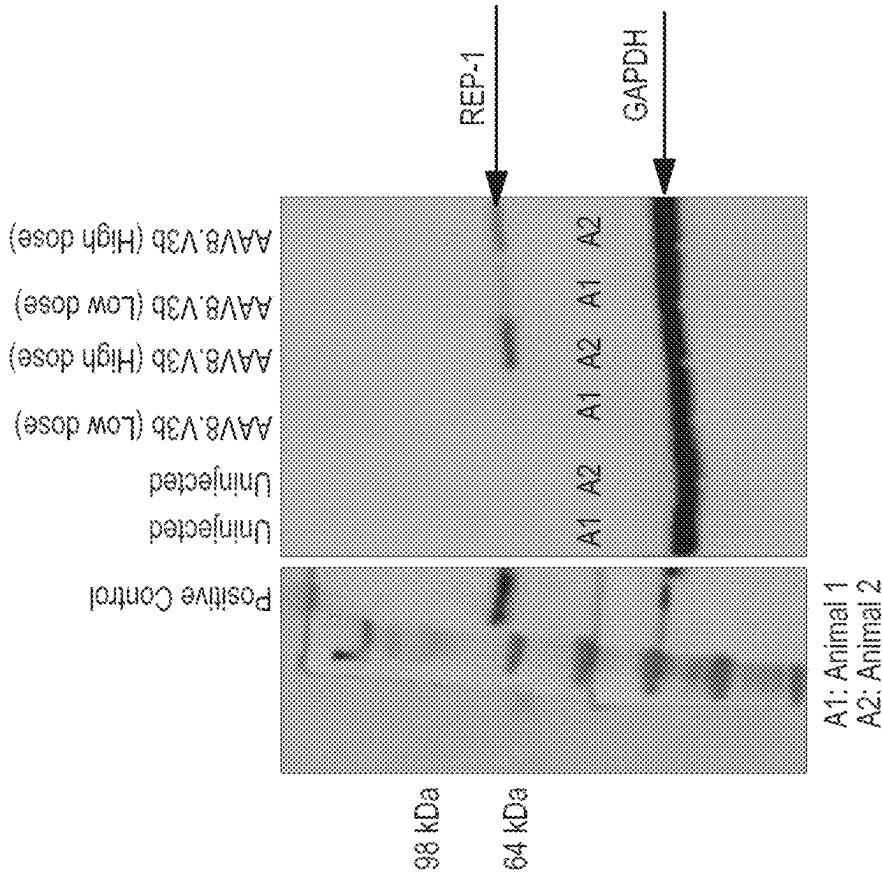


FIG. 12B

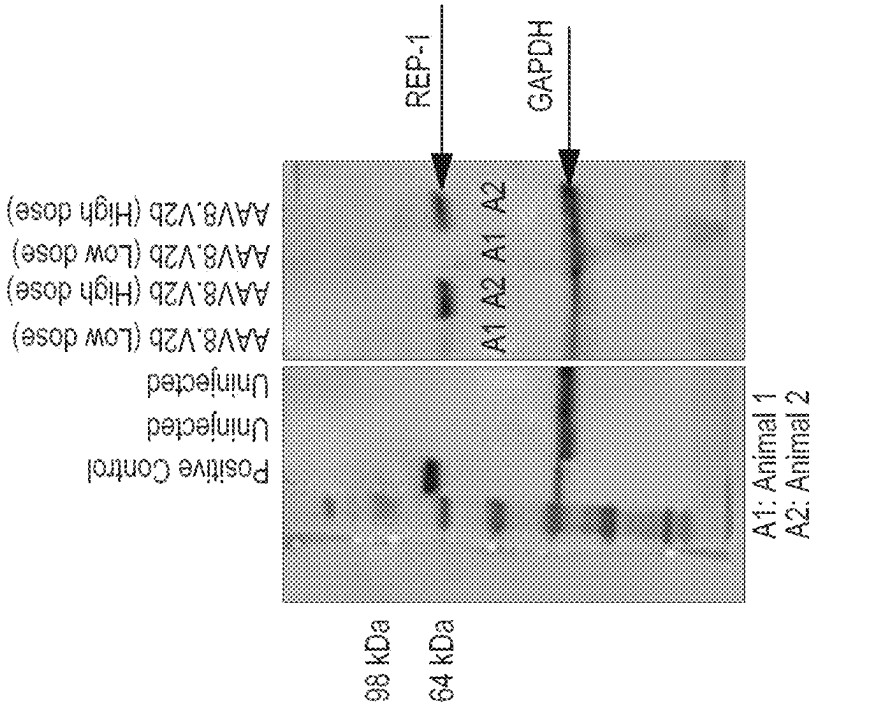


FIG. 12A

GENE THERAPY FOR OCULAR DISORDERS**CROSS-REFERENCE TO RELATED APPLICATIONS**

This Application is a divisional of U.S. patent application Ser. No. 16/061,530, filed Jun. 12, 2018, which is a US National Stage of PCT/US2016/066402, filed Dec. 13, 2016, and which claims priority to U.S. Provisional Patent Application No. 62/266,789, filed Dec. 14, 2015. Each of these applications are incorporated herein by reference.

INCORPORATION-BY-REFERENCE OF MATERIAL SUBMITTED IN ELECTRONIC FORM

Applicant hereby incorporates by reference the Sequence Listing material filed in electronic form herewith. This file is labeled "16-7660_Seq_Listing_new_ST25.txt".

BACKGROUND OF THE INVENTION

Choroideremia (CHM) is an X-linked inherited retinal disease characterized by the degeneration of photoreceptors, retinal pigment epithelium (RPE) and choriocapillaris. Symptoms develop in the 1st or 2nd decade of life with complaints of poor night vision (nyctalopia) and progressive loss of peripheral vision. Visual fields constrict as the disease progresses. This culminates with loss of central vision (visual acuity) and blindness as early as the fourth decade of life. More than 140 mutations in the CHM gene have been found to cause choroideremia. Mutations may lead to the production of an abnormally small, nonfunctional and/or unstable Rab escort protein-1 (REP-1) protein, a decrease in the protein's function or loss of REP-1 protein production. Lack of normal REP-1 disrupts the ability of Rab proteins to aid in intracellular trafficking. The immobility of proteins and organelles within the cell causes the cells to function poorly and to die prematurely.

The choroideremia gene, CHM, encodes Rab Escort Protein-1 (REP-1), a 653 amino acid protein involved in regulation of membrane trafficking. Since the CHM locus is on the X-chromosome, choroideremia is typically only diagnosed in males. Although female carriers of the disease are usually asymptomatic, retinal exams often reveal a patchy degeneration of the retina and RPE and female individuals can be affected depending on the extent of X-inactivation of the normal X chromosome (lyonization). Coussa, RG, Traboulsi, EI (2012) Choroideremia: a review of general findings and pathogenesis, *Ophthalmic Genet* 33(2): 57-65, which is incorporated herein by reference. See also, Vasireddy et al, AAV-mediated gene therapy for choroideremia: preclinical studies in personalized models. *PLoS One*. 2013 May 7;8(5):e61396, which is incorporated herein by reference.

Achromatopsia is a heterogeneous group of autosomal recessive inherited retinal diseases characterized by early onset reduced visual acuity, impaired or complete color blindness, nystagmus, photoaversion and loss of cone photoreceptor function. About 80% of achromatopsia patients show mutations in the alpha or beta subunit (A3 and B3) of the cGMP controlled cation channel cyclic nucleotide-gated channel (CNG) of cone photoreceptors. Homologous to the human disease, *Cnga3* deficient mice reveal a loss of cone specific functionality leading to malfunction and degeneration of affected cone photoreceptors.

Therefore, compositions useful for expressing *CNGA3* or *CNGB3* in human subjects are needed.

SUMMARY OF THE INVENTION

Choroideremia (CHM) is an X-linked retinal degeneration that is symptomatic in the 1st or 2nd decade of life causing nyctalopia and loss of peripheral vision. The disease progresses through mid-life, when most patients become blind. CHM is a favorable target for gene augmentation therapy, as the disease is due to loss of function of a protein necessary for retinal cell health, Rab Escort Protein 1 (REP1), which is encoded by the CHM gene. The CHM cDNA can be packaged in recombinant adeno-associated virus (rAAV), which has an established track record in human gene therapy studies. In addition, there are sensitive and quantitative assays to document REP1 activity, including its ability to prenylate Rab proteins such as Rab27 and to correct a defect in Rab27 localization and trafficking due to lack of prenylation in REP-1 deficient cells.

In one aspect, a codon optimized cDNA sequence encoding Rab Escort Protein-1 (REP-1) is provided. In one embodiment, the codon optimized cDNA sequence is a variant of SEQ ID NO: 3. In another embodiment, the codon optimized cDNA sequence is SEQ ID NO: 1. In another embodiment, the cDNA sequence is codon optimized for expression in humans.

In another aspect, an expression cassette includes a codon optimized nucleic acid sequence that encodes REP-1. In one embodiment, the expression cassette includes the cDNA sequence of SEQ ID NO: 1. In still other embodiments, the REP-1 encoding sequence is positioned between 5' and 3' AAV ITR sequences.

In another embodiment, an adeno-associated virus (AAV) vector is provided. The AAV vector includes an AAV capsid and a nucleic acid sequence comprising AAV inverted terminal repeat sequences and a nucleic acid sequence encoding human Rab Escort Protein-1 (REP-1), and expression control sequences that direct expression of the REP-1 in a host cell. In one embodiment, the REP-1 sequence encodes a full length REP-1 protein. In one embodiment, the REP-1 sequence is the protein sequence of SEQ ID NO: 2.

In one aspect, a codon optimized cDNA sequence encoding cyclic nucleotide gated channel alpha 3 (*CNGA3*) is provided. In one embodiment, the codon optimized cDNA sequence is a variant of SEQ ID NO: 13 or SEQ ID NO: 15. In another embodiment, the codon optimized cDNA sequence is SEQ ID NO: 9 or SEQ ID NO: 11. In another embodiment, the cDNA sequence is codon optimized for expression in humans.

In another aspect, an expression cassette includes a codon optimized nucleic acid sequence that encodes cyclic nucleotide gated channel alpha 3 (*CNGA3*). In one embodiment, the expression cassette includes the cDNA sequence of SEQ ID NO: 9, SEQ ID NO: 11, SEQ ID NO: 13, or SEQ ID NO: 15. In still other embodiments, the *CNGA3* encoding sequence is positioned between 5' and 3' AAV ITR sequences.

In another aspect, an expression cassette includes a codon optimized nucleic acid sequence that encodes cyclic nucleotide gated channel alpha 3 (*CNGB3*). In one embodiment, the expression cassette includes the cDNA sequence of SEQ ID NO: 19 or SEQ ID NO: 21 or SEQ ID NO: 23. In still other embodiments, the *CNGB3* encoding sequence is positioned between 5' and 3' AAV ITR sequences.

In another embodiment, an adeno-associated virus (AAV) vector is provided. The AAV vector includes an AAV capsid

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and a nucleic acid sequence comprising AAV inverted terminal repeat sequences and a nucleic acid sequence encoding human CNGA3, and expression control sequences that direct expression of the CNGA3 in a host cell. In one embodiment, the CNGA3 sequence encodes a full length CNGA3 protein. In one embodiment, the CNGA3 sequence is the protein sequence of SEQ ID NO: 10, SEQ ID NO: 12 or SEQ ID NO: 14.

In another embodiment, an adeno-associated virus (AAV) vector is provided. The AAV vector includes an AAV capsid and a nucleic acid sequence comprising AAV inverted terminal repeat sequences and a nucleic acid sequence encoding human CNGB3, and expression control sequences that direct expression of the CNGB3 in a host cell. In one embodiment, the CNGB3 sequence encodes a full length CNGB3 protein. In one embodiment, the CNGB3 sequence is the protein sequence of SEQ ID NO: 20.

In another aspect, an adeno-associated virus (AAV) vector is provided which comprises an AAV8 capsid and an expression cassette, wherein said expression cassette comprises nucleic acid sequences encoding REP-1, inverted terminal repeat sequences and expression control sequences that direct expression of REP-1 in a host cell.

In yet another aspect, an adeno-associated virus (AAV) vector is provided which comprises an AAV8 capsid and an expression cassette, wherein said expression cassette comprises nucleic acid sequences encoding CNGA3, inverted terminal repeat sequences and expression control sequences that direct expression of CNGA3 in a host cell.

In yet another aspect, an adeno-associated virus (AAV) vector is provided which comprises an AAV8 capsid and an expression cassette, wherein said expression cassette comprises nucleic acid sequences encoding CNGB3, inverted terminal repeat sequences and expression control sequences that direct expression of CNGB3 in a host cell.

In another aspect, an adeno-associated virus (AAV) vector is provided which comprises an AAV2 capsid and an expression cassette, wherein said expression cassette comprises nucleic acid sequences encoding REP-1, inverted terminal repeat sequences and expression control sequences that direct expression of REP-1 in a host cell.

In yet another aspect, an adeno-associated virus (AAV) vector is provided which comprises an AAV2 capsid and an expression cassette, wherein said expression cassette comprises nucleic acid sequences encoding CNGA3, inverted terminal repeat sequences and expression control sequences that direct expression of CNGA3 in a host cell.

In yet another aspect, an adeno-associated virus (AAV) vector is provided which comprises an AAV2 capsid and an expression cassette, wherein said expression cassette comprises nucleic acid sequences encoding CNGB3, inverted terminal repeat sequences and expression control sequences that direct expression of CNGB3 in a host cell.

In another aspect, a pharmaceutical composition is provided which includes a pharmaceutically acceptable carrier, diluent, excipient and/or adjuvant and a least a viral vector as described herein.

In yet a further aspect a pharmaceutical composition comprises a pharmaceutically acceptable carrier, diluent, excipient and/or adjuvant and the nucleic acid sequence, a plasmid, a vector, or a viral vector, such as the rAAV, described specifically herein.

In another aspect, a method for treating choroideremia is provided. In one embodiment, the method includes administering a composition which includes the AAV vector which encodes REP-1, as described herein, to a subject in need thereof.

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In another aspect, a method for treating achromatopsia is provided. In one embodiment, the method includes administering a composition which includes the AAV vector which encodes CNGA3, as described herein, to a subject in need thereof.

In another aspect, a method for treating achromatopsia is provided. In one embodiment, the method includes administering a composition which includes the AAV vector which encodes CNGB3, as described herein, to a subject in need thereof.

In yet another aspect, a plasmid for producing an AAV vector is provided. In one embodiment, the plasmid includes the codon optimized cDNA sequence encoding REP-1 as described herein. In another embodiment, the plasmid includes the codon optimized cDNA sequence encoding CNGA3 as described herein. In another embodiment, the plasmid includes a codon optimized cDNA sequence encoding CNGB3 which is a sequence sharing at least 70% identity with SEQ ID NO: 19 or SEQ ID NO: 21. In one embodiment, the plasmid is modular.

In another aspect, a method of generating a rAAV virus is provided. The method includes culturing a packaging cell carrying the plasmid described herein in the presence of sufficient viral sequences to permit packaging of the gene expression cassette viral genome into an infectious AAV envelope or capsid. In another, aspect, a recombinant AAV produced according to the method is provided.

Other aspects and advantages of the invention will be readily apparent from the following detailed description of the invention.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1A and FIG. 1B are gels showing REP-1 protein expression in vitro after transfection of cultured 84-31 HEK cells. The first lane of each gel shows expression of codon-optimized REP-1 as described herein, expressed from plasmid p944. The second lane shows expression of native REP-1 from plasmid p742. The third lane shows endogenous expression of REP-1 by 84-31 cells that were not transfected with a plasmid. The last lane is a blank. The gels demonstrate that the codon-optimized REP-1 sequence, as described herein, results in a higher level of protein expression than the native REP-1 sequence, and that levels of expression from the exogenously transfected plasmids are many-fold higher than endogenous REP-1 expression.

FIG. 2 is an alignment of the native REP-1 coding sequence of SEQ ID NO: 1 vs. the codon optimized REP-1 coding sequence of SEQ ID NO: 3.

FIG. 3 is an alignment of the native CNGA3 coding sequence of SEQ ID NO: 13 vs. the codon optimized CNGA3 coding sequence of SEQ ID NO: 9.

FIG. 4 is an alignment of CNGB3 native ORF (SEQ ID NO: 19) vs. CNGB3 modified ORF (SEQ ID NO: 21) vs. CNGB3 modified orf with modified ends (SEQ ID NO: 23). Point mutations are highlighted.

FIG. 5 is a plasmid map of p584, described herein. The sequence of p584 is shown in SEQ ID NO: 7.

FIG. 6 is a plasmid map of AAV.hCHMco. Version 2a, described herein. The sequence of Version 2a is shown in SEQ ID NO: 25.

FIG. 7 is a plasmid map of AAV.hCHMco. Version 2b, described herein. The sequence of Version 2b is shown in SEQ ID NO: 26.

FIG. 8 is a plasmid map of AAV.hCHMco. Version 3a, described herein. The sequence of Version 3a is shown in SEQ ID NO: 27.

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FIG. 9 is a plasmid map of AAV.hCHMco. Version 3b, described herein. The sequence of Version 3b is shown in SEQ ID NO: 28.

FIG. 10 is a plasmid map of AAV.hCHM. Version 1, described herein. The sequence of Version 1 is shown in SEQ ID NO: 29. FIG. 11 is a graphic representation of the effect of lambda insert on AAV product impurity. All a-version (lambda containing) vectors have much reduced Kan^r signals from qPCR test.

FIG. 12A is a western blot showing human anti-REP-1 antibody detection of a protein of ~75-80 kDa in ocular tissues of CD-1 mice injected with AAV8.2b at 5E9 (High dose) vector genome copies. Animals injected with AAV8.2b at 5E8 (Low dose) showed a very faint protein band at ~75-80 kDa. FIG. 12B is a Western blot analysis of ocular tissues of AAV8.3b injected CD1 mice (2 mice/group) detected with anti-REP-1 antibody, which revealed the presence of a protein of ~75-80 kDa in one eye injected with low dose and in both eyes injected with high dose of AAV8.3b. In the ocular tissues of uninjected mice there was no REP-1 expression detected.

DETAILED DESCRIPTION OF THE INVENTION

The methods and compositions described herein involve compositions and methods for delivering optimized CHM encoding REP-1 to mammalian subjects for the treatment of ocular disorders, primarily blinding diseases such as choroideremia. In addition, methods and compositions described herein involve compositions and methods for delivering optimized CNGA3 or CNGB3 to mammalian subjects for the treatment of ocular disorders, primarily blinding diseases such as achromatopsia. In one embodiment, such compositions involve codon optimization of the REP-1, CNGA3 or CNGB3 coding sequence. It is believed that these features increase the efficacy of the product, and increase safety, since a lower dose of reagent is used. It is anticipated that this optimization of the transgene cassette could theoretically maximize the level of production of the experimental protein compared to levels that can be generated using the endogenous sequence. However, also encompassed herein are compositions which include the native REP1, CNGA3, and CNGB3 coding sequences, as shown in SEQ ID NO: 3, SEQ ID NO: 13 and SEQ ID NO: 19, respectively. It is to be understood that when an embodiment is described for either REP-1, CNGA3 or CNGB3, a similar embodiment is intended to be recited for the other.

Technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this invention belongs and by reference to published texts, which provide one skilled in the art with a general guide to many of the terms used in the present application. The definitions contained in this specification are provided for clarity in describing the components and compositions herein and are not intended to limit the claimed invention.

The choroideremia gene, CHM, encodes Rab Escort Protein-1 (REP-1), a 653 amino acid protein thought to be involved in membrane trafficking. As used herein, the terms "REP-1" and "CHM" are used interchangeably when referring to the coding sequence. Since the CHM locus is on the X-chromosome, choroideremia is typically only diagnosed in males. Although female carriers of the disease are usually asymptomatic, retinal exams often reveal a patchy degeneration of the retina and RPE and female individuals can be affected depending on the extent of X-inactivation of the

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normal X chromosome (lyonization). See, Coussa, cited above. The native amino acid sequence encoding human REP-1 is reported at GenBank accession number P24386, and reproduced here in SEQ ID NO: 2. The native human nucleic acid sequence of CHM is reproduced here at SEQ ID NO: 3 (accession no. NM_000390.2).

Cyclic nucleotide-gated (CNG) ion channels are key mediators underlying signal transduction in retinal and olfactory receptors. Genetic defects in CNGA3 and CNGB3, encoding two structurally related subunits of cone CNG channels, are known to lead to achromatopsia. CNGA3 is a 694 amino acid protein. CNGB is an 809 amino acid protein.

Achromatopsia is a heterogeneous group of congenital, autosomal recessive retinal disorders that manifest by early onset cone photoreceptor dysfunction, severely reduced visual acuity, impaired or complete color blindness and photophobia. The native nucleic acid sequence encoding human CNGA3 is reported at GenBank accession no. XM_011210554.1, and reproduced in SEQ ID NO: 13. The native nucleic acid sequence encoding human CNGA3 is reported at GenBank accession no. XM_011210554.1, and reproduced in SEQ ID NO: 13. The native nucleic acid sequence for the human CNGA3 X1 variant, which includes an additional exon, is reported at GenBank accession no. NM_001298.2, and reproduced in SEQ ID NO: 15. The native nucleic acid sequence encoding human CNGB3 is reproduced in SEQ ID NO: 19.

In certain embodiments of this invention, a subject has an "ocular disorder", for which the components, compositions and methods of this invention are designed to treat. As used herein, the term "subject" as used herein means a mammalian animal, including a human, a veterinary or farm animal, a domestic animal or pet, and animals normally used for clinical research. In one embodiment, the subject of these methods and compositions is a human. Still other suitable subjects include, without limitation, murine, rat, canine, feline, porcine, bovine, ovine, non-human primate and others. As used herein, the term "subject" is used interchangeably with "patient".

As used herein "ocular disorder" includes, cone-rod dystrophies and retinal diseases including, without limitation, Stargardt disease (autosomal dominant or autosomal recessive), retinitis pigmentosa, and pattern dystrophy. In one embodiment, the subject has achromatopsia. In another embodiment, the subject has choroideremia or an X-linked hereditary retinal degeneration. Clinical signs of such ocular diseases include, but are not limited to, decreased peripheral vision, decreased central (reading) vision, decreased night vision, loss of color perception, reduction in visual acuity, decreased photoreceptor function, pigmentary changes, and ultimately blindness.

As used herein, the term "treatment" or "treating" is defined encompassing administering to a subject one or more compounds or compositions described herein for the purposes of amelioration of one or more symptoms of an ocular disease. "Treatment" can thus include one or more of reducing onset or progression of an ocular disease, preventing disease, reducing the severity of the disease symptoms, or retarding their progression, including the progression of blindness, removing the disease symptoms, delaying onset of disease or monitoring progression of disease or efficacy of therapy in a given subject.

The term "exogenous" as used to describe a nucleic acid sequence or protein means that the nucleic acid or protein does not naturally occur in the position in which it exists in a chromosome, or host cell. An exogenous nucleic acid sequence also refers to a sequence derived from and inserted

into the same host cell or subject, but which is present in a non-natural state, e.g. a different copy number, or under the control of different regulatory elements.

The term “heterologous” as used to describe a nucleic acid sequence or protein means that the nucleic acid or protein was derived from a different organism or a different species of the same organism than the host cell or subject in which it is expressed. The term “heterologous” when used with reference to a protein or a nucleic acid in a plasmid, expression cassette, or vector, indicates that the protein or the nucleic acid is present with another sequence or subsequence which with which the protein or nucleic acid in question is not found in the same relationship to each other in nature.

The terms “percent (%) identity”, “sequence identity”, “percent sequence identity”, or “percent identical” in the context of nucleic acid sequences refers to the bases in the two sequences which are the same when aligned for correspondence. The percent identity is determined by comparing two sequences aligned under optimal conditions over the sequences to be compared. The length of sequence identity comparison may be over the full-length of the REP-1, CNGA3 or CNGB3 coding sequence, or a fragment of at least about 100 to 150 nucleotides, or as desired. However, identity among smaller fragments, e.g. of at least about nine nucleotides, usually at least about 20 to 24 nucleotides, at least about 28 to 32 nucleotides, at least about 36 or more nucleotides, may also be desired. Multiple sequence alignment programs are also available for nucleic acid sequences. Examples of such programs include, “Clustal W”, “CAP Sequence Assembly”, “BLAST”, “MAP”, and “MEME”, which are accessible through Web Servers on the internet. Other sources for such programs are known to those of skill in the art. Alternatively, Vector NTI utilities are also used. There are also a number of algorithms known in the art that can be used to measure nucleotide sequence identity, including those contained in the programs described above. As another example, polynucleotide sequences can be compared using Fasta™, a program in GCG Version 6.1. Commonly available sequence analysis software, more specifically, BLAST or analysis tools provided by public databases may also be used.

The term “isolated” means that the material is removed from its original environment (e.g., the natural environment if it is naturally occurring). For example, a naturally-occurring polynucleotide or polypeptide present in a living animal is not isolated, but the same polynucleotide or polypeptide, separated from some or all of the coexisting materials in the natural system, is isolated, even if subsequently reintroduced into the natural system. Such polynucleotides could be part of a vector and/or such polynucleotides or polypeptides could be part of a composition, and still be isolated in that such vector or composition is not part of its natural environment.

By “engineered” is meant that the nucleic acid sequences encoding the REP-1 or CNGA3 or CNGB3 protein described herein are assembled and placed into any suitable genetic element, e.g., naked DNA, phage, transposon, cosmid, episome, etc., which transfers the REP-1 or CNGA3 or CNGB3 sequences carried thereon to a host cell, e.g., for generating non-viral delivery systems (e.g., RNA-based systems, naked DNA, or the like) or for generating viral vectors in a packaging host cell and/or for delivery to a host cells in a subject. In one embodiment, the genetic element is a plasmid. The methods used to make such engineered constructs are known to those with skill in nucleic acid manipulation and include genetic engineering, recombinant

engineering, and synthetic techniques. See, e.g., Green and Sambrook, *Molecular Cloning: A Laboratory Manual*, Cold Spring Harbor Press, Cold Spring Harbor, NY (2012).

The term “transgene” as used herein means an exogenous or engineered protein-encoding nucleic acid sequence that is under the control of a promoter or expression control sequence in an expression cassette, rAAV genome, recombinant plasmid or production plasmid, vector, or host cell described in this specification. In certain embodiments, the transgene is a human CHM (REP-1) sequence, encoding a functional REP-1 protein. In some embodiments, the transgene is a codon optimized nucleic acid CHM (REP-1) encoding the REP-1 amino acid sequence set forth in SEQ ID NO: 2. In certain embodiments, the transgene is encoded by the sequence set forth in SEQ ID NO: 1. In certain embodiments, the REP-1 transgene is encoded by the sequence set forth in SEQ ID NO: 5. SEQ ID NO: 5 includes modified ends, which include restriction sites for cloning into a plasmid, such as a production plasmid described herein.

In certain embodiments, the transgene is a human CNGA3 sequence, encoding a functional CNGA3 protein. In certain embodiments, the transgene is a codon optimized CNGA3 encoding sequence SEQ ID NO: 10. In certain embodiments, the transgene is encoded by the sequence set forth in SEQ ID NO: 9. In one embodiment, the transgene includes modified ends, such as that shown in SEQ ID NO: 16, SEQ ID NO: 17 or SEQ ID NO: 18, which include restriction sites for cloning into a plasmid, such as a plasmid described herein. In certain embodiments, the transgene is a codon optimized CNGA3 encoding sequence SEQ ID NO: 12. In certain embodiments, the transgene is encoded by the sequence set forth in SEQ ID NO: 11. In certain embodiments, the transgene is encoded by the native coding sequence of CNGA3, which is set forth in SEQ ID NO: 13.

In certain embodiments, the transgene is a human CNGB3 sequence, encoding a functional CNGB3 protein. In certain embodiments, the transgene is a codon optimized CNGB3 encoding sequence which is a sequence sharing at least 70% identity with SEQ ID NO: 19 or 21. In certain embodiments, the transgene is encoded by the sequence set forth in SEQ ID NO: 23. SEQ ID NO: 23 includes modified ends, which include restriction sites for cloning into a plasmid, such as a production plasmid described herein. Nucleotides 13 to 2448 of SEQ ID NO: 23 provide the ORF for CNGB3. In certain embodiments, the transgene is a codon optimized CNGB3 encoding sequence SEQ ID NO: 20. In certain embodiments, the transgene is encoded by the sequence set forth in SEQ ID NO: 19. In certain embodiments, the transgene is encoded by the sequence set forth in SEQ ID NO: 21. In certain embodiments, the transgene includes modified ends for cloning into a plasmid, such as the plasmids described herein. SEQ ID NO: 21 is a novel cDNA sequence in which certain silent mutations have been made to the native coding sequence. Further modifications to the native sequence, as described herein, are contemplated by the invention.

In one embodiment, the nucleic acid sequence encoding REP-1, CNGA or CNGB further comprises a nucleic acid encoding a tag polypeptide covalently linked thereto. The tag polypeptide may be selected from known “epitope tags” including, without limitation, a myc tag polypeptide, a glutathione-S-transferase tag polypeptide, a green fluorescent protein tag polypeptide, a myc-pyruvate kinase tag polypeptide, a His6 tag polypeptide, an influenza virus hemagglutinin tag polypeptide, a flag tag polypeptide, and a maltose binding protein tag polypeptide.

A “vector” as used herein is a nucleic acid molecule into which an exogenous or heterologous or engineered nucleic acid transgene may be inserted which can then be introduced into an appropriate host cell. Vectors preferably have one or more origin of replication, and one or more site into which the recombinant DNA can be inserted. Vectors often have convenient means by which cells with vectors can be selected from those without, e.g., they encode drug resistance genes. Common vectors include plasmids, viral genomes, and (primarily in yeast and bacteria) “artificial chromosomes.” Certain plasmids are described herein.

“Virus vectors” are defined as replication defective viruses containing the exogenous or heterologous CHM (REP-1) or CNGA3 or CNGB3 nucleic acid transgene(s). In one embodiment, an expression cassette as described herein may be engineered onto a plasmid which is used for drug delivery or for production of a viral vector. Suitable viral vectors are preferably replication defective and selected from amongst those which target ocular cells. Viral vectors may include any virus suitable for gene therapy, including but not limited to adenovirus; herpes virus; lentivirus; retrovirus; parvovirus, etc. However, for ease of understanding, the adeno-associated virus is referenced herein as an exemplary virus vector.

A “replication-defective virus” or “viral vector” refers to a synthetic or recombinant viral particle in which an expression cassette containing a gene of interest is packaged in a viral capsid or envelope, where any viral genomic sequences also packaged within the viral capsid or envelope are replication-deficient; i.e., they cannot generate progeny virions but retain the ability to infect target cells. In one embodiment, the genome of the viral vector does not include genes encoding the enzymes required to replicate (the genome can be engineered to be “gutless”—containing only the transgene of interest flanked by the signals required for amplification and packaging of the artificial genome), but these genes may be supplied during production. Therefore, it is deemed safe for use in gene therapy since replication and infection by progeny virions cannot occur except in the presence of the viral enzyme required for replication.

In still another embodiment, the expression cassette, including any of those described herein is employed to generate a recombinant AAV genome.

As used herein, the term “host cell” may refer to the packaging cell line in which a recombinant AAV is produced from a production plasmid. In the alternative, the term “host cell” may refer to any target cell in which expression of the transgene is desired. Thus, a “host cell,” refers to a prokaryotic or eukaryotic cell that contains exogenous or heterologous DNA that has been introduced into the cell by any means, e.g., electroporation, calcium phosphate precipitation, microinjection, transformation, viral infection, transfection, liposome delivery, membrane fusion techniques, high velocity DNA-coated pellets, viral infection and protoplast fusion.

In certain embodiments herein, the term “host cell” refers to cultures of ocular cells of various mammalian species for in vitro assessment of the compositions described herein. In other embodiments herein, the term “host cell” refers to the cells employed to generate and package the viral vector or recombinant virus. Still in other embodiments, the term “host cell” is intended to reference the ocular cells of the subject being treated in vivo for the ocular disease.

As used herein, the term “ocular cells” refers to any cell in, or associated with the function of, the eye. The term may refer to any one of photoreceptor cells, including rod photoreceptors, cone photoreceptors and photosensitive gan-

glion cells, retinal pigment epithelium (RPE) cells, Mueller cells, choroidal cells, bipolar cells, horizontal cells, and amacrine cells. In one embodiment, the ocular cells are the photoreceptor cells. In another embodiment, the ocular cells are RPE cells.

“Plasmids” generally are designated herein by a lower case p preceded and/or followed by capital letters and/or numbers, in accordance with standard naming conventions that are familiar to those of skill in the art. Many plasmids and other cloning and expression vectors that can be used in accordance with the present invention are well known and readily available to those of skill in the art. Moreover, those of skill readily may construct any number of other plasmids suitable for use in the invention. The properties, construction and use of such plasmids, as well as other vectors, in the present invention will be readily apparent to those of skill from the present disclosure.

As used herein, the term “transcriptional control sequence” or “expression control sequence” refers to DNA sequences, such as initiator sequences, enhancer sequences, and promoter sequences, which induce, repress, or otherwise control the transcription of protein encoding nucleic acid sequences to which they are operably linked.

As used herein, the term “operably linked” or “operatively associated” refers to both expression control sequences that are contiguous with the nucleic acid sequence encoding the REP-1 or CNGA3 and/or expression control sequences that act in trans or at a distance to control the transcription and expression thereof.

The term “AAV” or “AAV serotype” as used herein refers to the dozens of naturally occurring and available adeno-associated viruses, as well as artificial AAVs. Among the AAVs isolated or engineered from human or non-human primates (NHP) and well characterized, human AAV2 is the first AAV that was developed as a gene transfer vector; it has been widely used for efficient gene transfer experiments in different target tissues and animal models. Unless otherwise specified, the AAV capsid, ITRs, and other selected AAV components described herein, may be readily selected from among any AAV, including, without limitation, AAV1, AAV2, AAV3, AAV4, AAV5, AAV6, AAV7, AAV8, AAV9, AAV8 bp, AAV7M8 and AAVAnc80, variants of any of the known or mentioned AAVs or AAVs yet to be discovered or variants or mixtures thereof. See, e.g., WO 2005/033321, which is incorporated herein by reference. In another embodiment, the AAV capsid is an AAV8 bp capsid, which preferentially targets bipolar cells. See, WO 2014/024282, which is incorporated herein by reference. In another embodiment, the AAV capsid is an AAV7m8 capsid, which has shown preferential delivery to the outer retina. See, Dalkara et al, In Vivo-Directed Evolution of a New Adeno-Associated Virus for Therapeutic Outer Retinal Gene Delivery from the Vitreous, *Sci Transl Med* 5, 189ra76(2013), which is incorporated herein by reference. In one embodiment, the AAV capsid is an AAV8 capsid. In another embodiment, the AAV capsid an AAV9 capsid. In another embodiment, the AAV capsid an AAV5 capsid. In another embodiment, the AAV capsid an AAV2 capsid.

As used herein, relating to AAV, the term variant means any AAV sequence which is derived from a known AAV sequence, including those sharing at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 97%, at least 99% or greater sequence identity over the amino acid or nucleic acid sequence. In another embodiment, the AAV capsid includes variants which may include up to about 10% variation from any described or known AAV capsid sequence. That is, the AAV capsid shares about

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90% identity to about 99.9% identity, about 95% to about 99% identity or about 97% to about 98% identity to an AAV capsid provided herein and/or known in the art. In one embodiment, the AAV capsid shares at least 95% identity with an AAV capsid. When determining the percent identity

of an AAV capsid, the comparison may be made over any of the variable proteins (e.g., vp1, vp2, or vp3). In one embodiment, the AAV capsid shares at least 95% identity with the AAV8 vp3. In another embodiment, a self-complementary AAV is used.

The ITRs or other AAV components may be readily isolated or engineered using techniques available to those of skill in the art from an AAV. Such AAV may be isolated, engineered, or obtained from academic, commercial, or public sources (e.g., the American Type Culture Collection, Manassas, VA). Alternatively, the AAV sequences may be engineered through synthetic or other suitable means by reference to published sequences such as are available in the literature or in databases such as, e.g., GenBank, PubMed, or the like. AAV viruses may be engineered by conventional molecular biology techniques, making it possible to optimize these particles for cell specific delivery of nucleic acid sequences, for minimizing immunogenicity, for tuning stability and particle lifetime, for efficient degradation, for accurate delivery to the nucleus, etc.

As used herein, “artificial AAV” means, without limitation, an AAV with a non-naturally occurring capsid protein. Such an artificial capsid may be generated by any suitable technique, using a selected AAV sequence (e.g., a fragment of a vp1 capsid protein) in combination with heterologous sequences which may be obtained from a different selected AAV, non-contiguous portions of the same AAV, from a non-AAV viral source, or from a non-viral source. An artificial AAV may be, without limitation, a pseudotyped AAV, a chimeric AAV capsid, a recombinant AAV capsid, or a “humanized” AAV capsid. Pseudotyped vectors, wherein the capsid of one AAV is replaced with a heterologous capsid protein, are useful in the invention. In one embodiment, AAV2/5 and AAV2/8 are exemplary pseudotyped vectors.

“Self-complementary AAV” refers a plasmid or vector having an expression cassette in which a coding region carried by a recombinant AAV nucleic acid sequence has been designed to form an intra-molecular double-stranded DNA template. Upon infection, rather than waiting for cell mediated synthesis of the second strand, the two complementary halves of scAAV will associate to form one double stranded DNA (dsDNA) unit that is ready for immediate replication and transcription. See, e.g., D M McCarty et al, “Self-complementary recombinant adeno-associated virus (scAAV) vectors promote efficient transduction independently of DNA synthesis”, *Gene Therapy*, (August 2001), Vol 8, Number 16, Pages 1248-1254. Self-complementary AAVs are described in, e.g., U.S. Pat. Nos. 6,596,535; 7,125,717; and 7,456,683, each of which is incorporated herein by reference in its entirety.

By “administering” as used in the methods means delivering the composition to the target selected cell which is characterized by the ocular disease. In one embodiment, the method involves delivering the composition by subretinal injection to the RPE, photoreceptor cells or other ocular cells. In another embodiment, intravitreal injection to ocular cells is employed. In still another method, injection via the palpebral vein to ocular cells may be employed. Still other methods of administration may be selected by one of skill in the art given this disclosure. By “administering” or “route of administration” is delivery of composition described herein,

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with or without a pharmaceutical carrier or excipient, of the subject. Routes of administration may be combined, if desired. In some embodiments, the administration is repeated periodically. The pharmaceutical compositions described herein are designed for delivery to subjects in need thereof by any suitable route or a combination of different routes. Direct delivery to the eye (optionally via ocular delivery, subretinal injection, intra-retinal injection, intravitreal, topical), or delivery via systemic routes, e.g., intraarterial, intraocular, intravenous, intramuscular, subcutaneous, intradermal, and other parental routes of administration. The nucleic acid molecules and/or vectors described herein may be delivered in a single composition or multiple compositions. Optionally, two or more different AAV may be delivered, or multiple viruses [see, e.g., WO20 2011/126808 and WO 2013/049493]. In another embodiment, multiple viruses may contain different replication-defective viruses (e.g., AAV and adenovirus), alone or in combination with proteins.

Certain compositions described herein are isolated, or synthetically or recombinantly engineered nucleic acid sequences that provide novel codon-optimized sequences encoding REP-1 or CNGA3 or CNGB3. In one embodiment, an isolated or engineered codon optimized nucleic acid sequence encoding human REP-1 is provided. In one embodiment, the codon-optimized sequence is SEQ ID NO: 1. In another embodiment, the codon optimized sequence includes N-terminal and C-terminal restriction sites for cloning. In one embodiment, such as that disclosed in SEQ ID NO: 5, the REP-1 coding sequence includes an N-terminal NotI restriction site and a C-terminal BamHI restriction site, in addition to a Kozak consensus sequence. In addition, the codon optimized sequence, in some embodiments, includes one or more additional restriction sites to allow for addition of markers, such as an epitope tag. When aligned with the native nucleic acid sequence, the codon optimized REP-1 may have a percent identity of at least 50%, or at least 60%, or at least 70%, or at least 80% or at least 90%, including any integer between any of those ranges. In one embodiment, the codon optimized REP-1 has a percent identity with the native sequence of at least 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98 or 99%. In one embodiment, when aligned with the native nucleic acid sequence SEQ ID NO: 3, it is revealed that codon optimized REP-1 (SEQ ID NO: 1) has a percent sequence identity of only 74% (see FIG. 2).

In another embodiment, an isolated or engineered codon optimized nucleic acid sequence encoding human CNGA3 is provided. In one embodiment, the codon-optimized sequence is SEQ ID NO: 9. In one embodiment, the codon-optimized sequence is a CNGA3 variant shown in SEQ ID NO: 11. In another embodiment, the codon optimized sequence includes N-terminal and C-terminal restriction sites for cloning. In one embodiment, the CNGA3 coding sequence includes an N-terminal NotI restriction site and a C-terminal BglII restriction site, in addition to a Kozak consensus sequence. Examples of CNGA3 sequences which include such modifications can be found in SEQ ID NO: 16, SEQ ID NO: 17 and SEQ ID NO: 18. In addition, the codon optimized sequence, in some embodiments, includes one or more additional restriction sites to allow for addition of markers, such as an epitope tag. When aligned with the native nucleic acid sequence, the codon optimized CNGA3 may have a percent identity of at least 50%, or at least 60%, or at least 70%, or at least 80% or at least 90%, including any

integer between any of those ranges. In one embodiment, the codon optimized CNGA3 has a percent identity with the native sequence of at least 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98 or 99%. In one embodiment, when aligned with the native nucleic acid sequence SEQ ID NO: 13, it is revealed that codon optimized CNGA3 (SEQ ID NO: 9) has a percent sequence identity of only 80% (see FIG. 3).

In another embodiment, an isolated or engineered codon optimized nucleic acid sequence encoding human CNGB3 is provided. In one embodiment, the codon-optimized sequence is a sequence sharing at least 70% identity with SEQ ID NO: 19 or SEQ ID NO 21. In another embodiment, the codon optimized sequence includes N-terminal and C-terminal restriction sites for cloning, for example, as shown in SEQ ID NO: 23. In addition, the codon optimized sequence, in some embodiments, includes one or more additional restriction sites to allow for addition of markers, such as an epitope tag. When aligned with the native nucleic acid sequence (as shown in SEQ ID NO: 19), the codon optimized CNGB3 may have a percent identity of at least 50%, or at least 60%, or at least 70%, or at least 80% or at least 90%, including any integer between any of those ranges. In one embodiment, the codon optimized CNGB3 has a percent identity with the native sequence of at least 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98 or 99%.

In one embodiment, the optimized nucleic acid sequences encoding the REP-1 or CNGA3 constructs described herein are engineered into any suitable genetic element, e.g., naked DNA, phage, transposon, cosmid, RNA molecule (e.g., mRNA), episome, etc., which transfers the REP-1 or CNGA3 sequences carried thereon to a host cell, e.g., for generating nanoparticles carrying DNA or RNA, viral vectors in a packaging host cell and/or for delivery to a host cells in subject. In one embodiment, the genetic element is a plasmid.

The selected genetic element may be delivered by any suitable method, including transfection, electroporation, liposome delivery, membrane fusion techniques, high velocity DNA-coated pellets, viral infection and protoplast fusion. The methods used to make such constructs are known to those with skill in nucleic acid manipulation and include genetic engineering, recombinant engineering, and synthetic techniques. See, e.g., Green and Sambrook, *Molecular Cloning: A Laboratory Manual*, Cold Spring Harbor Press, Cold Spring Harbor, NY (2012).

A variety of expression cassettes are provided which employ SEQ ID NOs. 1 or 5 for expression of the REP-1 protein. In one embodiment, an example of a plasmid containing such an expression cassette is shown in SEQ ID NO. 25. In one embodiment, an example of a plasmid containing such an expression cassette is shown in SEQ ID NO. 26. In one embodiment, an example of a plasmid containing such an expression cassette is shown in SEQ ID NO. 27. In one embodiment, an example of a plasmid containing such an expression cassette is shown in SEQ ID NO. 28. In another embodiment, a variety of expression cassettes are provided which employ SEQ ID NOs. 9, 11 or 13 for expression of the CNGA3 protein. In another embodiment, a variety of expression cassettes are provided which employ SEQ ID NOs. 19, 21 or 23 for expression of the CNGB3 protein. As used herein, an "expression cassette"

refers to a nucleic acid molecule which comprises coding sequences for the optimized REP-1 or CNGA3 or CNGB3 proteins, promoter, and may include other regulatory sequences therefor, which cassette may be engineered into a genetic element or plasmid, and/or packaged into the capsid of a viral vector (e.g., a viral particle). In one embodiment, an expression cassette comprises a codon optimized nucleic acid sequence that encodes REP-1. In one embodiment, the cassette provides the codon optimized REP-1 operatively associated with expression control sequences that direct expression of the codon optimized nucleic acid sequence that encodes REP-1 in a host cell.

In another embodiment, an expression cassette comprises a codon optimized nucleic acid sequence that encodes CNGA3. In one embodiment, the cassette provides the codon optimized CNGA3 operatively associated with expression control sequences that direct expression of the codon optimized nucleic acid sequence that encodes CNGA3 in a host cell.

In another embodiment, an expression cassette comprises a codon optimized nucleic acid sequence that encodes CNGB3. In one embodiment, the cassette provides the codon optimized CNGB3 operatively associated with expression control sequences that direct expression of the codon optimized nucleic acid sequence that encodes CNGB3 in a host cell.

In another embodiment, an expression cassette for use in an AAV vector is provided. In that embodiment, the AAV expression cassette includes at least one AAV inverted terminal repeat (ITR) sequence. In another embodiment, the expression cassette comprises 5' ITR sequences and 3' ITR sequences. In one embodiment, the 5' and 3' ITRs flank the codon optimized nucleic acid sequence that encodes REP-1 or CNGA3 or CNGB3, optionally with additional sequences which direct expression of the codon optimized nucleic acid sequence that encodes REP-1 or CNGA3 or CNGB3 in a host cell. Thus, as described herein, a AAV expression cassette is meant to describe an expression cassette as described above flanked on its 5' end by a 5' AAV inverted terminal repeat sequence (ITR) and on its 3' end by a 3' AAV ITR. Thus, this rAAV genome contains the minimal sequences required to package the expression cassette into an AAV viral particle, i.e., the AAV 5' and 3' ITRs. The AAV ITRs may be obtained from the ITR sequences of any AAV, such as described herein. These ITRs may be of the same AAV origin as the capsid employed in the resulting recombinant AAV, or of a different AAV origin (to produce an AAV pseudotype). In one embodiment, the ITR sequences from AAV2, or the deleted version thereof (Δ ITR), are used for convenience and to accelerate regulatory approval. However, ITRs from other AAV sources may be selected. Each rAAV genome can be then introduced into a production plasmid. In one embodiment, the production plasmid is that described herein, or as described in WO2012/158757, which is incorporated herein by reference. Various plasmids are known in the art for use in producing rAAV vectors, and are useful herein. The production plasmids are cultured in the host cells which express the AAV cap and/or rep proteins. In the host cells, each rAAV genome is rescued and packaged into the capsid protein or envelope protein to form an infectious viral particle.

One type of production plasmid is that shown in SEQ ID NO: 7, which is termed p584. This plasmid is used in the examples for generation of the rAAV-REP-1 vector. Such a plasmid is one that contains a 5' AAV ITR sequence; a selected promoter; a polyA sequence; and a 3' ITR; additionally, it also contains a stuffer sequence, such as lambda.

In one embodiment, a non-coding lambda stuffer region is included in the vector backbone. The nucleic acid sequence encoding REP-1, CNGA3 or CNGB2 are inserted in place of between the selected promoter and the polyA sequence, or a similar, plasmid. An example of p584 which includes the REP-1 encoding sequence can be found in SEQ ID NO: 8. In another embodiment, the production plasmid is modified to optimized vector plasmid production efficiency. Such modifications include addition of other neutral sequences, or deletion of portion(s) of or the entire lambda stuffer sequence to modulate the level of supercoil of the vector plasmid. Such modifications are contemplated herein. In other embodiments, terminator and other sequences are included in the plasmid.

In still a further embodiment, a recombinant adeno-associated virus (AAV) vector is provided for delivery of the REP-1, CNGA3 and CNGB3 constructs and optimized sequences described herein. An adeno-associated virus (AAV) viral vector is an AAV DNase-resistant particle having an AAV protein capsid into which is packaged nucleic acid sequences for delivery to target cells. An AAV capsid is composed of 60 capsid (cap) protein subunits, VP1, VP2, and VP3, that are arranged in an icosahedral symmetry in a ratio of approximately 1:1:10 to 1:1:20, depending upon the selected AAV. AAVs may be selected as sources for capsids of AAV viral vectors as identified above. See, e.g., US Published Patent Application No. 2007-0036760-A1; US Published Patent Application No. 2009-0197338-A1; EP 1310571. See also, WO 2003/042397 (AAV7 and other simian AAV), U.S. Pat. Nos. 7,790,449 and 7,282,199 (AAV8), WO 2005/033321 and U.S. Pat. No. 7,906,111 (AAV9), and WO 2006/110689, and WO 2003/042397 (rh.10). These documents also describe other AAV which may be selected for generating AAV and are incorporated by reference. In some embodiments, an AAV cap for use in the viral vector can be generated by mutagenesis (i.e., by insertions, deletions, or substitutions) of one of the aforementioned AAV capsids or its encoding nucleic acid. In some embodiments, the AAV capsid is chimeric, comprising domains from two or three or four or more of the aforementioned AAV capsid proteins. In some embodiments, the AAV capsid is a mosaic of Vp1, Vp2, and Vp3 monomers from two or three different AAVs or recombinant AAVs. In some embodiments, an rAAV composition comprises more than one of the aforementioned Caps.

In another embodiment, the AAV capsid includes variants which may include up to about 10% variation from any described or known AAV capsid sequence. That is, the AAV capsid shares about 90% identity to about 99.9% identity, about 95% to about 99% identity or about 97% to about 98% identity to an AAV capsid provided herein and/or known in the art. In one embodiment, the AAV capsid shares at least 95% identity with an AAV capsid. When determining the percent identity of an AAV capsid, the comparison may be made over any of the variable proteins (e.g., vp1, vp2, or vp3). In one embodiment, the AAV capsid shares at least 95% identity with the AAV8 vp3. In another embodiment, a self-complementary AAV is used. In one embodiment, it is desirable to utilize an AAV capsid, which shows tropism for the desired target cell, e.g., photoreceptors, RPE or other ocular cells. In one embodiment, the AAV capsid is a tyrosine capsid-mutant in which certain surface exposed tyrosine residues are substituted with phenylalanine (F). Such AAV variants are described, e.g., in Mowat et al, Tyrosine capsid-mutant AAV vectors for gene delivery to the

canine retina from a subretinal or intravitreal approach, Gene Therapy 21, 96-105 (January 2014), which is incorporated herein by reference.

For packaging an expression cassette or rAAV genome or production plasmid into virions, the ITRs are the only AAV components required in cis in the same construct as the transgene. In one embodiment, the coding sequences for the replication (rep) and/or capsid (cap) are removed from the AAV genome and supplied in trans or by a packaging cell line in order to generate the AAV vector. For example, as described above, a pseudotyped AAV may contain ITRs from a source which differs from the source of the AAV capsid. Additionally or alternatively, a chimeric AAV capsid may be utilized. Still other AAV components may be selected. Sources of such AAV sequences are described herein and may also be isolated or engineered obtained from academic, commercial, or public sources (e.g., the American Type Culture Collection, Manassas, VA). Alternatively, the AAV sequences may be obtained through synthetic or other suitable means by reference to published sequences such as are available in the literature or in databases such as, e.g., GenBank®, PubMed®, or the like.

Methods for generating and isolating AAV viral vectors suitable for delivery to a subject are known in the art. See, e.g., U.S. Pat. Nos. 7,790,449; 7,282,199; WO 2003/042397; WO 2005/033321, WO 2006/110689; and U.S. Pat. No. 7,588,772 B2]. In a one system, a producer cell line is transiently transfected with a construct that encodes the transgene flanked by ITRs and a construct(s) that encodes rep and cap. In a second system, a packaging cell line that stably supplies rep and cap is transiently transfected with a construct encoding the transgene flanked by ITRs. In each of these systems, AAV virions are produced in response to infection with helper adenovirus or herpesvirus, requiring the separation of the rAAVs from contaminating virus. More recently, systems have been developed that do not require infection with helper virus to recover the AAV—the required helper functions (i.e., adenovirus E1, E2a, VA, and E4 or herpesvirus UL5, UL8, UL52, and UL29, and herpesvirus polymerase) are also supplied, in trans, by the system. In these newer systems, the helper functions can be supplied by transient transfection of the cells with constructs that encode the required helper functions, or the cells can be engineered to stably contain genes encoding the helper functions, the expression of which can be controlled at the transcriptional or posttranscriptional level.

In yet another system, the transgene flanked by ITRs and rep/cap genes are introduced into insect cells by infection with baculovirus-based vectors. For reviews on these production systems, see generally, e.g., Zhang et al., 2009, “Adenovirus-adeno-associated virus hybrid for large-scale recombinant adeno-associated virus production,” Human Gene Therapy 20:922-929, the contents of which is incorporated herein by reference in its entirety. Methods of making and using these and other AAV production systems are also described in the following U.S. patents, the contents of each of which is incorporated herein by reference in its entirety: U.S. Pat. Nos. 5,139,941; 5,741,683; 6,057,152; 6,204,059; 6,268,213; 6,491,907; 6,660,514; 6,951,753; 7,094,604; 7,172,893; 7,201,898; 7,229,823; and 7,439,065. See generally, e.g., Grieger & Samulski, 2005, “Adeno-associated virus as a gene therapy vector: Vector development, production and clinical applications,” Adv. Biochem. Engin/Biotechnol. 99:119-145; Buning et al., 2008, “Recent developments in adeno-associated virus vector technology,”

J. Gene Med. 10:717-733; and the references cited below, each of which is incorporated herein by reference in its entirety.

The methods used to construct any embodiment of this invention are known to those with skill in nucleic acid manipulation and include genetic engineering, recombinant engineering, and synthetic techniques. See, e.g., Green and Sambrook et al, *Molecular Cloning: A Laboratory Manual*, Cold Spring Harbor Press, Cold Spring Harbor, NY (2012). Similarly, methods of generating rAAV virions are well known and the selection of a suitable method is not a limitation on the present invention. See, e.g., K. Fisher et al, (1993) *J. Virol.*, 70:520-532 and U.S. Pat. No. 5,478,745.

The rAAV vectors comprise an AAV capsid and an AAV expression cassette comprising sequences encoding REP-1 or CNGA3 or CNGB3, such as described above. In certain embodiments, the rAAV expression cassette comprises AAV inverted terminal repeat sequences and a codon optimized nucleic acid sequence that encodes REP-1 or CNGA3 or CNGB3, and expression control sequences that direct expression of the encoded proteins in a host cell. The rAAV expression cassette, in other embodiments, further comprises one or more of an intron, a Kozak sequence, a polyA, and post-transcriptional regulatory elements. Such rAAV vectors for use in pharmaceutical compositions for delivery to the eye, may employ a capsid from any of the many known AAVs identified above.

Other conventional components of the expression cassettes and vectors include other components that can be optimized for a specific species using techniques known in the art including, e.g. codon optimization, as described herein. The components of the cassettes, vectors, plasmids and viruses or other compositions described herein include a promoter sequence as part of the expression control sequences. In another embodiment, the promoter is cell-specific. The term "cell-specific" means that the particular promoter selected for the recombinant vector can direct expression of the optimized REP-1 or CNGA3 or CNGB3 transgene in a particular ocular cell type. In one embodiment, the promoter is specific for expression of the transgene in photoreceptor cells. In another embodiment, the promoter is specific for expression in the rods and cones. In another embodiment, the promoter is specific for expression in the rods. In another embodiment, the promoter is specific for expression in the cones. In one embodiment, the photoreceptor-specific promoter is a human rhodopsin kinase promoter. The rhodopsin kinase promoter has been shown to be active in both rods and cones. See, e.g., Sun et al, *Gene Therapy with a Promoter Targeting Both Rods and Cones Rescues Retinal Degeneration Caused by AIPL1 Mutations*, *Gene Ther.* 2010 January; 17(1): 117-131, which is incorporated herein by reference in its entirety. In one embodiment, the promoter is modified to add one or more restriction sites to facilitate cloning.

In another embodiment, the promoter is a human rhodopsin promoter. In one embodiment, the promoter is modified to include restriction on the ends for cloning. See, e.g. Nathans and Hogness, *Isolation and nucleotide sequence of the gene encoding human rhodopsin*, *PNAS*, 81:4851-5 (August 1984), which is incorporated herein by reference in its entirety. In another embodiment, the promoter is a portion or fragment of the human rhodopsin promoter. In another embodiment, the promoter is a variant of the human rhodopsin promoter.

Other exemplary promoters include the human G-protein-coupled receptor protein kinase 1 (GRK1) promoter (Genbank Accession number AY327580). In another embodi-

ment, the promoter is a 292 nt fragment (positions 1793-2087) of the GRK1 promoter (See, Beltran et al, *Gene Therapy* 2010 17:1162-74, which is hereby incorporated by reference in its entirety). In another preferred embodiment, the promoter is the human interphotoreceptor retinoid-binding protein proximal (IRBP) promoter. In one embodiment, the promoter is a 235 nt fragment of the hIRBP promoter. In one embodiment, the promoter is the RPGR proximal promoter (Shu et al, *IOVS*, May 2102, which is incorporated by reference in its entirety). Other promoters useful in the invention include, without limitation, the rod opsin promoter, the red-green opsin promoter, the blue opsin promoter, the cGMP- β -phosphodiesterase promoter (Qgueta et al, *IOVS*, *Invest Ophthalmol Vis Sci.* 2000 December;41 (13): 4059-63), the mouse opsin promoter (Beltran et al 2010 cited above), the rhodopsin promoter (Mussolino et al, *Gene Ther.* July 2011, 18(7): 637-45); the alpha-subunit of cone transducin (Morrissey et al, *BMC Dev. Biol.* January 2011, 11:3); beta phosphodiesterase (PDE) promoter; the retinitis pigmentosa (RP1) promoter (Nicord et al, *J. Gene Med.* December 2007, 9(12): 1015-23); the NXNL2/NXNL1 promoter (Lambard et al, *PLoS One*, October 2010, 5(10):e13025), the RPE65 promoter; the retinal degeneration slow/peripherin 2 (Rds/perph2) promoter (Cai et al, *Exp Eye Res.* 2010 August;91(2): 186-94); and the VMD2 promoter (Kachi et al, *Human Gene Therapy*, 2009(20:31-9)). Each of these documents is incorporated by reference herein in its entirety. In another embodiment, the promoter is selected from human human EF1 α promoter, rhodopsin promoter, rhodopsin kinase, interphotoreceptor binding protein (IRBP), cone opsin promoters (red-green, blue), cone opsin upstream sequences containing the red-green cone locus control region, cone transducing, and transcription factor promoters (neural retina leucine zipper (Nrl) and photoreceptor-specific nuclear receptor Nr2e3, bZIP).

In another embodiment, the promoter is a ubiquitous or constitutive promoter. An example of a suitable promoter is a hybrid chicken β -actin (CBA) promoter with cytomegalovirus (CMV) enhancer elements. In another embodiment, the promoter is the CB7 promoter. Other suitable promoters include the human β -actin promoter, the human elongation factor-1 α promoter, the cytomegalovirus (CMV) promoter, the simian virus 40 promoter, and the herpes simplex virus thymidine kinase promoter. See, e.g., Damdindorj et al, (August 2014) *A Comparative Analysis of Constitutive Promoters Located in Adeno-Associated Viral Vectors*. *PLoS ONE* 9(8):e106472. Still other suitable promoters include viral promoters, constitutive promoters, regulatable promoters [see, e.g., WO 2011/126808 and WO 2013/04943]. Alternatively a promoter responsive to physiologic cues may be utilized in the expression cassette, rAAV genomes, vectors, plasmids and viruses described herein. In one embodiment, the promoter is of a small size, under 1000 bp, due to the size limitations of the AAV vector. In another embodiment, the promoter is under 400 bp. Other promoters may be selected by one of skill in the art. In one embodiment, the REP-1 construct incorporates a ubiquitous promoter. In another embodiment, the CNGA3 construct incorporates a photoreceptor-specific promoter. In one embodiment, the REP-1 construct includes a CBA promoter with CMV enhancer elements.

In another embodiment, the promoter is an inducible promoter. The inducible promoter may be selected from known promoters including the rapamycin/rapalog promoter, the ecdysone promoter, the estrogen-responsive promoter, and the tetracycline-responsive promoter, or heterodimeric repressor switch. See, Sochor et al, *An Autogenously*

Regulated Expression System for Gene Therapeutic Ocular Applications. Scientific Reports, 2015 Nov. 24; 5:17105 and Daber R, Lewis M., A novel molecular switch. J Mol Biol. 2009 Aug. 28;391(4): 661-70, Epub 2009 Jun. 21 which are both incorporated herein by reference in their entirety.

In other embodiments, the cassette, vector, plasmid and virus constructs described herein contain other appropriate transcription initiation, termination, enhancer sequences, efficient RNA processing signals such as splicing and polyadenylation (polyA) signals; TATA sequences; sequences that stabilize cytoplasmic mRNA; sequences that enhance translation efficiency (i.e., Kozak consensus sequence); introns; sequences that enhance protein stability; and when desired, sequences that enhance secretion of the encoded product. The expression cassette or vector may contain none, one or more of any of the elements described herein. Examples of suitable polyA sequences include, e.g., SV40, bovine growth hormone (bGH), and TK polyA. Examples of suitable enhancers include, e.g., the CMV enhancer, the RSV enhancer, the alpha fetoprotein enhancer, the TTR minimal promoter/enhancer, LSP (TH-binding globulin promoter/alpha1-microglobulin/bikunin enhancer), amongst others. In one embodiment, a Kozak sequence is included upstream of the transgene coding sequence to enhance translation from the correct initiation codon. In another embodiment, CBA exon 1 and intron are included in the expression cassette. In one embodiment, the transgene is placed under the control of a hybrid chicken β actin (CBA) promoter. This promoter consists of the cytomegalovirus (CMV) immediate early enhancer, the proximal chicken β actin promoter, and CBA exon 1 flanked by intron 1 sequences.

In one embodiment, the expression cassette contains a 5' ITR, CBA promoter, CMV enhancer, CBA exon 1 and intron, kozak sequence, human codon optimized CHM sequence (SEQ ID NO: 1), bGH poly A and 3' ITR.

In yet other aspects, these nucleic acid sequences, vectors, expression cassettes and rAAV viral vectors are useful in a pharmaceutical composition, which also comprises a pharmaceutically acceptable carrier, buffer, diluent and/or adjuvant, etc. Such pharmaceutical compositions are used to express the optimized REP-1 or CNGA3 or CNGB3 in the ocular cells through delivery by such recombinantly engineered AAVs or artificial AAVs.

To prepare these pharmaceutical compositions containing the nucleic acid sequences, vectors, expression cassettes and rAAV viral vectors, the sequences or vectors or viral vector is preferably assessed for contamination by conventional methods and then formulated into a pharmaceutical composition suitable for administration to the eye. Such formulation involves the use of a pharmaceutically and/or physiologically acceptable vehicle or carrier, particularly one suitable for administration to the eye, such as buffered saline or other buffers, e.g., HEPES, to maintain pH at appropriate physiological levels, and, optionally, other medicinal agents, pharmaceutical agents, stabilizing agents, buffers, carriers, adjuvants, diluents, etc. For injection, the carrier will typically be a liquid. Exemplary physiologically acceptable carriers include sterile, pyrogen-free water and sterile, pyrogen-free, phosphate buffered saline. A variety of such known carriers are provided in U.S. Pat. No. 7,629,322, incorporated herein by reference. In one embodiment, the carrier is an isotonic sodium chloride solution. In another embodiment, the carrier is balanced salt solution. In one embodiment, the carrier includes tween. If the virus is to be stored long-term, it may be frozen in the presence of glycerol or Tween20.

In one exemplary specific embodiment, the composition of the carrier or excipient contains 180 mM NaCl, 10 mM NaPi, pH7.3 with 0.0001%-0.01% Pluronic F68 (PF68). The exact composition of the saline component of the buffer ranges from 160 mM to 180 mM NaCl. Optionally, a different pH buffer (potentially HEPES, sodium bicarbonate, TRIS) is used in place of the buffer specifically described. Still alternatively, a buffer containing 0.9% NaCl is useful.

Optionally, the compositions of the invention may contain, in addition to the rAAV and/or variants and carrier(s), other conventional pharmaceutical ingredients, such as preservatives, or chemical stabilizers. Suitable exemplary preservatives include chlorobutanol, potassium sorbate, sorbic acid, sulfur dioxide, propyl gallate, the parabens, ethyl vanillin, glycerin, phenol, and parachlorophenol. Suitable chemical stabilizers include gelatin and albumin.

The pharmaceutical compositions containing at least one replication-defective rAAV virus, as described herein, can be formulated with a physiologically acceptable carrier, diluent, excipient and/or adjuvant, for use in gene transfer and gene therapy applications. In the case of AAV viral vectors, quantification of the genome copies ("GC"), vector genomes ("VG"), or virus particles may be used as the measure of the dose contained in the formulation or suspension. Any method known in the art can be used to determine the genome copy (GC) number of the replication-defective virus compositions of the invention. One method for performing AAV GC number titration is as follows: Purified AAV vector samples are first treated with DNase to eliminate un-encapsidated AAV genome DNA or contaminating plasmid DNA from the production process. The DNase resistant particles are then subjected to heat treatment to release the genome from the capsid. The released genomes are then quantitated by real-time PCR using primer/probe sets targeting specific region of the viral genome (usually poly A signal). In another method the effective dose of a recombinant adeno-associated virus carrying a nucleic acid sequence encoding the optimized REP-1 or CNGA3 transgene is measured as described in S.K. McLaughlin et al, 1988 J. Virol., 62:1963, which is incorporated by reference in its entirety.

As used herein, the term "dosage" can refer to the total dosage delivered to the subject in the course of treatment, or the amount delivered in a single unit (or multiple unit or split dosage) administration. The pharmaceutical virus compositions can be formulated in dosage units to contain an amount of replication-defective virus carrying the codon optimized nucleic acid sequences encoding REP-1 or CNGA3 or CNGB3 as described herein that is in the range of about 1.0×10^9 GC to about 1.0×10^{15} GC including all integers or fractional amounts within the range. In one embodiment, the compositions are formulated to contain at least 1×10^9 , 2×10^9 , 3×10^9 , 4×10^9 , 5×10^9 , 6×10^9 , 7×10^9 , 8×10^9 , or 9×10^9 GC per dose including all integers or fractional amounts within the range. In another embodiment, the compositions are formulated to contain at least 1×10^{10} , 2×10^{10} , 3×10^{10} , 4×10^{10} , 5×10^{10} , 6×10^{10} , 7×10^{10} , 8×10^{10} , or 9×10^{10} GC per dose including all integers or fractional amounts within the range. In another embodiment, the compositions are formulated to contain at least 1×10^{11} , 2×10^{11} , 3×10^{11} , 4×10^{11} , 5×10^{11} , 6×10^{11} , 7×10^{11} , 8×10^{11} , or 9×10^{11} GC per dose including all integers or fractional amounts within the range. In another embodiment, the compositions are formulated to contain at least 1×10^{12} , 2×10^{12} , 3×10^{12} , 4×10^{12} , 5×10^{12} , 6×10^{12} , 7×10^{12} , 8×10^{12} , or 9×10^{12} GC per dose including all integers or fractional amounts within the range. In another embodiment, the compositions are formulated to contain at

least 1×10^{13} , 2×10^{13} , 3×10^{13} , 4×10^{13} , 5×10^{13} , 6×10^{13} , 7×10^{13} , 8×10^{13} , or 9×10^{13} GC per dose including all integers or fractional amounts within the range. In another embodiment, the compositions are formulated to contain at least 1×10^{14} , 2×10^{14} , 3×10^{14} , 4×10^{14} , 5×10^{14} , 6×10^{14} , 7×10^{14} , 8×10^{14} , or 9×10^{14} GC per dose including all integers or fractional amounts within the range. In another embodiment, the compositions are formulated to contain at least 1×10^{15} , 2×10^{15} , 3×10^{15} , 4×10^{15} , 5×10^{15} , 6×10^{15} , 7×10^{15} , 8×10^{15} , or 9×10^{15} GC per dose including all integers or fractional amounts within the range. In one embodiment, for human application the dose can range from 1×10^{10} to about 1×10^{12} GC per dose including all integers or fractional amounts within the range. All dosages may be measured by any known method, including as measured by qPCR or digital droplet PCR (ddPCR) as described in, e.g., M. Lock et al, Hum Gene Ther Methods. 2014 April;25(2): 115-25. doi: 10.1089/hgtb.2013.131, which is incorporated herein by reference.

These above doses may be administered in a variety of volumes of carrier, excipient or buffer formulation, ranging from about 25 to about 1000 microliters, including all numbers within the range, depending on the size of the area to be treated, the viral titer used, the route of administration, and the desired effect of the method. In one embodiment, the volume of carrier, excipient or buffer is at least about 25 μ L. In one embodiment, the volume is about 50 μ L. In another embodiment, the volume is about 75 μ L. In another embodiment, the volume is about 100 μ L. In another embodiment, the volume is about 125 μ L. In another embodiment, the volume is about 150 μ L. In another embodiment, the volume is about 175 μ L. In yet another embodiment, the volume is about 200 μ L. In another embodiment, the volume is about 225 μ L. In yet another embodiment, the volume is about 250 μ L. In yet another embodiment, the volume is about 275 μ L. In yet another embodiment, the volume is about 300 μ L. In yet another embodiment, the volume is about 325 μ L. In another embodiment, the volume is about 350 μ L. In another embodiment, the volume is about 375 μ L. In another embodiment, the volume is about 400 μ L. In another embodiment, the volume is about 450 μ L. In another embodiment, the volume is about 500 μ L. In another embodiment, the volume is about 550 μ L. In another embodiment, the volume is about 600 μ L. In another embodiment, the volume is about 650 μ L. In another embodiment, the volume is about 700 μ L. In another embodiment, the volume is between about 700 and 1000 μ L.

In one embodiment, the viral constructs may be delivered in doses of from at least 1×10^9 to about 1×10^{11} GCs in volumes of about 1 μ L to about 3 μ L for small animal subjects, such as mice. For larger veterinary subjects having eyes about the same size as human eyes, the larger human dosages and volumes stated above are useful. See, e.g., Diehl et al, J. Applied Toxicology, 21:15-23 (2001) for a discussion of good practices for administration of substances to various veterinary animals. This document is incorporated herein by reference.

It is desirable that the lowest effective concentration of virus or other delivery vehicle be utilized in order to reduce the risk of undesirable effects, such as toxicity, retinal dysplasia and detachment. Still other dosages in these ranges may be selected by the attending physician, taking into account the physical state of the subject, preferably human, being treated, the age of the subject, the particular ocular disorder and the degree to which the disorder, if progressive, has developed.

Yet another aspect described herein is a method for treating, retarding or halting progression of blindness in a mammalian subject having, or at risk of developing, choroideremia. In one embodiment, a rAAV carrying the REP-1 codon optimized sequences, preferably suspended in a physiologically compatible carrier, diluent, excipient and/or adjuvant, may be administered to a desired subject including a human subject. This method comprises administering to a subject in need thereof any of the nucleic acid sequences, expression cassettes, rAAV genomes, plasmids, vectors or rAAV vectors or compositions containing them. In one embodiment, the composition is delivered subretinally. In another embodiment, the composition is delivered intravitreally. In still another embodiment, the composition is delivered using a combination of administrative routes suitable for treatment of ocular diseases, and may also involve administration via the palpebral vein or other intravenous or conventional administration routes.

Yet another aspect described herein is a method for treating, retarding or halting progression of blindness in a mammalian subject having, or at risk of developing, achromatopsia. In one embodiment, an rAAV carrying the CNGA3 or CNGB3 native, modified or codon optimized sequence, preferably suspended in a physiologically compatible carrier, diluent, excipient and/or adjuvant, may be administered to a desired subject including a human subject. This method comprises administering to a subject in need thereof any of the nucleic acid sequences, expression cassettes, rAAV genomes, plasmids, vectors or rAAV vectors or compositions containing them. In one embodiment, the composition is delivered subretinally. In another embodiment, the composition is delivered intravitreally. In still another embodiment, the composition is delivered using a combination of administrative routes suitable for treatment of ocular diseases, and may also involve administration via the palpebral vein or other intravenous or conventional administration routes.

For use in these methods, the volume and viral titer of each dosage is determined individually, as further described herein, and may be the same or different from other treatments performed in the same, or contralateral, eye. The dosages, administrations and regimens may be determined by the attending physician given the teachings of this specification. In one embodiment, the composition is administered in a single dosage selected from those above listed in a single affected eye. In another embodiment, the composition is administered as a single dosage selected from those above listed in a both affected eyes, either simultaneously or sequentially. Sequential administration may imply a time gap of administration from one eye to another from intervals of minutes, hours, days, weeks or months. In another embodiment, the method involves administering the compositions to an eye two or more dosages (e.g., split dosages). In another embodiment, multiple injections are made in different portions of the same eye. In another embodiment, a second administration of an rAAV including the selected expression cassette (e.g., CHM containing cassette) is performed at a later time point. Such time point may be weeks, months or years following the first administration. Such second administration is, in one embodiment, performed with an rAAV having a different capsid than the rAAV from the first administration. In another embodiment, the rAAV from the first and second administration have the same capsid.

In still other embodiments, the compositions described herein may be delivered in a single composition or multiple compositions. Optionally, two or more different AAV may

be delivered, or multiple viruses [see, e.g., WO 2011/126808 and WO 2013/049493]. In another embodiment, multiple viruses may contain different replication-defective viruses (e.g., AAV and adenovirus).

In certain embodiments of the invention it is desirable to perform non-invasive retinal imaging and functional studies to identify areas of the rod and cone photoreceptors to be targeted for therapy. In these embodiments, clinical diagnostic tests are employed to determine the precise location(s) for one or more subretinal injection(s). These tests may include electroretinography (ERG), perimetry, topographical mapping of the layers of the retina and measurement of the thickness of its layers by means of confocal scanning laser ophthalmoscopy (cSLO) and optical coherence tomography (OCT), topographical mapping of cone density via adaptive optics (AO), functional eye exam, etc, depending upon the species of the subject being treated, their physical status and health and the dosage. In view of the imaging and functional studies, in some embodiments of the invention one or more injections are performed in the same eye in order to target different areas of the affected eye. The volume and viral titer of each injection is determined individually, as further described herein, and may be the same or different from other injections performed in the same, or contralateral, eye. In another embodiment, a single, larger volume injection is made in order to treat the entire eye. In one embodiment, the volume and concentration of the rAAV composition is selected so that only the region of damaged ocular cells is impacted. In another embodiment, the volume and/or concentration of the rAAV composition is a greater amount, in order reach larger portions of the eye, including non-damaged photoreceptors.

In one embodiment of the methods described herein, a one-time intra-ocular delivery of a composition as described herein, e.g., an AAV delivery of an optimized REP-1 cassette, is useful in preventing vision loss and blindness in a subject at risk of developing choroideremia. In another embodiment of the methods described herein, a one-time intra-ocular delivery of a composition as described herein, e.g., an AAV delivery of an optimized CNGA3 or CNGB3 cassette, is useful in preventing vision loss and blindness in a subject at risk of developing achromatopsia.

Thus, in one embodiment, the composition is administered before disease onset. In another embodiment, the composition is administered prior to the initiation of vision impairment or loss. In another embodiment, the composition is administered after initiation of vision impairment or loss. In yet another embodiment, the composition is administered when less than 90% of the rod and/or cones or photoreceptors are functioning or remaining, as compared to a non-diseased eye.

In another embodiment, the method includes performing additional studies, e.g., functional and imaging studies to determine the efficacy of the treatment. For examination in animals, such tests include retinal and visual function assessment via electroretinograms (ERGs) looking at rod and cone photoreceptor function, optokinetic nystagmus, pupillometry, water maze testing, light-dark preference, optical coherence tomography (to measure thickness of various layers of the retina), histology (retinal thickness, rows of nuclei in the outer nuclear layer, immunofluorescence to document transgene expression, cone photoreceptor counting, staining of retinal sections with peanut agglutinin-which identifies cone photoreceptor sheaths).

Specifically for human subjects, following administration of a dosage of a compositions described in this specification, the subject is tested for efficacy of treatment using elec-

troretinograms (ERGs) to examine rod and cone photoreceptor function, pupillometry visual acuity, contrast sensitivity color vision testing, visual field testing (Humphrey visual fields/Goldmann visual fields), perimetry mobility test (obstacle course), and reading speed test. Other useful post-treatment efficacy test to which the subject is exposed following treatment with a pharmaceutical composition described herein are functional magnetic resonance imaging (fMRI), full-field light sensitivity testing, retinal structure studies including optical coherence tomography, fundus photography, fundus autofluorescence, adaptive optics laser scanning ophthalmoscopy, mobility testing, test of reading speed and accuracy, micropertimetry and/or ophthalmoscopy. These and other efficacy tests are described in U.S. Pat. No. 8,147,823; in co-pending International patent application publication WO 2014/011210 or WO 2014/124282, incorporated by reference.

In yet another embodiment, any of the above described methods is performed in combination with another, or secondary, therapy. In still other embodiments, the methods of treatment of these ocular diseases involve treating the subject with the composition described in detail herein in combination with another therapy, such as antibiotic treatment, palliative treatment for pain, and the like. The additional therapy may be any now known, or as yet unknown, therapy which helps prevent, arrest or ameliorate these mutations or defects or any of the effects associated therewith. The secondary therapy can be administered before, concurrent with, or after administration of the compositions described above. In one embodiment, a secondary therapy involves non-specific approaches for maintaining the health of the retinal cells, such as administration of neurotrophic factors, anti-oxidants, anti-apoptotic agents. The non-specific approaches are achieved through injection of proteins, recombinant DNA, recombinant viral vectors, stem cells, fetal tissue, or genetically modified cells. The latter could include genetically modified cells that are encapsulated.

In one embodiment, a method of generating a recombinant rAAV comprises obtaining a plasmid containing an AAV expression cassette as described above and culturing a packaging cell carrying the plasmid in the presence of sufficient viral sequences to permit packaging of the AAV viral genome into an infectious AAV envelope or capsid. Specific methods of rAAV vector generation are described above and may be employed in generating a rAAV vector that can deliver the codon optimized REP-1 or CNGA3 or CNGB3 in the expression cassettes and genomes described above and in the examples below.

In yet another embodiment, a vector comprising any of the expression cassettes described herein is provided. As described above, such vectors can be plasmids of variety of origins and are useful in certain embodiments for the generation of recombinant replication defective viruses as described further herein.

In one another embodiment, the vector is a plasmid that comprises an expression cassette, wherein the expression cassette comprises AAV inverted terminal repeat sequences and a codon optimized nucleic acid sequence that encodes REP-1, and expression control sequences that direct expression of the encoded protein in a host cell.

In another embodiment, the vector is a plasmid that comprises an expression cassette, wherein the expression cassette comprises AAV inverted terminal repeat sequences and a codon optimized nucleic acid sequence that encodes CNGA3, and expression control sequences that direct expression of the encoded protein in a host cell.

In another embodiment, the vector is a plasmid that comprises an AAV expression cassette, wherein the expression cassette comprises AAV inverted terminal repeat sequences and a codon optimized nucleic acid sequence that encodes CNGB3, and expression control sequences that direct expression of the encoded protein in a host cell.

It is to be noted that the term “a” or “an” refers to one or more. As such, the terms “a” (or “an”), “one or more,” and “at least one” are used interchangeably herein.

The words “comprise”, “comprises”, and “comprising” are to be interpreted inclusively rather than exclusively. The words “consist”, “consisting”, and its variants, are to be interpreted exclusively, rather than inclusively. While various embodiments in the specification are presented using “comprising” language, under other circumstances, a related embodiment is also intended to be interpreted and described using “consisting of” or “consisting essentially of” language.

As used herein, the term “about” means a variability of 10% from the reference given, unless otherwise specified.

The term “regulation” or variations thereof as used herein refers to the ability of a composition to inhibit one or more components of a biological pathway.

Unless defined otherwise in this specification, technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art and by reference to published texts, which provide one skilled in the art with a general guide to many of the terms used in the present application.

The following examples are illustrative only and are not intended to limit the present invention.

Example 1—Differentiation of Pluripotent Stem Cells into RPE

Choroideremia lacks a relevant mouse model and there is no canine model, therefore, transduction and expression is tested in a human retinal cell model of the disease. Because it is impossible to obtain retinal cells from a living patient, RPE are generated from induced pluripotent stem cells. Pluripotent stem cells are directed to RPE using the protocol described by Buchholz et al, Rapid and Efficient Directed Differentiation of Human Pluripotent Stem Cells Into Retinal Pigmented Epithelium, Stem Cells Translational Medicine, 2013; 2:384-393 which is incorporated by reference in its entirety. See also, Cereso et al, Proof of concept for AAV2/5-mediated gene therapy in iPSC-derived retinal pigment epithelium of a choroideremia patient, Molecular Therapy—Methods & Clinical Development (2014) 1, 14011, which is incorporated by reference in its entirety. Other methods for producing RPE are known in the art.

Briefly, the human induced pluripotent stem cell line is maintained in Dulbecco’s Modified Eagle’s Medium: Nutrient Mixture F-12 (DMEM/F12) containing 2mM GlutaMAX-I, 20% knockout serum replacement, 0.1 mM Modified Eagle’s Medium Non-Essential Amino Acids (MEM NEAA), 0.1 mM β -mercaptoethanol and 4 ng/ml bFGF on a mitomycin C-treated or irradiated mouse embryonic fibroblast feeder layer.

Pluripotent stem cells are passaged directly onto Matrigel (BD Biosciences) in DMEM/F12 with 1 $\times$ B27, 1 $\times$ N2, and 1 $\times$ NEAA (Invitrogen). From days 0 to 2, 50 ng/ml Noggin, 10 ng/ml Dkk1, 10 ng/ml IGF1 and 10 mM nicotinamide are added to the base medium. From days 2 to 4, 10 ng/ml Noggin, 10 ng/ml Dkk1, 10 ng/ml IGF1, 5 ng/ml bFGF and 10mMnicotinamide are added to the base medium. From days 4 to 6, 10 ng/ml Dkk1, 10 ng/ml IGF1 and 100 ng/ml

Activin A (R&D Systems) are added to the base medium. From days 6 to 14, 100 ng/ml Activin A, 10 μ M SU5402 (EMD Millipore, Darmstadt, Germany), and 1 mM VIP are added to the base medium. Control experiments are performed in base media alone (DMEM/F12, B27, N2, and NEAA).

The cells are mechanically enriched by scraping away cells with non-RPE morphology. Subsequently, the remaining RPE are digested using TrypLE Express (Invitrogen) for 5 minutes at 37° C. The cells are passed through a 30- μ m single-cell strainer and seeded onto Matrigel-coated tissue culture plastic, Transwell membranes or CC2-treated chambered slides. Enriched cells are cultured in DMEM-high glucose with 1% fetal bovine serum (FBS), GlutaMAX, and sodium pyruvate for 30 days.

Example 2—Cells Transduced with AAV-REP-1

Briefly, AAV2/8CMV.CBA-REP-1 viral vector incorporating REP-1 codon optimized sequences are produced by transient transfection of HEK293 cells, and the viral particles are precipitated from either the supernatant using polyethylene glycol. See, e.g., Guo et al, Rapid and simplified purification of recombinant adeno-associated virus, J Virol Methods. 2012 August; 183(2): 139-146, which is incorporated herein by reference. The vectors are purified by double CsCl centrifugation, dialyzed, and titered by dot blot assay.

For the prenylation experiments, RPE are seeded in 24-well plates, and 1.2 $\times$ 10⁶ cells are estimated at confluence. Cells are transduced with 100,000 vg per cell, and prenylation assays are performed at 4 weeks posttransduction. Experiments are performed in triplicate.

Example 3—Prenylation

An in vitro prenylation assay is performed as described in Vasireddy et al, PLoS One. 2013 May 7;8(5): c61396, cited above, using [3H]-geranylgeranyl pyrophosphate (GGPP) (Perkin Elmer, Boston, MA, USA) as a prenyl group donor, in the presence of recombinant Rab geranylgeranyl transferase and RAB27. Incorporation of radiolabeled prenyl groups into the RAB27 protein is measured by scintillation counting. For consistency the control values are normalized to 100 and used as the base value. All experiments are performed in triplicate, and statistical comparison of prenylation between experimental and control groups is evaluated using the two-tailed unpaired student’s t-test.

Briefly, 48 hr post transduction, transduced REP cells are washed with cold PBS. Cell pellets are collected and washed thoroughly with cold PBS. Cells are lysed on ice for 30 min using RIPA+Protease inhibitors. In an alternative protocol, cells are sonicated. Cytosolic fractions are collected by centrifuging the lysate at 75,000-100,000 g for 1-2 h at 4° C.

Stocks are prepared for the prenylation reaction as follows.

	STOCKS prepared	FINAL CONCENTRATION REQUIRED
Rab GGTase	16.63 μ M - (GGTase-a) 14.59 μ M - (GGTase-b)	0.05 μ M
Rab 27a	25.93 μ M	4 μ M
3H GGPP	22.2	5 μ M
NP40		1 mM
DTT	10 mM	1 mM

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-continued

STOCKS prepared		FINAL CONCENTRATION REQUIRED
HEPES	1M	50 mM
Mgcl2	100 mM	5 mM

Final reaction volume used for prenylation is 25 μ L

Rab GGTase a	0.075 μ l
Rab Ggtaseb	0.085 μ l
[³ H]-geranylgeranyl pyrophosphate (GGPP)	5.68 μ l
NP40	0.15 μ l
DTT	2.5 μ l
HEPES	1.25 μ l
MgCl2	1.25 μ l
Rab 27a	3.12 μ l
Cytosolic Fraction (Cell lysate)	10.89 μ l

The reaction mixture is incubated at 37° C. for 30 min. The reaction is stopped by adding 9:1 ethanol/HCL, and incubated for 30 minutes. The proteins are collected on glass fiber filter papers (Whatman papers) by vacuum filtration (0.1 ml). The filters are washed carefully with cold phosphate buffer—3 times to remove unbound material. The membranes are dried carefully. The filters are placed in 5 ml scintillation cocktail and scintillation counting is performed. See also, Tolmachova et al, CHM/REP1 cDNA delivery by lentiviral vectors provides functional expression of the transgene in the retinal pigment epithelium of choroideremia mice, *The Journal of Gene Medicine*, 2012; 14:158-68, which is incorporated herein by reference in its entirety.

Assays for CNGA3 or CNGB3 proof-of-concept may include use of a spontaneous mutant animal model (for example, the *Cnga3*^{-/-} mouse or the Awassi sheep). The mouse model could be bred with an “all-cone” photoreceptor mouse, the *Nrl*^{-/-} mouse, to obtain double knockouts. The latter (*Cnga3*^{-/-}*Nrl*^{-/-}) mouse may expedite identification of efficacy. Efficacy could be measured by pupillometry, measures of visual acuity and contrast (for example, using optokinetics), electroretinograms, and visual behavior. Ultimately, histology will document expression of the transgene with improved outcomes on the other measures. Histologic approaches will also be used to document any effects of the intervention on cone photoreceptors (total number of cone photoreceptors, density, location, etc).

Similar to choroideremia as discussed above, assays for proof-of-concept for gene augmentation therapy for

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CNGA3- or CNGB3-associated achromatopsia may include use of induced pluripotent stem cell (iPSC) models. The iPSC models, generated from patients with achromatopsia due to CNGA3 or CNGB3 mutations, will be differentiated into retinal precursors and/or photoreceptor cells in vitro. The wildtype CNGA3 (or CNGB3) cDNA will be delivered to these cells using recombinant AAV and the cells will be analyzed for biogenesis and preservation of function of the relevant (Cyclic nucleotide-gated, CNG) channel comprised by these subunits. Channel function will be assessed by electrophysiology on membrane patches. Restoration of the channel should rescue cGMP-activated currents. Additional studies can test for sensitivity of channel function before and after delivery of the wildtype CNG cDNA to physiological ligands.

Example 4: In Vitro Expression of AAV.Codon-Optimized Human CHM

The objective of this study was to evaluate the ability of AAV mediated CHM expression after gene delivery using a series of next generation AAV 2 and AAV8 vectors encoding the codon optimized CHM gene (SEQ ID NO: 1) in 84-31 and COS-7 cell lines.

To maximize the expression of CHM, a codon optimized CHM sequence was produced (SEQ ID NO: 1). The codon optimized plasmid was synthesized and used in the creation of all the next generation CHM transgene expression cassettes. To overcome the potential problem of contamination of non-functional AAV genomes, a non-coding lambda stuffer region was included in the vector backbone. Incorporation of stuffer not only increases the length of the plasmid, but also diminishes the possibility of plasmid DNA backbone contamination while packaging the AAV. The impact of incorporating a stuffer region in the vector backbone to eliminate the plasmid DNA impurities was carried out as an independent study. Two recombinant AAV proviral plasmids (high and low copy) backbones were used to generate the different constructs. The high copy plasmid was designed based on the pUC vector origin. The low copy plasmid was designed based on the p15A origin. To further enhance the translation from the correct initiation codon, a Kozak sequence upstream of the start codon was incorporated.

A total of four plasmids have been engineered for the current study and those described in the following examples (Table 1). In addition, a plasmid carrying the CHM native sequence, which is currently being used in a clinical trial, was also generated (version 1). Plasmid maps for each of Version 2a, 2b, 3a and 3b, and Version 1 are shown in FIGS. 6-10, respectively.

TABLE 1

Plasmid features						
Name	REP Sequence	Lambda insert	Kozak Sequence	Origin	Copy number	Promoter/Intron
Version 1 (V1)	Native	Present	Absent	pUC	High copy	CMV-CBA promoter + Enhancer extension
Version 2a (V2a)	Codon-optimized	Present	present	pUC	High copy	CMV-CBA promoter + Enhancer extension
Version 2b (V2b)	Codon-optimized	Not Present	present	pUC	High copy	CMV-CBA promoter + Enhancer extension
Version 3a (V3a)	Codon-optimized	Present	present	p15A	Low copy	CMV-CBA promoter + Enhancer extension
Version 3b (V3b)	Codon-optimized	Not Present	present	p15A	Low copy	CMV-CBA promoter + Enhancer extension

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The in vitro expression of these constructs was tested in COS-7 and 84-31 cell lines. The engineered features of the next-generation CHM constructs are depicted in Table 1.

Recombinant AAV proviral high and low copy plasmids were generated by cloning the codon optimized human CHM cDNA (hCHM) (SEQ ID NO: 1) into the transgene cassette. The transgene was placed under the control of a hybrid chicken β actin (CBA) promoter. This promoter consists of the cytomegalovirus (CMV) immediate early enhancer, the proximal chicken β actin promoter, and CBA exon 1 flanked by intron 1 sequences. The proviral high and low copy number plasmids also contain AAV inverted terminal repeats and a PolyA sequence. The next generation plasmid backbones used in the current study contain a lambda phage fragment stuffer followed by the kanamycin bacterial selection gene. Additional plasmids lack the stuffer but contain the kanamycin selection gene. The high-copy number vector is similar to that of pUC plasmids (~300 copies/bacterial cell). The low copy number plasmid (~10 copies/bacterial cell) has an origin of p15A. To enhance translation from the correct initiation codon, all next generation constructs contain a Kozak consensus sequence upstream of the start codon, ATG. The generated plasmids are sequence verified using primers that can specifically target either the promoter-enhancer extension sequence or the codon optimized CHM sequence. The plasmid maps and sequences of all five constructs are shown in FIGS. 6-10.

Standard triple transfection with calcium phosphate was used to generate AAV vectors listed below (see Table 2 for vector qualification). Both AAV2 and AAV8 serotypes of the vectors were generated to ensure the results are serotype-independent.

TABLE 2

Summary of AAV2 and AAV8 vectors generated and concentration of viral stocks				
Name	Serotype	Plasmid name	Lot number	Stock Conc. (vg/ml)
AAV2.V1	AAV 2	Version 1 (V1)	KA 892*	4.47E+12
AAV2.V2a	AAV 2	Version 2a (V2a)	CT 239	2.16E+12
AAV2.V2b	AAV 2	Version 2b (V2b)	CT 238	7.40E+12
AAV2.V3a	AAV 2	Version 3a (V3a)	CT 258	4.82E+12
AAV2.V3b	AAV 2	Version 3b (V3b)	CT 256	5.91E+12
AAV8.V1	AAV 8	Version 1 (V1)	KA 808*	1.39E+13
AAV8.V2a	AAV 8	Version 2a (V2a)	CT 245	1.04E+13
AAV8.V2b	AAV 8	Version 2b (V2b)	CT 244	1.11E+13
AAV8.V3a	AAV 8	Version 3a (V3a)	CT 259	8.67E+12
AAV8.V3b	AAV 8	Version 3b (V3b)	CT 255	1.36E+13

The 84-31 cell line is a subclone of 293 HEK cell line (human embryonic kidney cells) and constitutively expresses adenovirus E4 proteins to enhance transduction of AAV virus. COS-7 cells are fibroblast like cell lines that are derived from monkey kidney tissues. Both 84-31 cells and COS-7 cells were plated, separately, in 6-well cell culture plates and transduced with one of the ten test articles (either AAV2 or AAV8) at five different multiplicity of infection (MOIs). After 36-48 hours, cells were harvested, lysed and protein samples were prepared for SDS-PAGE followed by Western blot analysis to detect the expression of exogenous CHM.

Both 84-31 and COS-7 cells were cultured in Dulbecco's modified Eagle medium (DMEM)-high glucose with 10% fetal bovine serum, and 1% penicillin/streptomycin at 37° C. in an environment supplied with 5% CO₂. The day before transduction (18-24 h before) cells at a density of 3E5 were seeded in 2 ml of cell culture media in each well of a 6-well

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cell culture dish. Seeded cells were incubated at 37° C. in an environment supplied with 5% CO₂. Wells of both COS-7 and 84-31 cells were infected with AAV vectors listed below at various multiplicities of infection (MOI) (Table 3 and Table 4). No virus was added to negative control cells (untransduced cells). Briefly, the tissue culture media was removed and a fresh 2 ml aliquot of media was added to each well of the 6 well culture dish. Then the predetermined amount of AAV vector was measured (directly from the stock) and added to each well (Table 3 and Table 4). For an MOI of 1E4, 1 μ L of respective virus stock was diluted to 10 μ L with cell culture media. From this solution, the predetermined volume of the virus was added to respective well (Table 3 and 4). Cells were incubated with the AAV virus for 36-48 hours at 37° C. with 5% CO₂ till harvesting. Cells were observed under microscope before harvesting to check for abnormalities.

TABLE 3

Infection dose of four next generation AAV2 and AAV8 hCHM vectors in COS-7 cells.				
Vector Used	Cell Line	Cell Density	Vector Used (μ L)	MOI
No AAV	COS-7	3E5	—	—
AAV2.V2a	COS-7	3E5	1.5	1E4
	COS-7	3E5	15	1E5
	COS-7	3E5	45	3E5
	COS-7	3E5	75	5E5
	COS-7	3E5	150	1E6
AAV2.V2b	COS-7	3E5	4.2 (from a 1 to 10 dilution of the stock)	1E4
	COS-7	3E5	4.2	1E5
	COS-7	3E5	12.6	3E5
	COS-7	3E5	21	5E5
	COS-7	3E5	42	1E6
AAV8.V2a	COS-7	3E5	2.9 (from a 1 to 10 dilution of the stock)	1E4
	COS-7	3E5	2.88	1E5
	COS-7	3E5	8.65	3E5
	COS-7	3E5	14.42	5E5
	COS-7	3E5	28.85	1E6
AAV8.V2b	COS-7	3E5	2.7 (from a 1 to 10 dilution of the stock)	1E4
	COS-7	3E5	2.7	1E5
	COS-7	3E5	8.1	3E5
	COS-7	3E5	13.5	5E5
	COS-7	3E5	27	1E6

TABLE 4

Infection rates of four next generation AAV2 and AAV8 hCHM vectors in 84-31 cells.				
Vector Used	Cell Line	Cell Density	Vector Used (μ L)	MOI
No AAV	84-31	3E5	—	—
AAV2.V2a	84-31	3E5	1.5	1E4
	84-31	3E5	15	1E5
	84-31	3E5	45	3E5
	84-31	3E5	75	5E5
	84-31	3E5	150	1E6
AAV2.V2b	84-31	3E5	4.2 (from a 1 to 10 dilution of the stock)	1E4
	84-31	3E5	4.2	1E5
	84-31	3E5	12.6	3E5
	84-31	3E5	21	5E5
	84-31	3E5	42	1E6

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TABLE 4-continued

Infection rates of four next generation AAV2 and AAV8 hCHM vectors in 84-31 cells.				
Vector Used	Cell Line	Cell Density	Vector Used (μL)	MOI
AAV8.V2a	84-31	3E5	2.9 (from a 1 to 10 dilution of the stock)	1E4
	84-31	3E5	2.88	1E5
	84-31	3E5	8.65	3E5
	84-31	3E5	14.42	5E5
	84-31	3E5	28.85	1E6
AAV8.V2b	84-31	3E5	2.7 (from a 1 to 10 dilution of the stock)	1E4
	84-31	3E5	2.7	1E5
	84-31	3E5	8.1	3E5
	84-31	3E5	13.5	5E5
	84-31	3E5	27	1E6

First, both, the COS7 and 84-31 cell lines were used to test if the in vitro expression of CHM is cell line independent. Once the independence was established, all subsequent experiments were carried out only in 84-31 cells, which have shown superior transduction efficiency with AAV. Wells of 84-31 cells were infected with the AAV vectors listed below at various MOI (see table 3 and 4).

Western blot analysis: 1. Cell lysates were prepared. The AAV transduced cells along with the untreated control cells, were harvested 36-48 h post-infection after a thorough PBS wash. Cells were then lysed on ice using RIPA buffer with protease inhibitors. Cell lysates were cleared by centrifuging at 13,000 rpm for 10 min. 2. Quantification and preparation of proteins. Protein quantification of the cell lysates was carried out using ThermoFisher Micro BCA™ Protein Assay Kit following manufacturer's instructions. Protein concentration was determined by taking OD reading at 562 nm. To evaluate the in vitro expression of CHM, between 40-60 ug of measured protein was loaded on 4-12% Bis-Tris gels. 3. SDS-PAGE and blotting SDS-PAGE and western blot analysis were carried out according to known protocols. Briefly, the protein gels were transferred on to a nitrocellulose membrane, blocked in milk and incubated with the primary antibodies. Antihuman REP-1 2F1 antibody (2F1, 1:1000 dilution) and one of the following: anti-GAPDH antibody (1:1000 dilution), anti actin antibody (1:1000 dilution) or anti-Tubulin antibody (1:5000 dilution) was used as primary antibodies for each blot. After washing the blot, HRP conjugated anti-mouse IgG antibody and/or anti-rabbit IgG antibody at a concentration of 1:5000 were used as secondary antibodies. The blots were developed by chemiluminescence using ECL reagents according to the manufacturer's instructions. Controls: 1. Loading controls: One of the following: anti-Actin antibody, anti-tubulin antibody or anti-GAPDH antibody was used as a loading control to demonstrate equal loading of protein in each well of the gels. Anti-Tubulin antibody detects a protein of ~51 kDa. Anti-Actin antibody detects a protein of ~42 kDa, and anti-GAPDH antibody detects a protein of ~39 kDa. Initial blots were probed with either anti-tubulin antibody or anti-Actin antibody or anti-GAPDH antibody depending up on their availability. After initial experiments, to be consistent, anti-GAPDH antibody was used as the loading control. 2. Positive control: After the production of hREP-1 protein was established in AAV2.V2a transduced COS-7 cells, the AAV2.V2a-Cos-7 cell lysates were used as positive control in later western blot experiments. 3. Negative control:

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Untreated cells were used as negative control. Analyses of western blot results of REP-1 protein production in various cell lines are summarized in Table. 5.

TABLE 5

Name	Serotype	Cell Line	FIG.	MOI Used	CHM Expression (Observation)
AAV2.V2a	AAV2	COS-7	FIG. 1	1E4-1E6	Detectable
AAV2.V2a	AAV2	84-31	FIG. 1	1E4-1E6	expression of
AAV2.V2b	AAV2	COS-7	FIG. 2	1E4-1E6	CHM at all
AAV2.V2b	AAV2	84-31	FIG. 2	1E4-1E6	MOIs tested
AAV2.V3a	AAV2	84-31	FIG. 3	1E4-1E6	
AAV2.V3b	AAV2	84-31	FIG. 3	1E4-1E6	
AAV2.V2a	AAV8	COS-7	FIG. 4	1E4-1E6	Detectable
AAV2.V2a	AAV8	84-31	FIG. 4	1E4-1E6	expression of
					CHM at MOI of
					1E5-1E6.
AAV2.V2b	AAV8	COS-7	FIG. 5	1E4-1E6	Detectable
AAV2.V2b	AAV8	84-31	FIG. 5	1E4-1E6	expression of
					CHM above
					MOI of
					3E5-1E6
AAV2.V3a	AAV8	84-31	FIG. 6	1E4-1E6	Detectable
AAV8.V3b	AAV8	84-31	FIG. 6	1E4-1E6	expression of
					CHM above
					MOI of
					3E5-1E6

Monoclonal human REP-1-specific antibody, detected one single ~75-80 kDa hREP-1 protein in cells transduced with next generation AAV2.copt.CHM/AAV8.copt.CHM. A 75-80 kDa band was not observed in cell lysates of untreated control cells. Probing of the blots with either anti-Actin/antitubulin/anti-GAPDH antibody showed a band of equal density in all lanes of the western blot including in untreated controls. Anti-actin antibody detected a protein molecular weight band at ~42 kDa, anti-tubulin antibody detected a protein at ~51 kDa, and anti-GAPDH antibody detected a protein at ~39 kDa. All antibodies detected only specific bands of expected size molecular weight. No nonspecific bands were observed in any of the blots. A pre-stained molecular weight marker was used to compare the molecular weights of protein of interest.

Briefly, REP-1 protein was observed at the expected size in COS-7 and 84-31 cells transduced with AAV2.V2a, AAV2.V2b, AAV2.hCHM.V3a and AAV2.hCHM.V3b. Untreated controls did not reveal the presence of expected size human REP-1 protein. Labeling the blot with anti-actin antibody detected a protein band of equal intensity in all lanes of the gel at ~42 kDa. Pre-stained protein ladder was used to compare the molecular weights of REP-1 and Actin. Data not shown.

The results indicate that AAV2 and AAV8 serotype vectors containing next generation plasmids are able to transduce 84-31 and COS-7 cells efficiently. Expression of CHM in the next generation plasmids was in the detectable range, and demonstrated a dose dependent trend. Transduction of cells with the next generation hCHM viruses resulted in production of REP-1 protein of the predicted size.

Example 5: Comparison of in Vitro Protein Expression of AAV.Codon-Optimized.Human CHM with AAV Native.Human CHM

The objective of this study was to delineate transduction efficiency of AAV vectors (serotype 2 and 8) containing various versions of the CHM-containing transgene cassettes by measuring levels of REP-1 protein in a 84-31 cell line based study model.

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Plasmids and Vectors: A total of 5 transgene plasmids were compared either in AAV2 or AAV8: Version 1 (previously being used in an on-going clinical trial) and four next generation versions (V2a, V2b, V3a, and V3b). The plasmids were engineered as described in Example 4, and the features thereof are shown in Table 1. Table 2 above shows a summary of AAV2 and AAV8 vectors generated and concentration of viral stocks.

Study Design (e.g. Treatment Groups)

1. In a pilot experiment, COS-7 and 84-31 cells were transduced with AAV2.hCHM.Version1, Version2a and Version 2b. Western blot was performed to compare transduction efficiency levels in the two cells lines.

2. 84-31 cells, plated in 6-well plates were transduced with one of the 10 test articles (Version 1, 2a, 2b, 3a and 3b in either AAV2 or AAV8 background) at an MOI of 3E5. After 36-48 hours, cells were harvested and lysed. The lysate was loaded on SDS-PAGE, and subjected to further Western blot analyses. Levels of REP-1 protein are compared amongst all construct versions. Two separate plates were setup for each AAV2.CHM or AAV8.CHM experiments were analyzed, separately.

Test Material Administration

3.4.1 Cell Culture

83-41 cells and COS-7, both were cultured in Dulbecco's modified Eagle medium (DMEM)-high glucose with 10% fetal bovine serum, and 1% penicillin/streptomycin at 37° C. in an environment supplied with 5% CO₂.

3.4.2 Preparation of Cells for Transduction:

The day before transduction (18-24 h before) 83-41 and COS-7 cells were seeded at a density of 3E5 in 2 ml of cell culture media per well in a 6-well cell culture dish. The seeded cells were incubated at 37° C. in an environment supplied with 5% CO₂.

3.4.3 Transduction:

Wells of 84-31 cells and Cos-7 were infected with AAV vectors as described below at an MOI of 3E5 (see Table 6 for the pilot experiment and Table 7 for the second set of experiments). No virus was added to the negative (untransduced) control. Briefly, first, the tissue culture media was removed and replaced with 2 ml fresh media/well in each the wells in the 6 well cell culture dish. Then the predetermined amount of AAV vector (see table 2 for vector volumes used for transduction) was measured (from the stock) and directly added to each well. Cells were incubated with the AAV virus for 36-48 hours at 37° C. with 5% CO₂ until harvesting. Cells were observed under microscope before harvesting to check any abnormality. Western blot analysis was performed as described in Example 4.

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TABLE 6

Pilot Experiment: Infection doses of AAV2.hCHM.V1, 2a, 2b in 84-31 and COS-7 cells.				
Vector Used	Cell Line	Cell density	Vector used (μL)	MOI
None	84-31	3E5	0	0
AAV2.V1	84-31	3E5	13.42	3E5
AAV2.V2b	84-31	3E5	8.11	3E5
AAV2.V2a	84-31	3E5	27.78	3E5
AAV2.V1	COS-7	3E5	13.42	3E5
AAV2.V2b	COS-7	3E5	8.11	3E5
AAV2.V2a	COS-7	3E5	27.78	3E5
None	COS-7	3E5	0	0

TABLE 7

Infection doses of AAV.hCHM next generation vectors and V1 (AAV2 and AAV8) in 84-31 cells				
Vector Used	Cell Line	Cell density	Vector used (μL)	MOI
None	84-31	3E5	0	0
AAV8.V2a	84-31	3E5	5.77	3E5
AAV8.V2b	84-31	3E5	5.41	3E5
AAV8.V3a	84-31	3E5	6.92	3E5
AAV8.V3b	84-31	3E5	4.41	3E5
AAV8.V1	84-31	3E5	4.32	3E5
None	84-31	3E5	0	0
AAV2.V2a	84-31	3E5	27.78	3E5
AAV2.V2b	84-31	3E5	8.11	3E5
AAV2.V3a	84-31	3E5	12.45	3E5
AAV2.V3b	84-31	3E5	10.15	3E5
AAV2.V1	84-31	3E5	13.42	3E5

Results: Comparison of the Expression of Native hCHM (AAV2.hCHM.V1) Versus Codon-Optimized CHM AAV2a and 2b Vectors in 84-31 and COS-7 Cells

In this experiment 84-31 and COS-7 cells were transduced with either no vector (untreated control), AAV2.hCHM.Version1, AAV2.hCHM.Version2a or AAV2.hCHM.Version2b. Western blot analysis with an anti-human REP-1 antibody, showed that REP-1 protein levels were detectable at ~75-80 kDa in all AAV2 (V1, V2a, V2b) transduced samples and in both cells lines (Data not shown). A slightly better protein expression was seen in 84-31 cell line (Table 8). Anti-REP1 antibody detected negligible amount of REP-1 protein in untreated cells. Labeling of the blot with GAPDH antibody detected a band at ~39 kDa in all cell lysates, including the untreated cells.

Densitometric evaluation (quantification of the protein level) of the blots using ImageJ software demonstrated that after normalizing the values to the expression of endogenous GAPDH protein the transduction efficiency was similar in 84-31 and COS-7 cells. (See Table 8 for results.) Based on this, the 84-31 cell line, which is from human origin was used for further experiments.

In conclusion, AAV2.V1, AAV2.V2a and AAV2.V2b induced the production of REP-1 protein in both, 84-31 and COS-7 cells with similar transduction efficiency.

		GAPDH		REP-1	REP-1
		NORMALIZED		NORMALIZED	NORMALIZED
		TO GAPDH		TO RESPECTIVE	TO REP-1 OF V1
		V1		GAPDH	(%)
84-31	AAV2.V1	23416.844	19350.773	1	23416.844
	AAV2.V2b	36626.765	20357.894	1.011	36203.838
	AAV2.V2a	31114.844	20315.945	1.009	30819.0684
					131.610

-continued

				GAPDH NORMALIZED TO GAPDH OF V1	REP-1 NORMALIZED TO RESPECTIVE GAPDH	REP-1 NORMALIZED TO REP-1 OF V1 (%)
	REP1	GAPDH				
			Molecular weight	Marker		
COS7	AAV2.V1	12880.459	15479.288	1	12880.459	100
	AAV2.V2b	19209.823	14321.167	0.711	26991.925	209.557
	AAV2.V2a	15132.602	13145.924	0.849	17818.595	138.338

Comparison of the expression of native CHM versus codon-optimized CHM AAV2 vectors in 84-31 cells: Using an anti-human REP-1 antibody, the Western blot analysis of the 84-31 cells transduced with AAV2.hCHM.V2a, V3a, V2b, V3b and V1 detected a band at ~75-80 kDa in all conditions (Data not shown). Anti-REP1 antibody detected negligible amount of REP-1 protein in untreated cells. Labeling of the blot with GAPDH antibody detected a band at ~39 kDa in all cell lysates, including the untreated cells. Densitometric evaluation (quantification of the expression level) of the blots using ImageJ software demonstrated an increase in the expression of AAV2.hCHM.V2a, 3a, 2b, and 3b compared to AAV2.hCHM.V1 after normalizing the values to the production of endogenous GAPDH protein. See Table 9 and 10 for results.

Comparison of the expression of native CHM versus codon-optimized CHM in AAV8.V1, V2a, V3a, V2b, V3b vectors in 84-31 cells: Western blot analysis of cells transduced with AAV8.V1, AAV8.V2a, AAV8.V3a, AAV8.2b, AAV8.3b, with anti-human REP-1 antibody detected a band at ~75-80 kDa in all transduced cells (Data not shown). Anti-REP1 antibody detected negligible amount of REP-1 protein in untreated cells. Labeling of the blot with GAPDH antibody detected a band at ~39 kDa in all cell lysates, including the untreated cells. Densitometric evaluation of the blots using ImageJ software demonstrated higher expression of AAV8.hCHM.V2a; 3a; 2b; 3b compared to AAV8.V1. Values are obtained after normalizing the CHM values first, to the expression of the respective endogenous GAPDH protein and then normalized to the expression level of the average of Version 1. See Table 11 and Table 12 for results.

TABLE 9

Values of REP-1 protein in 84-31 cells after transduction with AAV2.hCHM. V1, Va, V2b, V3a or V3b for PLATE 1 (Western Blot not shown)						
CONSTRUCT		GAPDH NORMALIZED		REP-1 NORMALIZED TO	REP-1 NORMALIZED	
LANE		RAW VALUE		TO GAPDH OF	RESPECTIVE	TO REP-1 OF V1
NAME	NUMBER	REP-1	GAPDH	V1	GAPDH	(%)
AAV2.V1	1	23367.593	15155.602	1	23367.593	100
AAV2.V2a	3	26949.421	10969.581	0.723797115	37233.39103	159.3377248
AAV2.V3a	5	29867.714	14595.894	0.963069233	31013.0497	132.7182038
AAV2.V2b	7	32728.128	14133.551	0.932562824	35094.82381	150.1858741
AAV2.V3b	9	33986.543	13670.066	0.901981063	37679.88531	161.2484662

TABLE 10

Values of REP-1 protein in 84-31 cells after transduction with AAV2.hCHM. V1, V2a, V2b, V3a or V3b for PLATE 2 (Western blot not shown)						
CONSTRUCT		GAPDH NORMALIZED		REP-1 NORMALIZED TO	REP-1 NORMALIZED	
LANE		RAW VALUE		TO GAPDH OF	RESPECTIVE	TO REP-1
NAME	NUMBER	REP-1	GAPDH	V1	GAPDH	OF V1 (%)
AAV2.V1	2	23128.593	11993.823	1	23128.593	100
AAV2.V2a	4	23623.836	10982.798	0.915704526	25798.53582	111.5439051
AAV2.V3a	6	28832.543	13176.359	1.098595419	26244.91465	113.473892
AAV2.V2b	8	31349.229	16028.329	1.336381986	23458.28463	101.4254721
AAV2.V3b	10	33273.856	14760	1.230633469	27037.99047	116.9028763

TABLE 11

Values of REP-1 protein expression in 84-31 cells after transduction with AAV8 hCHM Version 1, 2a, 2b, 3a and 3b - PLATE 1 (Western blot not shown)						
CONSTRUCT		GAPDH NORMALIZED TO		REP-1 NORMALIZED TO		REP-1 NORMALIZED
NAME	LANE NUMBER	RAW VALUE		GAPDH OF V1	RESPECTIVE GAPDH	TO REP-1 OF V1 (%)
AAV8.V1	11	3630.589	20309.924	1	3630.589	100
AAV8.V2a	13	7133.439	17051.48	0.839	8496.599	234.028
AAV8.V3a	15	5828.418	15801.045	0.777	7491.575	206.346
AAV8.V2b	17	11411.702	19249.681	0.947	12040.241	331.633
AAV8.V3b	19	17610.066	18727.024	0.922	19098.555	526.045

TABLE 12

Values of REP-1 protein expression in 84-31 cells after transduction with AAV8 hCHM Version 1, 2a, 2b, 3a and 3b - PLATE 2 (Western blot not shown)						
CONSTRUCT		GAPDH NORMALIZED TO		REP-1 NORMALIZED TO		REP-1 NORMALIZED
NAME	LANE NUMBER	RAW VALUE		TO GAPDH OF V1	RESPECTIVE GAPDH	TO REP-1 OF V1 (%)
AAV8.V1	12	3507.468	19082.681	1	3507.468	100
AAV8.V2a	14	4359.296	13274.731	0.695642871	6266.571801	178.6636913
AAV8.V3a	16	6533.246	20720.246	1.0858142	6016.909708	171.5456765
AAV8.V2b	18	13962.045	17842.167	0.934992677	14932.7854	425.7425983
AAV8.V3b	20	16049.823	13836.368	0.725074637	22135.40811	631.0936581

Conclusion: Comparative expression studies demonstrated that application of AAV vectors carrying the next generation AAV.hCHM. Version 2a, 2b, 3a and 3b induced increased production of REP-1 protein compared with Version 1 (currently used in clinical trials) in both AAV2 and AAV8 serotype vectors in 84-31 cells.

Example 6: Evaluation of Lambda Stuffer's Effect on AAV Vector Production by qPCR Titer Analysis

A single qPCR (quantitative polymerase chain reaction) run was performed on all 8 AAV vectors shown in Table 2, above, in order to assess the effect of lambda stuffer sequences on the amount of DNA impurities. Linearized AAV plasmid standard was used to generate the assay standard. Primer-probe sets were designed on either the CMV/CBA promoter region for quantitation of properly packaged AAV genome or the Kanamycin resistance (KanR) encoding region for reverse packaging. Standards and vector samples were run in two sets, one with the CMV/CBA primer-probe set and the other with the KanR set. Vector sample values (viral genome copy per mL) were determined from each respective standard curve. The effect of the stuffer sequence was assessed by comparing the relative amount of KanR-containing impurities in each vector lot against CMV/CBA containing sequences.

Reagents:

Transgene-Containing Viral Vector Titer:

Reference: CMV-CBA Promoter

Primers: CMV-F: CCC ACT TGG CAG TAC ATC AA

-continued

CMV-R: GCC AAG TAG GAA AGT CCC ATA A

FAM-Probe: /56-FAM/CA TAA TGC C/ZEN/A GGC GGG CCA

TTT AC/3IABkFQ/

Impurity-Containing Viral Vector Titer
Reference: Kanamycin Resistance Gene

Primers: KAN-F: GAT GGT CGG AAG TGG CAT AA

KAN-R: TGC GCC AGA GTT GTT TCT

FAM-Probe: /56-FAM/CC GTC AGC C/ZEN/A GTT TAG TCT

GAC CA/3IABkFQ/

Dilution Reagent: Diluent Q (0.001% PF-68 in nuclease free water); Diluted 1% PF-68 solution 1000-folds with sterile water. Diluent S: Diluent Q+2 ng/μL salmon sperm DNA (Agilent technologies Cat #201190)
ABI TaqMan™ Universal Master Mix (Applied Biosystems 4304437/4326708)

Qiagen PCR Product Purification Kit (Qiagen 28104)·
ABI QuantStudio 6 Flex System

1.2. Sample Preparation

DNase digest solution was prepared by combining the following: DNase buffer (10×) 5 μL, Nuclease-free H₂O 30 μL, DNase I (Invitrogen, 18068-015) 5 μL

Ten μL of each AAV vector sample was mixed in and incubated at ambient temperature for 10 minutes. The digest mix was inactivated by adding 50 μL of SDS/EDTA/NaCl solution (0.2% SDS/5 mM EDTA/0.2M NaCl) and incubating at 95° C. for 10 minutes. Each AAV vector sample was diluted 10-100,000 fold in Diluent S for qPCR analysis.

1.3. qPCR Standard Preparation

Reference standard DNA (linearized) was prepared by digesting plasmid p1008 (low copy transgene plasmid without stuffer) with XhoI and purifying with Qiagen PCR purification kit. Purified material was analyzed on agarose gel to confirm identity, and quantified by Nanodrop. DNA copy number was determined from the stock concentration using the following equivalence: 1 bp=1.096E-21 g. qPCR Standards were prepared according to the following table:

TABLE 13

DNA Standard	Concentration [Copies/10 μ L]	Dilution
Standard Stock	2×10^9	NA
S1	1×10^8	10 μ L Stock + 190 μ L Diluent S
S2	1×10^7	10 μ L S1 + 90 μ L Diluent S
S3	1×10^6	10 μ L S2 + 90 μ L Diluent S
S4	1×10^5	10 μ L S3 + 90 μ L Diluent S
S5	1×10^4	10 μ L S4 + 90 μ L Diluent S
S6	1×10^3	10 μ L S5 + 90 μ L Diluent S

PCR Reaction Setup

Extracted DNA samples were analyzed in triplicate (3 wells) in a single qPCR run. The run included reference DNA standards in triplicate, ranging from 103 to 108 copies per well. No-template-control (NTC) was included as negative control. Each AAV vector preparation was analyzed with both CMV/CBA and KanR primer/probe sets. Similarly, for quantitation of each set, the standards were also analyzed with both CMV/CBA and KanR primer/probe sets.

TABLE 14

PCR Reaction Setup		
Reagents	Final conc. in Reaction	Volume per 25 μ L Reaction
Universal Master Mix (2x)	1x	12.5 μ L
Optimized Primer Mix (20 μ M)	0.8 μ M	1.0 μ L
Taqman Probe (10 μ M)	0.2 μ M	0.5 μ L
Nuclease-Free Water	NA	1.0 μ L
Sample/Standard	NA	10.0 μ L

PCR reaction sequence was set up as follows: 50° C. 2 minutes 1 cycle; 95° C. 10 minutes 1 cycle; 95° C. 15 seconds 40 cycles; 60° C. 1 minutes 40 cycles

Run performance. Standards were prepared and run at 103 to 108 DNA copies per well. Lower limit of the assay was set at 1000 copies since assay sensitivity was not an important factor for this experiment. A standard curve was generated for the run using the standard copy numbers and CT (threshold cycle) values of the standards. Linear regression of the standards was performed using the ABI software (data not shown). Standard curve fit had a correlation coefficient (R2 value) of 0.998 or greater indicating a reliable fit model. The slope of the standard curves was -3.5. Slope was used to calculate the efficiency of the amplification reaction, and values between -3.2 and -3.6 represented amplification efficiency between 90% and 110%. Both standard reactions were run with 92.6~93.8% efficiency. Precision of triplicate wells ranged from 2~10%, indicating good agreement among replicates. No-template-control (NTC) resulted in non-quantifiable amplification below the lower limit of the assay.

TABLE 15

Summary of standard curve fit					
Reporter	Target	Slope	Y-intercept	R	Efficiency (%)
FAM	CMV	-3.513	41.896	0.998	92.597
FAM	KAN	-3.481	39.968	1.000	93.761

Results:

Sample value determination: The sample values (AAV genome and reverse-packaging copy number) were interpolated from each matching standard curve (CMV/CBA or KanR), using CT values. Interpolated DNA copy number was corrected for initial dilution and/or digest dilution. Additional correction factor of 2 was applied to account for the difference between double-stranded DNA standards and single-stranded DNA in samples.

Analysis results for 8 AAV vectors are summarized in the table below, with quantitative comparison between the transgene-containing AAV concentration (CMV/CBA) and the KanR-containing impurity concentration. Analysis of results demonstrate that insertion of lambda stuffer into the transgene plasmid effectively reduced the occurrence of plasmid-backbone DNA (i.e. KanR) packaging during AAV production from ~7-20 folds (FIG. 11).

TABLE 16

qPCR amplification of kanamycin versus CMV/CBA expressed as percentage				
Sample Name	Lambda Stuffer	CMV/CBA qPCR (vg/ml)	KanR qPCR (vg/ml)	Kan vs. CMV/CBA (%)
AAV2.V2b	No	1.23E+13	6.46E+11	5.25%
AAV2.V2a	Yes	4.61E+12	3.60E+10	0.78%
AAV8.V2b	No	3.19E+13	1.43E+12	4.48%
AAV8.V2a	Yes	2.90E+13	1.19E+11	0.41%
AAV2.V3b	No	1.26E+13	6.69E+11	5.31%
AAV2.V3a	Yes	6.33E+12	4.56E+10	0.72%
AAV8.V3b	No	5.19E+13	3.07E+12	5.92%
AAV8.V3a	Yes	2.60E+13	8.00E+10	0.31%

Example 6: In Vitro Expression of Next Generation AAV8 Vectors in iPS Cells by Western Blot

The objective of this study was to evaluate the ability of AAV mediated CHM expression after gene delivery using a series of next generation AAV2 and AAV8 vectors carrying the codon optimized REP-1-encoding gene in induced pluripotent cell lines (iPSC).

Induced pluripotent stem (iPS) cell technology has been successfully utilized as a platform for testing gene therapy vectors in several proof-of-concept and gene therapy studies including ocular diseases. These patient-specific iPS cells provide a valuable in vitro model system to study disease pathogenesis and establish a model to test proof-of-concept of gene therapy where relevant animal models are unavailable. As a preliminary step to test our AAV-mediated gene augmentation therapy for Choroideremia (CHM), we have generated iPS cells from human patients harboring mutations in the causative gene, CHM, which encodes Rab Escort Protein 1 (REP-1) (See example 1) (Method is described in NCP.003). The generated iPS cells were used to evaluate the in vitro expression of our next generation AAV.codon optimized.CHM constructs.

Plasmids and vectors were as described in Example 4. Induced pluripotent stem (iPS) cells are stem cells generated in the laboratory from somatic cells, peripheral blood mononuclear cells, that were reprogrammed back to a pluripotent state. Reprogramming of blood cells enables the development of personalized in vitro cellular models for therapeutic applications. In this report, iPS cells from individuals affected by CHM were used to test the in vitro production of REP-1 protein through western blot analysis. The following table (Table 17) describes the details of iPS cells studied and their respective CHM disease-causing mutations. Table 17: An overview of the iPS cells generated from patients with CHM mutations

Cell Line	Affected	Mutation in CHM	Method of iPSC generation
JB 588	Affected	Arg 555 stop	Sendai virus mediated reprogramming
JB 527*	Affected	Exon 2-4 deletion	Sendai virus mediated reprogramming
JB 500*	Affected	Ex 10 c.1327_1327 del AT (Needs confirmation)	Sendai virus mediated reprogramming

*iPS cell line qualification tests are on-going.

Study Design (e.g. Treatment Groups)

1. iPS cells plated on a 12 well cell culture plate are infected with AAV2. hCHM Version 1, Version 2a; Version 2b; Version 3a; Version 3b (AAV2.V1; V2a; V2b; V3a; V3b) at an MOI of either 1E5 or 3E5. After 24 hours of transduction, 1 ml of iPS cell culture media was added to the cells. 36-48 hours of transduction, cells were harvested, lysed and processed for SDS-PAGE followed by Western blot analysis. Production of REP-1 protein was evaluated in cells transduced with all versions of the constructs and compared with untreated controls.

2. As a pilot experiment, three different iPS cell lines plated on a 12 well cell culture plate are transduced with AAV8.hCHM Version 1 and AAV8.hCHM Version 2a (AAV8.V1; AAV8.V2a) at an MOI of 1E6. The iPS cell lines were derived from three CHM affected individuals with unrelated mutations in REP1 gene and were plated in separate plates for this purpose. After 36-48 hours, cells were harvested and lysed and subjected to Western blot analyses compared with untreated cell lysate.

Test Material Administration

3.4.1 Cell Culture

Culturing of iPS cells from CHM patient. In brief, the iPS cells were cultured on Mouse Embryonic Fibroblasts (MEFs, feeders) in iPS cell culture media at 37° C. in an environment supplied with 5% CO₂ and 5% O₂.

3.4.2 Preparation of Cells for Transduction

The day before seeding the cells, 12-well dishes were coated with Matrigel as described in reference NCP.003 (NCP.003: Culturing of iPS cells from CHM patient and controls). Before transduction of iPS cells with respective AAV2 or AAV8 viral vectors, the cells that are cultured on MEFs were seeded on Matrigel without MEFs (feeder free culturing). Cells were seeded at a density of 4.5+E5 to 6+E5 in 1 ml of iPS cell culture media in each well of a 12-well cell culture dish. Seeded cells were incubated at 37° C. in an environment supplied with 5% CO₂, 5% O₂.

3.4.3 Transduction

To infect the iPS cells with viral vectors, cells were grown to approximately 50-60% confluence. (This can take 2-4 days in feeder free conditions). Once 50-60% confluence is achieved, one well of the 12-wells is dissociated and cell

counts were performed to determine the total number of cells per well. Wells of the iPS cells were then infected with AAV vectors listed below at the predetermined MOI (see Table 18 and 19). Before transduction, the old iPS cell culture media from the plates was removed and a fresh 1 ml of iPS cell culture media was added in each well. Predetermined volumes of the virus from the stock were directly added to each well. See Table 18. and Table 19. For the information on total number of cells infected, MOI and the volume of virus used for infection. Cells were then incubated at 37° C. in an environment supplied with 5% CO₂, 5% O₂ for 18-24 hours. After 18-24 h of transduction, cells were observed under microscope for any abnormalities or cell death. At this point, another 1 ml of fresh iPS cell culture media was added to each well containing infected and uninfected cells and were further incubated for additional 18-24 hours at 37° C. in an environment supplied with 5% CO₂, 5% O₂. Cells were observed under the microscope before harvesting to evaluate any cell death or abnormal appearance.

TABLE 18

Infection details and MOIs of next generation AAV2.hCHMV2a, 2B, 3a, 3b Vectors and AAV2.hCHM.V1 vectors in CHM patient-derived iPS cells.

Vector Used	Cell Line	Cell line number	Cell density	Viral stock concentration (vg/ml)	Vector used (μL)	MOI
AAV2.V2a	iPSC	JB 588	6E5	2.16E+12	30	1E5
	iPSC	JB 588	6E5	2.16E+12	90	3E5
AAV2.V2b	iPSC	JB 588	6E5	7.4E+12	8.1	1E5
	iPSC	JB 588	6E5	7.4E+12	24.3	3E5
AAV2.V3a	iPSC	JB 588	6E5	4.82E+12	12.4	1E5
	iPSC	JB 588	6E5	4.82E+12	37.3	3E5
AAV2.V3b	iPSC	JB 588	6E5	5.91E+12	10.2	1E5
	iPSC	JB 588	6E5	5.91E+12	30.5	3E5
AAV2.V1	iPSC	JB 588	6E5	4.47E+12	30.6	1E5
	iPSC	JB 588	6E5	4.47E+12	40.9	3E5

TABLE 19

Infection dose of AAV8.V2a and AAV8.V1 vectors in three iPS cell lines derived from 3 different CHM patients.

Vector Used	Cell Line	Cell line number	Cell density	Viral stock concentration (vg/ml)	Vector used (μL)	MOI
Untreated	iPSC	JB 588	4.5E5			
AAV8.V2a	iPSC	JB 588	4.5E5	1.04E+13	43	1E6
Untreated	iPSC	JB 500	4.5E5			
AAV8.V1	iPSC	JB 500	4.5E5	1.39E+13	32	1E6
Untreated	iPSC	JB 527	4.5E5			
AAV8.V1	iPSC	JB 527	4.5E5	1.39E+13	32	1E6
AAV8.V2a	iPSC	JB 527	4.5E5	1.04E+13	43	1E6

Outcome measurement method-Western blot analysis was performed as described herein.

Results

5.1 Expression of AAV2-hCHM V1, V2a, V2b, V3a, V3b in JB588 iPS cell line: Monoclonal human REP-1-specific antibody, detected one single ~75-80 kDa hREP-1 protein in the transduced JB 588 iPS cells (Data not shown). No band was observed in the case of the untreated control, confirming presence of the disease (data not shown). The intensity of REP-1 protein band at an MOI of 3E5 observed to be stronger in all vectors compared to an MOI of 1E5. Recombinant AAV2. hCHM viral mediated delivery of the hCHM gene, to iPS cells, resulted in a dose-dependent production

of REP-1 protein. Probing of the blots with GAPDH antibody showed a band of equal density in all lysates. GAPDH detected a protein at ~39 kDa. Both REP-1 and GAPDH antibodies detected only specific bands of expected molecular weight. No nonspecific bands were observed in the blots.

Expression of AAV8-hCHM. V1, V2a in iPS cells: Monoclonal human REP-1-specific antibody, detected one single ~75-80 kDa REP-1 protein in the transduced JB527, JB500 and JB588 patient derived iPS cells (Data not shown). No protein band was observed in the case of the untreated control. (Data not shown). Probing of the blots with GAPDH antibody showed a band of equal density in all cell lysates including the cell lysates from untreated cells. Anti-GAPDH antibody detected a specific ~39 kDa protein band. Both REP-1 and GAPDH antibodies detected only specific protein bands at the expected size molecular weight. No detectable nonspecific protein bands were observed in the blot.

CONCLUSIONS

The preliminary results presented in the current report revealed the following observations: Western blot analysis confirmed presence of CHM (lack of REP-1 protein) in each one of the three patient-derived iPSCs (JB588, JB500, JB527). In vitro expression studies demonstrated that infecting iPS cells from CHM patients with AAV2.hCHM. Version 2a, 2b, 3a, 3b and AAV2.hCHM Version1 (a current clinical trial candidate) induced the production of REP-1 protein at all tested MOIs. Infecting iPS cells with AAV8.hCHM. Version 2a and AAV8.hCHMVersion1 at an MOI of 1E6 resulted in production of REP1 protein in all three CHM iPS cell lines. Level of REP1 production was higher in the iPSCs infected with AAV8.hCHM.V2a than with AAV8.hCHM.V1.

Example 7: Comparison of In Vivo Expression of AAV8.Codon Optimized.Human CHM Versus AAV.Native.Human CHM

Gene therapy for a number of retinal diseases depends on efficient transduction of the appropriate target cells, which for choroideremia, are retinal pigment epithelium (RPE) cells and photoreceptor cells. This study report focuses on the comparison of in vivo expression induced by the native CHM sequence based construct, (Version 1) and four next generation transgene cassettes packaged into an AAV8 backbone in wild type mice. Here we evaluated AAV8 serotype for the purpose of improving gene transfer to photoreceptor cells.

Our experiments were designed to answer the following questions: a. How would these vectors compare for in vivo transduction of photoreceptors: In particular, how efficiently would the next generation AAV8. CHM transduce photoreceptors after subretinal injection of the respective test article compared to version.1. b. Dose response: Would the next generation AAV8. CHM and AAV8. CHM-Version1 vectors differ in dose response of gene expression.

Experimental Details:

Plasmids and vectors were as described in Example 4. Mice (Animals): Wild type, CD1 mice were used to test the in vivo expression of CHM as assessed by production of REP-1 protein. CD1 mouse strain is an outbred Swiss mouse

strain which colony we maintain in house. The details of the study are described under CAROT study protocol PCPR02.01.

3.3 Study Design (e.g. Treatment Groups)

3.3.1 Animal Husbandry: Both male and female mice (~3-4 months old) weighing ~20-30 gm were injected with the described test articles. Animals were housed in the University of Pennsylvania's John Morgan University Laboratory Animal Resources (ULAR) facility according to University of Pennsylvania's ULAR regulations. Mice were maintained on a 12-hour light/12-hour dark cycle. Food and water were provided ad libitum. All animals were identified with ear tag numbers.

3.4 Test material administration: The test article formulation provided by the CAROT Vector Core was used for dose administration. The test material was stored at -60 to -80° C. The test material was thawed on ice prior to dosing. For intra-ocular injections, the test article is diluted to the target concentration with phosphate-buffered saline as described in the formulation Table 20. A total of 60 µl of master mix was prepared.

TABLE 20

Dose Formulation table for subretinal injections of test articles.

Identification	Lot #	Vector Conc (Vg/ml)	Volume (µl) For a total of 60 µl	Volume of PBS (µl) for a total of 60 µl	In-jected Conc (vg)	Total vol. inj. (µl)
AAV8.V2a	CT245	1.04E+13	1.92	58.1	5E8	1.5 µ3
			19.2	40.8	5E9	1.5 µ3
AAV8.V2b	CT244	1.11E+13	1.8	58.2	5E8	1.5 µ3
			18	42	5E9	1.5 µ3
AAV8.V3a	CT259	8.67E+12	2.31	57.7	5E8	1.5 µ3
			23.1	36.9	5E9	1.5 µ3
AAV8.V3b	CT255	1.36E+13	1.47	58.53	5E8	1.5 µ3
			14.7	45.3	5E9	1.5 µ3
AAV8.V1	KA808	1.39E+13	1.44	58.6	5E8	1.5 µ3
			14.4	45.6	5E9	1.5 µ3

Preparation of Injection Log Before Subretinal Injections:

An injection log was maintained with the following information before subretinal injection of the test articles:

Cage Number/mouse number
Study Identification
Strain
Date of Birth
Date of injection
Name of investigator/injector
Eye injected into (left or right)
Injection material (vector/serotype)
Dose and Volume
Route of Administration (ROA)

Subretinal injections: Injections were performed by Subretinal Injection by the Surgeon. In brief, animals were anaesthetized before injection. Subretinal injection of the test article was performed using Hamilton 33G syringe. The details of test articles and injections are described in Table 21. From the prepared injection master mix, a volume of 1.5 µl was administered, per injection. One eye per animal was injected with 5E8 vg/eye and the contralateral eye was injected with 5E9 vg/eye.

TABLE 21

Subretinal injection scheme and injection doses								
Gr. No.	Number of Animals	ROA	Test Material Identification	Vector Dose (vg/eye)		Dose Volume (μL/eye)		Dosing Day (PD)
				Right Eye	Left Eye	Right Eye	Left Eye	
1	2	Subretinal	AAV8.V2a	5+E8	5+E9	1.5 μl	1.5 μl	Day 90-120
2	2	Subretinal	AAV8.V2b	5+E8	5+E9	1.5 μl	1.5 μl	Day 90-120
3	2	Subretinal	AAV8.V3a	5+E8	5+E9	1.5 μl	1.5 μl	Day 90-120
4	2	Subretinal	AAV8.V3b	5+E8	5+E9	1.5 μl	1.5 μl	Day 90-120
5	2	Subretinal	AAV8.V1	5+E8	5+E9	1.5 μl	1.5 μl	Day 90-120
6	2	Uninjected	Not Applicable (N/A)	N/A	N/A	N/A	N/A	N/A

Outcome Measurement Methods

Animal Sacrifice: a. After injecting the animals with the test articles, all animals were observed for 48 hours for any post injection related abnormalities. b. 21-35 days of post injection, the animals were observed for ocular abnormalities using ophthalmoscopy. c. 90-12-days post injection, the animals were sacrificed and eye tissues were collected for evaluating the production of exogenous REP-1 protein by SDS-PAGE followed by western blot analysis.

Collection of Eye Tissue: Eye tissue for western blot analysis was collected after removing the lens from the eye using a sharp surgical blade. The eye (without the lens) was collected in freezer tubes that are labelled appropriately.

Western Blot Analysis

Briefly: 1. Preparation of Tissue Lysate

a. Ocular tissue of animals injected with 2 different doses of next generation AAV8. CHM and AAV8.V1 along with the uninjected control animal tissues, were collected after 21-35 days of injection by sacrificing the animals. b. Tissues were then lysed on ice using RIPA buffer with protease inhibitors.

c. Tissue lysates were cleared by centrifuging at 13,000 rpm for 10 min.

2. Quantification and Preparation of Proteins

a. Protein quantification of the cell lysates was carried out using ThermoFisher Micro BCA™ Protein Assay Kit following manufacturer's instructions. b. Protein concentration was determined by taking OD reading at 562 nm. c. To evaluate the in vivo expression of CHM, between 20-40 ug of measured protein was loaded on 4-12% Bis-Tris gels.

3. SDS-PAGE and Western Blot

The protein gels were transferred on to a nitrocellulose membrane, blocked in milk and incubated with the primary antibodies. Anti-human REP-1 2F1 antibody (2F1, 1:1000 dilution) and/or anti-GAPDH antibody (1:1000 dilution); were used as primary antibodies. After washing the blot, HRP conjugated anti-mouse IgG antibody and/or anti-rabbit IgG antibody at a concentration of 1:5000 were used as secondary antibodies. The blots were developed by chemiluminescence using ECL reagents according to the manufacturer's instructions.

4. Controls

a) Loading controls: Anti-GAPDH antibody was used as loading controls to demonstrate equal loading of protein in each well of the gels. Anti-GAPDH antibody detects a protein of ~39 kDa. b) Positive control: AAV2. V2a trans-

duced COS-7 cell lysates were used as positive controls. c) Negative control: Ocular tissues of uninjected animals were used as negative control.

Sample Value Determination

Quantification of Western blot analysis using Image J software. In brief, densitometric evaluations presented in this report are normalized first, to the levels of endogenous expression of GAPDH protein of the corresponding sample. Then the expression levels are normalized again, to the average REP-1 expression level of uninjected control.

The details of densitometric evaluations and fold change calculations to represent the expression of REP-1 protein are presented as Table 22 and 23.

The description in brief:

1. In table 22 and 23, Column 2 shows the raw values of REP-1 protein and column 3 shows the raw values of GAPDH protein.

2. GAPDH value of each samples was first normalized to the GAPDH values of animal-1 of AAV8.V1 and are shown in Table 22 in 4th column.

3. The values of each sample were also normalized to the GAPDH values of animal-2 of AAV8.V1 and are shown in Table 22 in 5th column.

4. The REP-1 values (Column 2) are then normalized to either to the GAPDH normalized to animal 1 (column 4) or to the GAPDH previously normalized to animal 2 (column 5). These are represented in column 6 and 7 respectively.

5. The normalized REP-1 values are then converted to fold change.

6. The respective REP-1 values are normalized to expression of REP-1 either in animal 1 or animal 2 of the AAV8.V1 injected group and are expressed as fold change (column 8 and 9)

7. Column 10 represents the average fold change in REP-1 protein expression.

Results

Comparison of the CHM expression using the native CHM AAV8.V1 versus the codon optimized CHM vectors: AAV8. V2a, V2b, V3a and V3b. Wild type CD1 mice were injected with two different doses of the each AAV8 vector: a high dose of 5E9 vg/eye and a low dose of 5E8 vg/eye. Following results describe the levels of REP1 protein after injection with high and low doses of AAV8.V1, AAV8.V2a and AAV8.V3a.

Comparison of the expression of AAV8. V1 versus AAV8. V2a and AAV8.V3a (vectors with stuffer) in animals

injected with high dose (5E9 vg/eye) of viral vector. Western blot analysis with human anti REP-1 antibody detected a ~75-80 kDa hREP-1 protein band in both (low and high dose injected) ocular tissues of each animal treated with either the next generation AAV8. V2a or V3a or the original AAV8. Version1. A very faint (minimal) band is observed in the case of the uninjected control mice, both. A band of increased intensity was observed in tissues that were transduced with next generation vectors (AAV8V2a and AAV8.V3a) compared to the tissues transduced with Version.1. Anti-GAPDH antibodies showed a ~39 kDa band of equal density in all lanes of the western blot including the uninjected controls. Pre-stained protein marker is used to compare the molecular weights of protein of interest. Densitometric evaluation (quantification of the expression level) of the blots using ImageJ software demonstrated that production of REP-1 was increased in animals injected with one of the next generation AAV8. high and low doses constructs (V2a or V3a). (See Table 22 for values.)

Comparison of the expression of AAV8.V1 versus AAV8.V2a and AAV8.V3a in animals injected with low dose (5E8 vg/eye) of viral vector

Human anti REP-1 antibody, the Western blot analysis of the ocular tissues of animals injected with next generation AAV8.V2a, V3a and AAV8. Version1 at a dose of 5E8 detected a ~75-80 kDa hREP-1 protein band in tissues of injected mice. A faint (minimal) band of REP-1 was observed in ocular tissue lysates of the uninjected control mice, both. A band of increased intensity was observed in tissue lysates that are transduced with next generation vectors compared to the lysates that are transduced with Version1. Anti-GAPDH antibody detected an equal intensity protein band at ~39 kDa in all cell lysates. This data demonstrates that delivery of next generation V2a CHM through AAV8 results in robust levels of REP-1 protein in comparison with levels produced after injection of AAV8.V3a or AAV8.V1.

TABLE 22

Quantified REP-1 protein production results for treatment with high dose (5E9vg) AAV8 V1, V2a and V2b									
	GAPDH	REP-1	GAPDH Normalized to GAPDH of Version 1 Animal 1	GAPDH Normalized to GAPDH of Version 2 Animal 2	REP1 Normalized to respective GAPDH (normalized to animal 1)	REP1 Normalized to respective GAPDH (normalized to animal 2)	Fold change in REP-1 (normalized to animal 1)	Fold change in REP-1 (normalized to animal 2)	Average fold change in REP-1 expression
AAV8.V2a	12758.589	10058.359	1.055	1.190	9533.571	8450.521	2.335	3.742	3.038
AAV8.V2a	11885.518	13247.510	0.982	1.108	13489.242	11956.812	3.303	5.294	4.299
AAV8.V3a	12139.418	15542.551	1.003	1.132	15495.152	13734.843	3.794	6.081	4.938
AAV8V3a	11113.640	7274.388	0.918	1.036	7921.575	7021.653	1.940	3.109	2.524
AAV8.V1	12102.397	4083.761	1.000	1.128	4083.761	3619.830	1.000		
AAV8.V1	10727.518	2258.477	0.886	1.000	2547.932	2258.477		1.000	
REP-1 expression values untreated animal was negligible see the values below:									
Name					REP-1		GAPDH		
Uninjected			Animal-1		651.678		16633.539		
			Animal-2		253.778		13025.397		

TABLE 23

Quantified REP-1 protein production results for treatment with low dose (5E8 vg) AAV8 V1, V2a and V2b									
	GAPDH	REP-1	GAPDH Normalized to GAPDH of Version 1 Animal 1	GAPDH Normalized to GAPDH of Version 2 Animal 2	REP1 Normalized to respective GAPDH (normalized to animal 1)	REP1 Normalized to respective GAPDH (normalized to animal 2)	Fold change in REP-1 (normalized to animal 1)	Fold change in REP-1 (normalized to animal 2)	Average fold change in REP-1 expression
AAV8.V2a	11815.489	11194.037	0.809	0.934	13844.203	11984.286	11.123	29.133	20.128
AAV8.V2a	12889.418	7162.924	0.882	1.019	8120.634	7029.658	6.524	17.089	11.806
AAV8.V3a	13088.418	1516.506	0.896	1.035	1693.128	1465.663	1.360	3.563	2.462
AAV8V3a	9201.075	593.192	0.630	0.727	942.084	815.519	0.757	1.982	1.370
AAV8.V1	14612.782	1244.678	1.000	1.155	1244.678	1077.460	1.000		
AAV8.V1	12649.610	411.364	0.866	1.000	475.206	411.364		1.000	
REP-1 expression values untreated animal was negligible see the values below:									
Name					REP-1		GAPDH		
Uninjected			Animal-1		694.263		15930.368		
			Animal-2		254.364		13896.246		

Densitometric evaluation (quantification of the expression level) of the blots using ImageJ software further demonstrate an increased production of REP-1 in animals injected with next generation AAV8.CHM constructs (especially V2a) compared with Version 1. See Table 23 for values.

Expression of AAV8. V2b in CD1 Mice

This current study and the evaluation of lambda stuffer's effect on AAV vector production by qPCR titer analysis were carried out simultaneously. We performed all the animal injections for the in vivo expression study as described in the study protocol PCPR.02 and all samples were harvested. After the qPCR study on the lambda stuffer element was concluded (described above), we decided to carry out the Western blot experiments only to test the expression of AAV vectors without the stuffer such as AAV8.2b and AAV8.3b and exclude them from further analysis (such as comparison with Version 1).

Human anti-REP-1 antibody detected a protein of ~75-80 kDa in ocular tissues of CD-1 mice injected with AAV8.2b at 5E9 (High dose) vector genome copies (FIG. 12A). Animals injected with AAV8.2b at 5E8 (Low dose) showed a very faint protein band at ~75-80 kDa (FIG. 12A). Lysates of ocular tissues from uninjected control animals did not show the presence of REP-1 protein. Anti-GAPDH antibody detected a protein of ~39 kDa in all ocular tissue lysates including the uninjected controls. This data may establish the minimal dose for AAV8.2b.

Expression of AAV8.V3b in CD1 Mice

We performed a Western blot analysis on ocular tissues of AAV8.3b injected CD1 mice (2 mice/group) with anti-REP-1 antibody, which revealed the presence of a protein of ~75-80 kDa in one eye injected with low dose and in both eyes injected with high dose of AAV8.3b. In the ocular tissues of uninjected mice there was no REP-1 expression detected (FIG. 12B). The level of REP-1 produced was dose dependent in animals injected with AAV8.3b. Injection with high dose of AAV8.3b (5E9 vector genomes) induced a higher amount of REP-1 compared with the low dose injected eyes (5E8 vector genomes). Anti-GAPDH antibody detected a protein of ~39 kDa in ocular tissue lysates of all injected and uninjected animals.

These results revealed the following observations:

1) The next generation vectors AAV8. Version2a, 2b, 3a and 3b are able to transduce ocular tissues efficiently. 2) Expression of the transgene (codon optimized CHM) was detectable for all of the next generation vectors. 3) Expression of transgene (codon optimized CHM) is dose dependent. 4) AAV8. Version2a and AAV8. Version.2b induced an

increased production of REP-1 protein compared to AAV8. Version 1 in ocular tissues of CD-1 mice. 5) There is variation in the exact level of production of the transgenic protein between eyes injected with the same dose reflecting the variability in the surgical delivery procedure. However, differences in levels are large between the low (5E8) and high (5E9) doses. 6) AAV8.CHM.V2a and AAV8. V3a result in much higher levels of REP-1 protein production than AAV8.V1 after in vivo administration of high dose (5E9 vg) vector subretinally in mice.

All publications cited in this specification are incorporated herein by reference. Similarly, the SEQ ID NOs which are referenced herein and which appear in the appended Sequence Listing are incorporated by reference. While the invention has been described with reference to particular embodiments, it will be appreciated that modifications can be made without departing from the spirit of the invention. Such modifications are intended to fall within the scope of the appended claims.

SEQ ID NO (SID)	Sequence
1	codon optimized REP-1
2	AA from SID 1
3	native REP-1
4	AA from SID 3
5	codon optimized REP-1 with modifications
6	AA from SID 5
7	p584
8	p584-REP-1 optimized
9	codon optimized CNGA3
10	AA from SID 9
11	codon optimized CNGA variant X1
12	AA from SID 11
13	native CNGA3
14	AA from SID 13
15	native CNGA3 variant X1
16	codon optimized CNGA3 with modifications
17	codon optimized CNGA3 variant with modifications
18	native CNGA3 with modifications
19	native CNGB3
20	AA from SID 19
21	CNGB3 with modified ORF
22	AA from SID21
23	CNGB3 with modified ORF with modified ends
24	AA from SID 23
25	Plasmid Version 2a
26	Plasmid version 2b
27	Plasmid version3a
28	Plasmid version 3b
29	Plasmid version 1

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 <222> LOCATION: (1)..(1962)

<400> SEQUENCE: 1

atg gct gat acc ctg ccc tct gaa ttc gac gtg att gtg att gga acc
 Met Ala Asp Thr Leu Pro Ser Glu Phe Asp Val Ile Val Ile Gly Thr
 1 5 10 15

48

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gga ctc cct gaa tcg atc atc gcc gcg gcc tgt tcc cgg tcc ggt cgg	96
Gly Leu Pro Glu Ser Ile Ile Ala Ala Ala Cys Ser Arg Ser Gly Arg	
20 25 30	
cgc gtg ctg cac gtc gat tcg aga agc tac tac gga ggg aat tgg gcc	144
Arg Val Leu His Val Asp Ser Arg Ser Tyr Tyr Gly Gly Asn Trp Ala	
35 40 45	
tca ttc tcc ttc tcc gga ctg ctc tcc tgg ctg aag gag tat cag gag	192
Ser Phe Ser Phe Ser Gly Leu Leu Ser Trp Leu Lys Glu Tyr Gln Glu	
50 55 60	
aac tcc gac att gtc tcc gac tca cct gtg tgg cag gac cag atc ctg	240
Asn Ser Asp Ile Val Ser Asp Ser Pro Val Trp Gln Asp Gln Ile Leu	
65 70 75 80	
gaa aac gag gaa gca ata gcc ctg agc cgg aag gac aag acc atc cag	288
Glu Asn Glu Glu Ala Ile Ala Leu Ser Arg Lys Asp Lys Thr Ile Gln	
85 90 95	
cac gtg gag gtg ttc tgt tat gcc tcc caa gac ctc cat gag gac gtg	336
His Val Glu Val Phe Cys Tyr Ala Ser Gln Asp Leu His Glu Asp Val	
100 105 110	
gaa gag gct gga gcg ttg cag aag aat cat gcc ctc gtg acc tcc gct	384
Glu Glu Ala Gly Ala Leu Gln Lys Asn His Ala Leu Val Thr Ser Ala	
115 120 125	
aac tcc acc gag gca gcc gac agc gcc ttc ctg ccg acc gag gat gaa	432
Asn Ser Thr Glu Ala Ala Asp Ser Ala Phe Leu Pro Thr Glu Asp Glu	
130 135 140	
tcc ctg tca act atg tcg tgc gaa atg ctg acc gaa cag act ccg agc	480
Ser Leu Ser Thr Met Ser Cys Glu Met Leu Thr Glu Gln Thr Pro Ser	
145 150 155 160	
tcc gac ccc gaa aac gcc ctg gaa gtg aac gga gcg gaa gtg acc ggc	528
Ser Asp Pro Glu Asn Ala Leu Glu Val Asn Gly Ala Glu Val Thr Gly	
165 170 175	
gaa aag gag aac cat tgc gac gac aag act tgt gtc cca tcc act tcc	576
Glu Lys Glu Asn His Cys Asp Asp Lys Thr Cys Val Pro Ser Thr Ser	
180 185 190	
gcg gag gac atg tcc gag aat gtg cct atc gcc gag gac acc acc gaa	624
Ala Glu Asp Met Ser Glu Asn Val Pro Ile Ala Glu Asp Thr Thr Glu	
195 200 205	
cag ccc aag aag aac aga atc acg tac agc cag atc atc aag gag ggg	672
Gln Pro Lys Lys Asn Arg Ile Thr Tyr Ser Gln Ile Ile Lys Glu Gly	
210 215 220	
cgg agg ttt aac atc gat ctg gtg tcg aag ctg ctg tac agc cgc ggt	720
Arg Arg Phe Asn Ile Asp Leu Val Ser Lys Leu Leu Tyr Ser Arg Gly	
225 230 235 240	
ctg ctg atc gat ctg ctc att aag tcg aac gtg tcg aga tac gcc gag	768
Leu Leu Ile Asp Leu Leu Ile Lys Ser Asn Val Ser Arg Tyr Ala Glu	
245 250 255	
ttc aag aac atc aca agg att ctc gcc ttc cgg gaa gga aga gtg gaa	816
Phe Lys Asn Ile Thr Arg Ile Leu Ala Phe Arg Glu Gly Arg Val Glu	
260 265 270	
caa gtg ccg tgc tcc cgg gcc gac gtg ttc aac tca aag caa ctt acc	864
Gln Val Pro Cys Ser Arg Ala Asp Val Phe Asn Ser Lys Gln Leu Thr	
275 280 285	
atg gtg gaa aag cgc atg ctg atg aaa ttc ctg acc ttc tgc atg gag	912
Met Val Glu Lys Arg Met Leu Met Lys Phe Leu Thr Phe Cys Met Glu	
290 295 300	
tac gaa aag tac cct gat gag tac aag ggt tac gaa gaa att act ttc	960
Tyr Glu Lys Tyr Pro Asp Glu Tyr Lys Gly Tyr Glu Glu Ile Thr Phe	
305 310 315 320	
tac gag tac ctc aag acc cag aag ctg acc ccg aat ctg cag tac att	1008
Tyr Glu Tyr Leu Lys Thr Gln Lys Leu Thr Pro Asn Leu Gln Tyr Ile	

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325					330					335					
gtg atg cac tca atc gca atg acc tcc gaa acc gcc tcc tcg acc atc	1056														
Val Met His Ser Ile Ala Met Thr Ser Glu Thr Ala Ser Ser Thr Ile															
340					345					350					
gac ggg ctc aag gcc acc aag aac ttc ctg cac tgt ttg ggg cgc tac	1104														
Asp Gly Leu Lys Ala Thr Lys Asn Phe Leu His Cys Leu Gly Arg Tyr															
355					360					365					
ggc aac act ccg ttc ctc ttc ccg ctg tac ggc cag gga gag ctg cct	1152														
Gly Asn Thr Pro Phe Leu Phe Pro Leu Tyr Gly Gln Gly Glu Leu Pro															
370					375					380					
cag tgt ttc tgc ccg atg tgc gcc gtg ttc ggc gga atc tac tgt ctc	1200														
Gln Cys Phe Cys Arg Met Cys Ala Val Phe Gly Gly Ile Tyr Cys Leu															
385					390					395					400
cgc cac tcg gtc cag tgc ctg gtg gtg gac aag gaa tcc agg aag tgc	1248														
Arg His Ser Val Gln Cys Leu Val Val Asp Lys Glu Ser Arg Lys Cys															
405					410					415					
aaa gcc att att gac cag ttc gga caa ccg atc att tcc gag cac ttt	1296														
Lys Ala Ile Ile Asp Gln Phe Gly Gln Arg Ile Ile Ser Glu His Phe															
420					425					430					
ctt gtg gag gac tca tac ttc ccg gag aac atg tgc tct ccg gtc cag	1344														
Leu Val Glu Asp Ser Tyr Phe Pro Glu Asn Met Cys Ser Arg Val Gln															
435					440					445					
tat cga cag att tcc agg gcg gtg ctc att act gac ccg agc gtc ctc	1392														
Tyr Arg Gln Ile Ser Arg Ala Val Leu Ile Thr Asp Arg Ser Val Leu															
450					455					460					
aag acc gat agc gac cag cag atc tcc atc ctg acc gtg ccg gcg gaa	1440														
Lys Thr Asp Ser Asp Gln Gln Ile Ser Ile Leu Thr Val Pro Ala Glu															
465					470					475					480
gaa ccc ggc act ttt gcc gtg cgc gtg atc gag ctt tgc tca tcc acc	1488														
Glu Pro Gly Thr Phe Ala Val Arg Val Ile Glu Leu Cys Ser Ser Thr															
485					490					495					
atg act tgc atg aaa ggc act tac ctg gtg cac ctg acg tgc acc tca	1536														
Met Thr Cys Met Lys Gly Thr Tyr Leu Val His Leu Thr Cys Thr Ser															
500					505					510					
tcg aaa acc gct aga gag gac ctg gaa tcc gtc gtc caa aag ctg ttc	1584														
Ser Lys Thr Ala Arg Glu Asp Leu Glu Ser Val Val Gln Lys Leu Phe															
515					520					525					
gtg cct tac acc gag atg gaa att gaa aac gaa caa gtg gag aag ccc	1632														
Val Pro Tyr Thr Glu Met Glu Ile Glu Asn Glu Gln Val Glu Lys Pro															
530					535					540					
cgc atc ctt tgg gcc ctg tac ttt aac atg cgc gat tcc tcc gat atc	1680														
Arg Ile Leu Trp Ala Leu Tyr Phe Asn Met Arg Asp Ser Ser Asp Ile															
545					550					555					560
tcg ccg tcc tgc tat aac gac ttg cct tcg aac gtc tac gtc tgc tcc	1728														
Ser Arg Ser Cys Tyr Asn Asp Leu Pro Ser Asn Val Tyr Val Cys Ser															
565					570					575					
ggg cca gac tgc ggt ctt ggc aac gac aat gcc gtg aag cag gcg gaa	1776														
Gly Pro Asp Cys Gly Leu Gly Asn Asp Asn Ala Val Lys Gln Ala Glu															
580					585					590					
aca ctg ttc caa gag atc tgc cct aac gag gat ttt tgc ccg ccc ccc	1824														
Thr Leu Phe Gln Glu Ile Cys Pro Asn Glu Asp Phe Cys Pro Pro Pro															
595					600					605					
cca aac ccc gag gat atc atc ttg gac gga gac agc ctg cag cca gaa	1872														
Pro Asn Pro Glu Asp Ile Ile Leu Asp Gly Asp Ser Leu Gln Pro Glu															
610					615					620					
gca tcc gag tcc agc gcc atc ccg gag gcc aac agc gaa acc ttc aag	1920														
Ala Ser Glu Ser Ser Ala Ile Pro Glu Ala Asn Ser Glu Thr Phe Lys															
625					630					635					640
gag agc act aac ctg ggc aac ctg gaa gag tcc agc gaa tga															
1962															

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Glu Ser Thr Asn Leu Gly Asn Leu Glu Glu Ser Ser Glu
645 650

<210> SEQ ID NO 2
 <211> LENGTH: 653
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 2

Met Ala Asp Thr Leu Pro Ser Glu Phe Asp Val Ile Val Ile Gly Thr
1 5 10 15
 Gly Leu Pro Glu Ser Ile Ile Ala Ala Ala Cys Ser Arg Ser Gly Arg
20 25 30
 Arg Val Leu His Val Asp Ser Arg Ser Tyr Tyr Gly Gly Asn Trp Ala
35 40 45
 Ser Phe Ser Phe Ser Gly Leu Leu Ser Trp Leu Lys Glu Tyr Gln Glu
50 55 60
 Asn Ser Asp Ile Val Ser Asp Ser Pro Val Trp Gln Asp Gln Ile Leu
65 70 75 80
 Glu Asn Glu Glu Ala Ile Ala Leu Ser Arg Lys Asp Lys Thr Ile Gln
85 90 95
 His Val Glu Val Phe Cys Tyr Ala Ser Gln Asp Leu His Glu Asp Val
100 105 110
 Glu Glu Ala Gly Ala Leu Gln Lys Asn His Ala Leu Val Thr Ser Ala
115 120 125
 Asn Ser Thr Glu Ala Ala Asp Ser Ala Phe Leu Pro Thr Glu Asp Glu
130 135 140
 Ser Leu Ser Thr Met Ser Cys Glu Met Leu Thr Glu Gln Thr Pro Ser
145 150 155 160
 Ser Asp Pro Glu Asn Ala Leu Glu Val Asn Gly Ala Glu Val Thr Gly
165 170 175
 Glu Lys Glu Asn His Cys Asp Asp Lys Thr Cys Val Pro Ser Thr Ser
180 185 190
 Ala Glu Asp Met Ser Glu Asn Val Pro Ile Ala Glu Asp Thr Thr Glu
195 200 205
 Gln Pro Lys Lys Asn Arg Ile Thr Tyr Ser Gln Ile Ile Lys Glu Gly
210 215 220
 Arg Arg Phe Asn Ile Asp Leu Val Ser Lys Leu Leu Tyr Ser Arg Gly
225 230 235 240
 Leu Leu Ile Asp Leu Leu Ile Lys Ser Asn Val Ser Arg Tyr Ala Glu
245 250 255
 Phe Lys Asn Ile Thr Arg Ile Leu Ala Phe Arg Glu Gly Arg Val Glu
260 265 270
 Gln Val Pro Cys Ser Arg Ala Asp Val Phe Asn Ser Lys Gln Leu Thr
275 280 285
 Met Val Glu Lys Arg Met Leu Met Lys Phe Leu Thr Phe Cys Met Glu
290 295 300
 Tyr Glu Lys Tyr Pro Asp Glu Tyr Lys Gly Tyr Glu Glu Ile Thr Phe
305 310 315 320
 Tyr Glu Tyr Leu Lys Thr Gln Lys Leu Thr Pro Asn Leu Gln Tyr Ile
325 330 335
 Val Met His Ser Ile Ala Met Thr Ser Glu Thr Ala Ser Ser Thr Ile
340 345 350

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agt ttt agc ttt tca gga cta ttg tcc tgg cta aag gaa tac cag gaa Ser Phe Ser Phe Ser Gly Leu Leu Ser Trp Leu Lys Glu Tyr Gln Glu 50 55 60	192
aac agt gac att gta agt gac agt cca gtg tgg caa gac cag atc ctt Asn Ser Asp Ile Val Ser Asp Ser Pro Val Trp Gln Asp Gln Ile Leu 65 70 75 80	240
gaa aat gaa gaa gcc att gct ctt agc agg aag gac aaa act att caa Glu Asn Glu Glu Ala Ile Ala Leu Ser Arg Lys Asp Lys Thr Ile Gln 85 90 95	288
cat gtg gaa gta ttt tgt tat gcc agt cag gat ttg cat gaa gat gtc His Val Glu Val Phe Cys Tyr Ala Ser Gln Asp Leu His Glu Asp Val 100 105 110	336
gaa gaa gct ggt gca ctg cag aaa aat cat gct ctt gtg aca tct gca Glu Glu Ala Gly Ala Leu Gln Lys Asn His Ala Leu Val Thr Ser Ala 115 120 125	384
aac tcc aca gaa gct gca gat tct gcc ttc ctg cct acg gag gat gag Asn Ser Thr Glu Ala Ala Asp Ser Ala Phe Leu Pro Thr Glu Asp Glu 130 135 140	432
tca tta agc act atg agc tgt gaa atg ctc aca gaa caa act cca agc Ser Leu Ser Thr Met Ser Cys Glu Met Leu Thr Glu Gln Thr Pro Ser 145 150 155 160	480
agc gat cca gag aat gcg cta gaa gta aat ggt gct gaa gtg aca ggg Ser Asp Pro Glu Asn Ala Leu Glu Val Asn Gly Ala Glu Val Thr Gly 165 170 175	528
gaa aaa gaa aac cat tgt gat gat aaa act tgt gtg cca tca act tca Glu Lys Glu Asn His Cys Asp Asp Lys Thr Cys Val Pro Ser Thr Ser 180 185 190	576
gca gaa gac atg agt gaa aat gtg cct ata gca gaa gat acc aca gag Ala Glu Asp Met Ser Glu Asn Val Pro Ile Ala Glu Asp Thr Thr Glu 195 200 205	624
caa cca aag aaa aac aga att act tac tca caa att att aaa gaa ggc Gln Pro Lys Lys Asn Arg Ile Thr Tyr Ser Gln Ile Ile Lys Glu Gly 210 215 220	672
agg aga ttt aat att gat tta gta tca aag ctg ctg tat tct cga gga Arg Arg Phe Asn Ile Asp Leu Val Ser Lys Leu Leu Tyr Ser Arg Gly 225 230 235 240	720
tta cta att gat ctt cta atc aaa tct aat gtt agt cga tat gca gag Leu Leu Ile Asp Leu Leu Ile Lys Ser Asn Val Ser Arg Tyr Ala Glu 245 250 255	768
ttt aaa aat att acc agg att ctt gca ttt cga gaa gga cga gtg gaa Phe Lys Asn Ile Thr Arg Ile Leu Ala Phe Arg Glu Gly Arg Val Glu 260 265 270	816
cag gtt ccg tgt tcc aga gca gat gtc ttt aat agc aaa caa ctt act Gln Val Pro Cys Ser Arg Ala Asp Val Phe Asn Ser Lys Gln Leu Thr 275 280 285	864
atg gta gaa aag cga atg cta atg aaa ttt ctt aca ttt tgt atg gaa Met Val Glu Lys Arg Met Leu Met Lys Phe Leu Thr Phe Cys Met Glu 290 295 300	912
tat gag aaa tat cct gat gaa tat aaa gga tat gaa gag atc aca ttt Tyr Glu Lys Tyr Pro Asp Glu Tyr Lys Gly Tyr Glu Glu Ile Thr Phe 305 310 315 320	960
tat gaa tat tta aag act caa aaa tta acc ccc aac ctc caa tat att Tyr Glu Tyr Leu Lys Thr Gln Lys Leu Thr Pro Asn Leu Gln Tyr Ile 325 330 335	1008
gtc atg cat tca att gca atg aca tca gag aca gcc agc agc acc ata Val Met His Ser Ile Ala Met Thr Ser Glu Thr Ala Ser Ser Thr Ile 340 345 350	1056
gat ggt ctc aaa gct acc aaa aac ttt ctt cac tgt ctt ggg cgg tat Asp Gly Leu Lys Ala Thr Lys Asn Phe Leu His Cys Leu Gly Arg Tyr	1104

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355	360	365	
ggc aac act cca ttt ttg ttt cct tta tat ggc caa gga gaa ctc ccc Gly Asn Thr Pro Phe Leu Phe Pro Leu Tyr Gly Gln Gly Glu Leu Pro 370 375 380			1152
cag tgt ttc tgc agg atg tgt gct gtg ttt ggt gga att tat tgt ctt Gln Cys Phe Cys Arg Met Cys Ala Val Phe Gly Gly Ile Tyr Cys Leu 385 390 395 400			1200
cgc cat tca gta cag tgc ctt gta gtg gac aaa gaa tcc aga aaa tgt Arg His Ser Val Gln Cys Leu Val Val Asp Lys Glu Ser Arg Lys Cys 405 410 415			1248
aaa gca att ata gat cag ttt ggt cag aga ata atc tct gag cat ttc Lys Ala Ile Ile Asp Gln Phe Gly Gln Arg Ile Ile Ser Glu His Phe 420 425 430			1296
ctc gtg gag gac agt tac ttt cct gag aac atg tgc tca cgt gtg caa Leu Val Glu Asp Ser Tyr Phe Pro Glu Asn Met Cys Ser Arg Val Gln 435 440 445			1344
tac agg cag atc tcc agg gca gtg ctg att aca gat aga tct gtc cta Tyr Arg Gln Ile Ser Arg Ala Val Leu Ile Thr Asp Arg Ser Val Leu 450 455 460			1392
aaa aca gat tca gat caa cag att tcc att ttg aca gtg cca gca gag Lys Thr Asp Ser Asp Gln Gln Ile Ser Ile Leu Thr Val Pro Ala Glu 465 470 475 480			1440
gaa cca gga act ttt gct gtt cgg gtc att gag tta tgt tct tca acg Glu Pro Gly Thr Phe Ala Val Arg Val Ile Glu Leu Cys Ser Ser Thr 485 490 495			1488
atg aca tgc atg aaa ggc acc tat ttg gtt cat ttg act tgc aca tct Met Thr Cys Met Lys Gly Thr Tyr Leu Val His Leu Thr Cys Thr Ser 500 505 510			1536
tct aaa aca gca aga gaa gat tta gaa tca gtt gtg cag aaa ttg ttt Ser Lys Thr Ala Arg Glu Asp Leu Glu Ser Val Val Gln Lys Leu Phe 515 520 525			1584
gtt cca tat act gaa atg gag ata gaa aat gaa caa gta gaa aag cca Val Pro Tyr Thr Glu Met Glu Ile Glu Asn Glu Gln Val Glu Lys Pro 530 535 540			1632
aga att ctg tgg gct ctt tac ttc aat atg aga gat tgc tca gac atc Arg Ile Leu Trp Ala Leu Tyr Phe Asn Met Arg Asp Ser Ser Asp Ile 545 550 555 560			1680
agc agg agc tgt tat aat gat tta cca tcc aac gtt tat gtc tgc tct Ser Arg Ser Cys Tyr Asn Asp Leu Pro Ser Asn Val Tyr Val Cys Ser 565 570 575			1728
ggc cca gat tgt ggt tta gga aat gat aat gca gtc aaa cag gct gaa Gly Pro Asp Cys Gly Leu Gly Asn Asp Asn Ala Val Lys Gln Ala Glu 580 585 590			1776
aca ctt ttc cag gaa atc tgc ccc aat gaa gat ttc tgt ccc cct cca Thr Leu Phe Gln Glu Ile Cys Pro Asn Glu Asp Phe Cys Pro Pro Pro 595 600 605			1824
cca aat cct gaa gac att atc ctt gat gga gac agt tta cag cca gag Pro Asn Pro Glu Asp Ile Ile Leu Asp Gly Asp Ser Leu Gln Pro Glu 610 615 620			1872
gct tca gaa tcc agt gcc ata cca gag gct aac tgc gag act ttc aag Ala Ser Glu Ser Ser Ala Ile Pro Glu Ala Asn Ser Glu Thr Phe Lys 625 630 635 640			1920
gaa agc aca aac ctt gga aac cta gag gag tcc tct gaa taa Glu Ser Thr Asn Leu Gly Asn Leu Glu Glu Ser Ser Glu 645 650			1962

<210> SEQ ID NO 4

<211> LENGTH: 653

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

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<400> SEQUENCE: 4

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Met Ala Asp Thr Leu Pro Ser Glu Phe Asp Val Ile Val Ile Gly Thr
1           5           10          15

Gly Leu Pro Glu Ser Ile Ile Ala Ala Ala Cys Ser Arg Ser Gly Arg
20          25          30

Arg Val Leu His Val Asp Ser Arg Ser Tyr Tyr Gly Gly Asn Trp Ala
35          40          45

Ser Phe Ser Phe Ser Gly Leu Leu Ser Trp Leu Lys Glu Tyr Gln Glu
50          55          60

Asn Ser Asp Ile Val Ser Asp Ser Pro Val Trp Gln Asp Gln Ile Leu
65          70          75          80

Glu Asn Glu Glu Ala Ile Ala Leu Ser Arg Lys Asp Lys Thr Ile Gln
85          90          95

His Val Glu Val Phe Cys Tyr Ala Ser Gln Asp Leu His Glu Asp Val
100         105         110

Glu Glu Ala Gly Ala Leu Gln Lys Asn His Ala Leu Val Thr Ser Ala
115         120         125

Asn Ser Thr Glu Ala Ala Asp Ser Ala Phe Leu Pro Thr Glu Asp Glu
130         135         140

Ser Leu Ser Thr Met Ser Cys Glu Met Leu Thr Glu Gln Thr Pro Ser
145         150         155         160

Ser Asp Pro Glu Asn Ala Leu Glu Val Asn Gly Ala Glu Val Thr Gly
165         170         175

Glu Lys Glu Asn His Cys Asp Asp Lys Thr Cys Val Pro Ser Thr Ser
180         185         190

Ala Glu Asp Met Ser Glu Asn Val Pro Ile Ala Glu Asp Thr Thr Glu
195         200         205

Gln Pro Lys Lys Asn Arg Ile Thr Tyr Ser Gln Ile Ile Lys Glu Gly
210         215         220

Arg Arg Phe Asn Ile Asp Leu Val Ser Lys Leu Leu Tyr Ser Arg Gly
225         230         235         240

Leu Leu Ile Asp Leu Leu Ile Lys Ser Asn Val Ser Arg Tyr Ala Glu
245         250         255

Phe Lys Asn Ile Thr Arg Ile Leu Ala Phe Arg Glu Gly Arg Val Glu
260         265         270

Gln Val Pro Cys Ser Arg Ala Asp Val Phe Asn Ser Lys Gln Leu Thr
275         280         285

Met Val Glu Lys Arg Met Leu Met Lys Phe Leu Thr Phe Cys Met Glu
290         295         300

Tyr Glu Lys Tyr Pro Asp Glu Tyr Lys Gly Tyr Glu Glu Ile Thr Phe
305         310         315         320

Tyr Glu Tyr Leu Lys Thr Gln Lys Leu Thr Pro Asn Leu Gln Tyr Ile
325         330         335

Val Met His Ser Ile Ala Met Thr Ser Glu Thr Ala Ser Ser Thr Ile
340         345         350

Asp Gly Leu Lys Ala Thr Lys Asn Phe Leu His Cys Leu Gly Arg Tyr
355         360         365

Gly Asn Thr Pro Phe Leu Phe Pro Leu Tyr Gly Gln Gly Glu Leu Pro
370         375         380

Gln Cys Phe Cys Arg Met Cys Ala Val Phe Gly Gly Ile Tyr Cys Leu
385         390         395         400

Arg His Ser Val Gln Cys Leu Val Val Asp Lys Glu Ser Arg Lys Cys

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405	410	415
Lys Ala Ile Ile Asp Gln Phe Gly Gln Arg Ile Ile Ser Glu His Phe 420 425 430		
Leu Val Glu Asp Ser Tyr Phe Pro Glu Asn Met Cys Ser Arg Val Gln 435 440 445		
Tyr Arg Gln Ile Ser Arg Ala Val Leu Ile Thr Asp Arg Ser Val Leu 450 455 460		
Lys Thr Asp Ser Asp Gln Gln Ile Ser Ile Leu Thr Val Pro Ala Glu 465 470 475 480		
Glu Pro Gly Thr Phe Ala Val Arg Val Ile Glu Leu Cys Ser Ser Thr 485 490 495		
Met Thr Cys Met Lys Gly Thr Tyr Leu Val His Leu Thr Cys Thr Ser 500 505 510		
Ser Lys Thr Ala Arg Glu Asp Leu Glu Ser Val Val Gln Lys Leu Phe 515 520 525		
Val Pro Tyr Thr Glu Met Glu Ile Glu Asn Glu Gln Val Glu Lys Pro 530 535 540		
Arg Ile Leu Trp Ala Leu Tyr Phe Asn Met Arg Asp Ser Ser Asp Ile 545 550 555 560		
Ser Arg Ser Cys Tyr Asn Asp Leu Pro Ser Asn Val Tyr Val Cys Ser 565 570 575		
Gly Pro Asp Cys Gly Leu Gly Asn Asp Asn Ala Val Lys Gln Ala Glu 580 585 590		
Thr Leu Phe Gln Glu Ile Cys Pro Asn Glu Asp Phe Cys Pro Pro Pro 595 600 605		
Pro Asn Pro Glu Asp Ile Ile Leu Asp Gly Asp Ser Leu Gln Pro Glu 610 615 620		
Ala Ser Glu Ser Ser Ala Ile Pro Glu Ala Asn Ser Glu Thr Phe Lys 625 630 635 640		
Glu Ser Thr Asn Leu Gly Asn Leu Glu Glu Ser Ser Glu 645 650		

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<210> SEQ ID NO 5
<211> LENGTH: 1985
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: constructed plasmid
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (1)..(8)
<223> OTHER INFORMATION: NotI restriction site for subcloning into
proviral plasmid
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (4)..(16)
<223> OTHER INFORMATION: Kozak consensus sequence
<220> FEATURE:
<221> NAME/KEY: CDS
<222> LOCATION: (13)..(1971)
<223> OTHER INFORMATION: codon-optimized open reading frame (ORF)
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (1972)..(1977)
<223> OTHER INFORMATION: BclI restriction site with embedded stop codon/
site to add optional epitope tag
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (1980)..(1985)
<223> OTHER INFORMATION: BamHI restriction site for subcloning into
proviral plasmid
<400> SEQUENCE: 5

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gcggcgccca cc atg gct gat acc ctg ccc tct gaa ttc gac gtg att gtg Met Ala Asp Thr Leu Pro Ser Glu Phe Asp Val Ile Val 1 5 10	51
att gga acc gga ctc cct gaa tcg atc atc gcc gcg gcc tgt tcc cgg Ile Gly Thr Gly Leu Pro Glu Ser Ile Ile Ala Ala Ala Cys Ser Arg 15 20 25	99
tcc ggt cgg cgc gtg ctg cac gtc gat tcg aga agc tac tac gga ggg Ser Gly Arg Arg Val Leu His Val Asp Ser Arg Ser Tyr Tyr Gly Gly 30 35 40 45	147
aat tgg gcc tca ttc tcc ttc tcc gga ctg ctc tcc tgg ctg aag gag Asn Trp Ala Ser Phe Ser Phe Ser Gly Leu Ser Trp Leu Lys Glu 50 55 60	195
tat cag gag aac tcc gac att gtc tcc gac tca cct gtg tgg cag gac Tyr Gln Glu Asn Ser Asp Ile Val Ser Asp Ser Pro Val Trp Gln Asp 65 70 75	243
cag atc ctg gaa aac gag gaa gca ata gcc ctg agc cgg aag gac aag Gln Ile Leu Glu Asn Glu Glu Ala Ile Ala Leu Ser Arg Lys Asp Lys 80 85 90	291
acc atc cag cac gtg gag gtg ttc tgt tat gcc tcc caa gac ctc cat Thr Ile Gln His Val Glu Val Phe Cys Tyr Ala Ser Gln Asp Leu His 95 100 105	339
gag gac gtg gaa gag gct gga gcg ttg cag aag aat cat gcc ctc gtg Glu Asp Val Glu Glu Ala Gly Ala Leu Gln Lys Asn His Ala Leu Val 110 115 120 125	387
acc tcc gct aac tcc acc gag gca gcc gac agc gcc ttc ctg ccg acc Thr Ser Ala Asn Ser Thr Glu Ala Ala Asp Ser Ala Phe Leu Pro Thr 130 135 140	435
gag gat gaa tcc ctg tca act atg tcg tgc gaa atg ctg acc gaa cag Glu Asp Glu Ser Leu Ser Thr Met Ser Cys Glu Met Leu Thr Glu Gln 145 150 155	483
act ccg agc tcc gac ccc gaa aac gcc ctg gaa gtg aac gga gcg gaa Thr Pro Ser Ser Asp Pro Glu Asn Ala Leu Glu Val Asn Gly Ala Glu 160 165 170	531
gtg acc ggc gaa aag gag aac cat tgc gac gac aag act tgt gtc cca Val Thr Gly Glu Lys Glu Asn His Cys Asp Asp Lys Thr Cys Val Pro 175 180 185	579
tcc act tcc gcg gag gac atg tcc gag aat gtg cct atc gcc gag gac Ser Thr Ser Ala Glu Asp Met Ser Glu Asn Val Pro Ile Ala Glu Asp 190 195 200 205	627
acc acc gaa cag ccc aag aag aac aga atc acg tac agc cag atc atc Thr Thr Glu Gln Pro Lys Lys Asn Arg Ile Thr Tyr Ser Gln Ile Ile 210 215 220	675
aag gag ggg cgg agg ttt aac atc gat ctg gtg tcg aag ctg ctg tac Lys Glu Gly Arg Arg Phe Asn Ile Asp Leu Val Ser Lys Leu Leu Tyr 225 230 235	723
agc cgc ggt ctg ctg atc gat ctg ctc att aag tcg aac gtg tcg aga Ser Arg Gly Leu Leu Ile Asp Leu Leu Ile Lys Ser Asn Val Ser Arg 240 245 250	771
tac gcc gag ttc aag aac atc aca agg att ctc gcc ttc cgg gaa gga Tyr Ala Glu Phe Lys Asn Ile Thr Arg Ile Leu Ala Phe Arg Glu Gly 255 260 265	819
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<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
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35 40 45

Ser Phe Ser Phe Ser Gly Leu Leu Ser Trp Leu Lys Glu Tyr Gln Glu
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Asn Ser Asp Ile Val Ser Asp Ser Pro Val Trp Gln Asp Gln Ile Leu
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Glu Asn Glu Glu Ala Ile Ala Leu Ser Arg Lys Asp Lys Thr Ile Gln
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His Val Glu Val Phe Cys Tyr Ala Ser Gln Asp Leu His Glu Asp Val
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Glu Glu Ala Gly Ala Leu Gln Lys Asn His Ala Leu Val Thr Ser Ala
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Ser Asp Pro Glu Asn Ala Leu Glu Val Asn Gly Ala Glu Val Thr Gly
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Glu Lys Glu Asn His Cys Asp Asp Lys Thr Cys Val Pro Ser Thr Ser
180 185 190

Ala Glu Asp Met Ser Glu Asn Val Pro Ile Ala Glu Asp Thr Thr Glu
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Gln Pro Lys Lys Asn Arg Ile Thr Tyr Ser Gln Ile Ile Lys Glu Gly
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Phe Lys Asn Ile Thr Arg Ile Leu Ala Phe Arg Glu Gly Arg Val Glu
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Gly Asn Thr Pro Phe Leu Phe Pro Leu Tyr Gly Gln Gly Glu Leu Pro
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Gln Cys Phe Cys Arg Met Cys Ala Val Phe Gly Gly Ile Tyr Cys Leu
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Arg His Ser Val Gln Cys Leu Val Val Asp Lys Glu Ser Arg Lys Cys
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Lys Ala Ile Ile Asp Gln Phe Gly Gln Arg Ile Ile Ser Glu His Phe
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Leu Val Glu Asp Ser Tyr Phe Pro Glu Asn Met Cys Ser Arg Val Gln
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Tyr Arg Gln Ile Ser Arg Ala Val Leu Ile Thr Asp Arg Ser Val Leu
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Lys Thr Asp Ser Asp Gln Gln Ile Ser Ile Leu Thr Val Pro Ala Glu
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Glu Pro Gly Thr Phe Ala Val Arg Val Ile Glu Leu Cys Ser Ser Thr
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Met Thr Cys Met Lys Gly Thr Tyr Leu Val His Leu Thr Cys Thr Ser
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Ser Lys Thr Ala Arg Glu Asp Leu Glu Ser Val Val Gln Lys Leu Phe
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Val Pro Tyr Thr Glu Met Glu Ile Glu Asn Glu Gln Val Glu Lys Pro
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Arg Ile Leu Trp Ala Leu Tyr Phe Asn Met Arg Asp Ser Ser Asp Ile
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Gly Pro Asp Cys Gly Leu Gly Asn Asp Asn Ala Val Lys Gln Ala Glu
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Thr Leu Phe Gln Glu Ile Cys Pro Asn Glu Asp Phe Cys Pro Pro Pro
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Pro Asn Pro Glu Asp Ile Ile Leu Asp Gly Asp Ser Leu Gln Pro Glu
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<223> OTHER INFORMATION: BamHI cloning site, cuts at 1806
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<223> OTHER INFORMATION: Kanamycin resistance (complementary)
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<222> LOCATION: (8485)..(9128)
<223> OTHER INFORMATION: Origin of replication (complementary)

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caactctttt tccgaaggta actggcttca gcagagcgca gataccaaat actgtccttc	10620
tagtgtagcc gtatgttagc caccacttca agaactctgt agcaccgct acatacctcg	10680
ctctgcta at cctgttacca gtggctgctg ccagtggcga taagtctgt cttaccgggt	10740
tggaactcaag acgatagta ccggataagg cgcagcggtc gggctgaacg gggggttcgt	10800
gcacacagcc cagcttgag cgaacgacct acaccgaact gagataccta cagcgtgagc	10860
tatgagaaag cgccacgctt ccgaaggga gaaaggcgga caggtatccg gtaagcggca	10920

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gggtcggaac aggagagcgc acgaggggagc ttccaggggg aaacgcctgg tatctttata 10980
gtcctgtcgg gtttcgccac ctctgacttg agcgtcgatt tttgtgatgc tcgtcagggg 11040
ggcggagcct atggaaaaac gccagcaacg cggccttttt acggttcctg gccttttgct 11100
ggccttttgc tcacatgtcc tgcaggcagc tgcgcgccag ctgcgcgc 11148

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<210> SEQ ID NO 9
<211> LENGTH: 2085
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: codon optimized sequence
<220> FEATURE:
<221> NAME/KEY: CDS
<222> LOCATION: (1)..(2085)
<223> OTHER INFORMATION: codon-optimized ORF

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<400> SEQUENCE: 9

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Met Ala Lys Ile Asn Thr Gln Tyr Ser His Pro Ser Arg Thr His Leu
1 5 10 15

aaa gtc aag acc tcc gat cgg gat ctg aac cgg gct gag aat ggg ctg 96
Lys Val Lys Thr Ser Asp Arg Asp Leu Asn Arg Ala Glu Asn Gly Leu
20 25 30

tcg cgc gcc cac tcg tcg tcc gag gaa acc agc agc gtg ctc cag ccg 144
Ser Arg Ala His Ser Ser Ser Glu Glu Thr Ser Ser Val Leu Gln Pro
35 40 45

ggc atc gcc atg gaa act agg ggg ctg gcg gac tcc gga cag gga tcc 192
Gly Ile Ala Met Glu Thr Arg Gly Leu Ala Asp Ser Gly Gln Gly Ser
50 55 60

ttc act gga cag ggt att gcc cgg ctg agc aga ctg atc ttc ctg ctt 240
Phe Thr Gly Gln Gly Ile Ala Arg Leu Ser Arg Leu Ile Phe Leu Leu
65 70 75 80

cgc cgc tgg gcg gcc aga cac gtg cac cat cag gac cag gga cct gat 288
Arg Arg Trp Ala Ala Arg His Val His His Gln Asp Gln Gly Pro Asp
85 90 95

agc ttc ccc gac cgc ttt agg gga gcc gag ctg aaa gaa gtg tca agc 336
Ser Phe Pro Asp Arg Phe Arg Gly Ala Glu Leu Lys Glu Val Ser Ser
100 105 110

cag gag tca aac gcg cag gcc aac gtc ggc agc caa gag cct gca gac 384
Gln Glu Ser Asn Ala Gln Ala Asn Val Gly Ser Gln Glu Pro Ala Asp
115 120 125

cgg gga cgc tcg gca tgg ccg ctc gca aag tgc aac act aac act tcc 432
Arg Gly Arg Ser Ala Trp Pro Leu Ala Lys Cys Asn Thr Asn Thr Ser
130 135 140

aac aac acc gaa gag gaa aag aaa acc aag aag aag gat gca att gtg 480
Asn Asn Thr Glu Glu Glu Lys Lys Thr Lys Lys Lys Asp Ala Ile Val
145 150 155 160

gtg gac cct tcc tcc aac ctg tac tac cgc tgg ttg acc gcc atc gcc 528
Val Asp Pro Ser Ser Asn Leu Tyr Tyr Arg Trp Leu Thr Ala Ile Ala
165 170 175

ctc ccg gtc ttt tac aat tgg tat ctc ctt atc tgc cgg gcc tgc ttc 576
Leu Pro Val Phe Tyr Asn Trp Tyr Leu Leu Ile Cys Arg Ala Cys Phe
180 185 190

gac gaa ctg caa tca gag tac ctg atg ctg tgg ctg gtg ctg gac tat 624
Asp Glu Leu Gln Ser Glu Tyr Leu Met Leu Trp Leu Val Leu Asp Tyr
195 200 205

agc gcc gat gtg ctc tac gtc ctg gat gtg ctc gtg cgc gcc cgg acc 672
Ser Ala Asp Val Leu Tyr Val Leu Asp Val Leu Val Arg Ala Arg Thr
210 215 220

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gga ttc ttg gaa caa ggc ctg atg gtg tcc gac acg aat aga ctg tgg	720
Gly Phe Leu Glu Gln Gly Leu Met Val Ser Asp Thr Asn Arg Leu Trp	
225 230 235 240	
cag cac tat aag acc aca acc cag ttc aag ctt gac gtg ctc agc ctt	768
Gln His Tyr Lys Thr Thr Thr Gln Phe Lys Leu Asp Val Leu Ser Leu	
245 250 255	
gtg ccg act gac ctg gcc tac ctg aaa gtc gga act aac tac ccg gaa	816
Val Pro Thr Asp Leu Ala Tyr Leu Lys Val Gly Thr Asn Tyr Pro Glu	
260 265 270	
gtc aga ttc aac cga ctc ctg aag ttc agc agg ctg ttc gag ttc ttt	864
Val Arg Phe Asn Arg Leu Leu Lys Phe Ser Arg Leu Phe Glu Phe Phe	
275 280 285	
gac cgc acc gag act cgg acc aac tac cct aac atg ttc cgg atc gga	912
Asp Arg Thr Glu Thr Arg Thr Asn Tyr Pro Asn Met Phe Arg Ile Gly	
290 295 300	
aat ctg gtg ctc tac ata ctg att atc atc cat tgg aac gcc tgt atc	960
Asn Leu Val Leu Tyr Ile Leu Ile Ile His Trp Asn Ala Cys Ile	
305 310 315 320	
tat ttc gcc att tcg aag ttc atc ggt ttc gga acc gat tcc tgg gtg	1008
Tyr Phe Ala Ile Ser Lys Phe Ile Gly Phe Gly Thr Asp Ser Trp Val	
325 330 335	
tac ccc aac atc tcg atc ccc gaa cac ggt cgc ctg tcc cgg aag tac	1056
Tyr Pro Asn Ile Ser Ile Pro Glu His Gly Arg Leu Ser Arg Lys Tyr	
340 345 350	
atc tac tcc ctg tac tgg tcc act ctg act ctg acc acg atc ggg gaa	1104
Ile Tyr Ser Leu Tyr Trp Ser Thr Leu Thr Leu Thr Ile Gly Glu	
355 360 365	
acc cct cca ccc gtg aag gac gaa gag tac ctg ttc gtg gtg gtg gac	1152
Thr Pro Pro Pro Val Lys Asp Glu Glu Tyr Leu Phe Val Val Val Asp	
370 375 380	
ttc ctg gtc gga gtg ttg att ttc gcc acc att gtg gga aac gtg ggc	1200
Phe Leu Val Gly Val Leu Ile Phe Ala Thr Ile Val Gly Asn Val Gly	
385 390 395 400	
tcc atg atc tcc aac atg aac gcg tcg aga gct gag ttc caa gcc aag	1248
Ser Met Ile Ser Asn Met Asn Ala Ser Arg Ala Glu Phe Gln Ala Lys	
405 410 415	
atc gac tcc att aag cag tac atg cag ttc aga aag gtc acc aag gac	1296
Ile Asp Ser Ile Lys Gln Tyr Met Gln Phe Arg Lys Val Thr Lys Asp	
420 425 430	
ctg gaa acc agg gtc atc cgc tgg ttc gac tac ctg tgg gcc aac aaa	1344
Leu Glu Thr Arg Val Ile Arg Trp Phe Asp Tyr Leu Trp Ala Asn Lys	
435 440 445	
aag act gtg gac gaa aag gaa gtg ctg aag tcg ctg ccg gat aag ctg	1392
Lys Thr Val Asp Glu Lys Glu Val Leu Lys Ser Leu Pro Asp Lys Leu	
450 455 460	
aag gcc gaa atc gcc att aac gtg cac ctt gac acc ctg aag aaa gtc	1440
Lys Ala Glu Ile Ala Ile Asn Val His Leu Asp Thr Leu Lys Lys Val	
465 470 475 480	
cgg atc ttc caa gac tgt gaa gcc ggc ctc ctg gtg gag ctc gtg ctc	1488
Arg Ile Phe Gln Asp Cys Glu Ala Gly Leu Leu Val Glu Leu Val Leu	
485 490 495	
aag ctg cgg ccc acc gtg ttc agc ccg gga gat tac att tgc aag aag	1536
Lys Leu Arg Pro Thr Val Phe Ser Pro Gly Asp Tyr Ile Cys Lys Lys	
500 505 510	
ggc gat atc ggc aaa gag atg tac atc atc aac gag gga aag ctg gcc	1584
Gly Asp Ile Gly Lys Glu Met Tyr Ile Ile Asn Glu Gly Lys Leu Ala	
515 520 525	
gtg gtc gcg gac gac ggc gtg acc cag ttc gtg gtg ctg tcc gac gga	1632
Val Val Ala Asp Asp Gly Val Thr Gln Phe Val Val Leu Ser Asp Gly	
530 535 540	

[illegible]

<400> SEQUENCE: 10

Met	Ala	Lys	Ile	Asn	Thr	Gln	Tyr	Ser	His	Pro	Ser	Arg	Thr	His	Leu
1			5						10					15	
Lys	Val	Lys	Thr	Ser	Asp	Arg	Asp	Leu	Asn	Arg	Ala	Glu	Asn	Gly	Leu
			20					25					30		
Ser	Arg	Ala	His	Ser	Ser	Ser	Glu	Glu	Thr	Ser	Ser	Val	Leu	Gln	Pro
		35					40					45			
Gly	Ile	Ala	Met	Glu	Thr	Arg	Gly	Leu	Ala	Asp	Ser	Gly	Gln	Gly	Ser
	50					55					60				
Phe	Thr	Gly	Gln	Gly	Ile	Ala	Arg	Leu	Ser	Arg	Leu	Ile	Phe	Leu	Leu
65					70					75					80
Arg	Arg	Trp	Ala	Ala	Arg	His	Val	His	His	Gln	Asp	Gln	Gly	Pro	Asp
				85					90					95	
Ser	Phe	Pro	Asp	Arg	Phe	Arg	Gly	Ala	Glu	Leu	Lys	Glu	Val	Ser	Ser
			100					105					110		
Gln	Glu	Ser	Asn	Ala	Gln	Ala	Asn	Val	Gly	Ser	Gln	Glu	Pro	Ala	Asp
		115					120					125			
Arg	Gly	Arg	Ser	Ala	Trp	Pro	Leu	Ala	Lys	Cys	Asn	Thr	Asn	Thr	Ser
	130					135					140				
Asn	Asn	Thr	Glu	Glu	Glu	Lys	Lys	Thr	Lys	Lys	Lys	Asp	Ala	Ile	Val

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145	150	155	160
Val Asp Pro Ser Ser Asn Leu Tyr Tyr Arg Trp Leu Thr Ala Ile Ala	165	170	175
Leu Pro Val Phe Tyr Asn Trp Tyr Leu Leu Ile Cys Arg Ala Cys Phe	180	185	190
Asp Glu Leu Gln Ser Glu Tyr Leu Met Leu Trp Leu Val Leu Asp Tyr	195	200	205
Ser Ala Asp Val Leu Tyr Val Leu Asp Val Leu Val Arg Ala Arg Thr	210	215	220
Gly Phe Leu Glu Gln Gly Leu Met Val Ser Asp Thr Asn Arg Leu Trp	225	230	235
Gln His Tyr Lys Thr Thr Thr Gln Phe Lys Leu Asp Val Leu Ser Leu	245	250	255
Val Pro Thr Asp Leu Ala Tyr Leu Lys Val Gly Thr Asn Tyr Pro Glu	260	265	270
Val Arg Phe Asn Arg Leu Leu Lys Phe Ser Arg Leu Phe Glu Phe Phe	275	280	285
Asp Arg Thr Glu Thr Arg Thr Asn Tyr Pro Asn Met Phe Arg Ile Gly	290	295	300
Asn Leu Val Leu Tyr Ile Leu Ile Ile Ile His Trp Asn Ala Cys Ile	305	310	315
Tyr Phe Ala Ile Ser Lys Phe Ile Gly Phe Gly Thr Asp Ser Trp Val	325	330	335
Tyr Pro Asn Ile Ser Ile Pro Glu His Gly Arg Leu Ser Arg Lys Tyr	340	345	350
Ile Tyr Ser Leu Tyr Trp Ser Thr Leu Thr Leu Thr Thr Ile Gly Glu	355	360	365
Thr Pro Pro Pro Val Lys Asp Glu Glu Tyr Leu Phe Val Val Val Asp	370	375	380
Phe Leu Val Gly Val Leu Ile Phe Ala Thr Ile Val Gly Asn Val Gly	385	390	395
Ser Met Ile Ser Asn Met Asn Ala Ser Arg Ala Glu Phe Gln Ala Lys	405	410	415
Ile Asp Ser Ile Lys Gln Tyr Met Gln Phe Arg Lys Val Thr Lys Asp	420	425	430
Leu Glu Thr Arg Val Ile Arg Trp Phe Asp Tyr Leu Trp Ala Asn Lys	435	440	445
Lys Thr Val Asp Glu Lys Glu Val Leu Lys Ser Leu Pro Asp Lys Leu	450	455	460
Lys Ala Glu Ile Ala Ile Asn Val His Leu Asp Thr Leu Lys Lys Val	465	470	475
Arg Ile Phe Gln Asp Cys Glu Ala Gly Leu Leu Val Glu Leu Val Leu	485	490	495
Lys Leu Arg Pro Thr Val Phe Ser Pro Gly Asp Tyr Ile Cys Lys Lys	500	505	510
Gly Asp Ile Gly Lys Glu Met Tyr Ile Ile Asn Glu Gly Lys Leu Ala	515	520	525
Val Val Ala Asp Asp Gly Val Thr Gln Phe Val Val Leu Ser Asp Gly	530	535	540
Ser Tyr Phe Gly Glu Ile Ser Ile Leu Asn Ile Lys Gly Ser Lys Ser	545	550	555
Gly Asn Arg Arg Thr Ala Asn Ile Arg Ser Ile Gly Tyr Ser Asp Leu	565	570	575

[illegible]

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<210> SEQ ID NO 11
<211> LENGTH: 2250
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: codon optimized sequence
<220> FEATURE:
<221> NAME/KEY: CDS
<222> LOCATION: (1)..(2250)
<223> OTHER INFORMATION: codon-optimized ORF
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<400> SEQUENCE: 11

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Met	Ala	Lys	Ile	Asn	Thr	Gln	Tyr	Ser	His	Pro	Ser	Arg	Thr	His	Leu	
1			5					10					15			
aaa	gtc	aag	acc	tcc	gat	cgg	gat	ctg	aac	cgg	gct	gag	aat	ggg	ctg	96
Lys	Val	Lys	Thr	Ser	Asp	Arg	Asp	Leu	Asn	Arg	Ala	Glu	Asn	Gly	Leu	
			20					25					30			
tcg	cgc	gcc	cac	tcg	tcg	tcc	gag	gaa	acc	agc	agc	gtg	ctc	cag	ccg	144
Ser	Arg	Ala	His	Ser	Ser	Ser	Glu	Glu	Thr	Ser	Ser	Val	Leu	Gln	Pro	
			35				40					45				
ggc	atc	gcc	atg	gaa	act	agg	ggg	ctg	gcg	gac	tcc	gga	cag	gga	tcc	192
Gly	Ile	Ala	Met	Glu	Thr	Arg	Gly	Leu	Ala	Asp	Ser	Gly	Gln	Gly	Ser	
	50					55					60					
ttc	act	gga	cag	ggg	att	gcc	cgg	ttc	ggg	cgg	att	cag	aag	aag	tcc	240
Phe	Thr	Gly	Gln	Gly	Ile	Ala	Arg	Phe	Gly	Arg	Ile	Gln	Lys	Lys	Ser	
65					70					75					80	
cag	ccg	gag	aag	gtc	gtg	cgg	gct	gcc	agc	agg	ggc	agg	cca	ctc	att	288
Gln	Pro	Glu	Lys	Val	Val	Arg	Ala	Ala	Ser	Arg	Gly	Arg	Pro	Leu	Ile	
				85					90				95			
ggg	tgg	aca	cag	tgg	tgc	gct	gag	gat	ggg	gga	gat	gaa	tgc	gaa	atg	336
Gly	Trp	Thr	Gln	Trp	Cys	Ala	Glu	Asp	Gly	Gly	Asp	Glu	Ser	Glu	Met	
			100					105				110				
gca	ctg	gcc	ggc	tct	ccc	gga	tgc	agc	tcg	ggc	ccc	caa	ggg	aga	ctg	384
Ala	Leu	Ala	Gly	Ser	Pro	Gly	Cys	Ser	Ser	Gly	Pro	Gln	Gly	Arg	Leu	
			115				120				125					
agc	aga	ctg	atc	ttc	ctg	ctt	cgc	cgc	tgg	gcg	gcc	aga	cac	gtg	cac	432
Ser	Arg	Leu	Ile	Phe	Leu	Leu	Arg	Arg	Trp	Ala	Ala	Arg	His	Val	His	
			130			135					140					
cat	cag	gac	cag	gga	ccc	gat	agc	ttc	ccc	gac	cgc	ttt	agg	gga	gcc	480
His	Gln	Asp	Gln	Gly	Pro	Asp	Ser	Phe	Pro	Asp	Arg	Phe	Arg	Gly	Ala	
145					150					155				160		

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gag ctg aaa gaa gtg tca agc cag gag tca aac gcg cag gcc aac gtc Glu Leu Lys Glu Val Ser Ser Gln Glu Ser Asn Ala Gln Ala Asn Val 165 170 175	528
ggc agc caa gag cct gca gac cgg gga cgc tcg gca tgg ccg ctc gca Gly Ser Gln Glu Pro Ala Asp Arg Gly Arg Ser Ala Trp Pro Leu Ala 180 185 190	576
aag tgc aac act aac act tcc aac aac acc gaa gag gaa aag aaa acc Lys Cys Asn Thr Asn Thr Ser Asn Asn Thr Glu Glu Glu Lys Lys Thr 195 200 205	624
aag aag aag gat gca att gtg gtg gac cct tcc tcc aac ctg tac tac Lys Lys Lys Asp Ala Ile Val Val Asp Pro Ser Ser Asn Leu Tyr Tyr 210 215 220	672
cgc tgg ttg acc gcc atc gcc ctc ccg gtc ttt tac aat tgg tat ctc Arg Trp Leu Thr Ala Ile Ala Leu Pro Val Phe Tyr Asn Trp Tyr Leu 225 230 235 240	720
ctt atc tgc cgg gcc tgc ttc gac gaa ctg caa tca gag tac ctg atg Leu Ile Cys Arg Ala Cys Phe Asp Glu Leu Gln Ser Glu Tyr Leu Met 245 250 255	768
ctg tgg ctg gtg ctg gac tat agc gcc gat gtg ctc tac gtc ctg gat Leu Trp Leu Val Leu Asp Tyr Ser Ala Asp Val Leu Tyr Val Leu Asp 260 265 270	816
gtg ctc gtg cgc gcc cgg acc gga ttc ttg gaa caa ggc ctg atg gtg Val Leu Val Arg Ala Arg Thr Gly Phe Leu Glu Gln Gly Leu Met Val 275 280 285	864
tcc gac acg aat aga ctg tgg cag cac tat aag acc aca acc cag ttc Ser Asp Thr Asn Arg Leu Trp Gln His Tyr Lys Thr Thr Gln Phe 290 295 300	912
aag ctt gac gtg ctc agc ctt gtg ccg act gac ctg gcc tac ctg aaa Lys Leu Asp Val Leu Ser Leu Val Pro Thr Asp Leu Ala Tyr Leu Lys 305 310 315 320	960
gtc gga act aac tac ccg gaa gtc aga ttc aac cga ctc ctg aag ttc Val Gly Thr Asn Tyr Pro Glu Val Arg Phe Asn Arg Leu Leu Lys Phe 325 330 335	1008
agc agg ctg ttc gag ttc ttt gac cgc acc gag act cgg acc aac tac Ser Arg Leu Phe Glu Phe Phe Asp Arg Thr Glu Thr Arg Thr Asn Tyr 340 345 350	1056
cct aac atg ttc cgg atc gga aat ctg gtg ctc tac ata ctg att atc Pro Asn Met Phe Arg Ile Gly Asn Leu Val Leu Tyr Ile Leu Ile Ile 355 360 365	1104
atc cat tgg aac gcc tgt atc tat ttc gcc att tcg aag ttc atc ggt Ile His Trp Asn Ala Cys Ile Tyr Phe Ala Ile Ser Lys Phe Ile Gly 370 375 380	1152
ttc gga acc gat tcc tgg gtg tac ccc aac atc tcg atc ccc gaa cac Phe Gly Thr Asp Ser Trp Val Tyr Pro Asn Ile Ser Ile Pro Glu His 385 390 395 400	1200
ggt cgc ctg tcc cgg aag tac atc tac tcc ctg tac tgg tcc act ctg Gly Arg Leu Ser Arg Lys Tyr Ile Tyr Ser Leu Tyr Trp Ser Thr Leu 405 410 415	1248
act ctg acc acg atc ggg gaa acc cct cca ccc gtg aag gac gaa gag Thr Leu Thr Thr Ile Gly Glu Thr Pro Pro Pro Val Lys Asp Glu Glu 420 425 430	1296
tac ctg ttc gtg gtg gtg gac ttc ctg gtc gga gtg ttg att ttc gcc Tyr Leu Phe Val Val Val Asp Phe Leu Val Gly Val Leu Ile Phe Ala 435 440 445	1344
acc att gtg gga aac gtg ggc tcc atg atc tcc aac atg aac gcg tcg Thr Ile Val Gly Asn Val Gly Ser Met Ile Ser Asn Met Asn Ala Ser 450 455 460	1392
aga gct gag ttc caa gcc aag atc gac tcc att aag cag tac atg cag Arg Ala Glu Phe Gln Ala Lys Ile Asp Ser Ile Lys Gln Tyr Met Gln 465 470 475 480	1440

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ttc aga aag gtc acc aag gac ctg gaa acc agg gtc atc cgc tgg ttc	1488
Phe Arg Lys Val Thr Lys Asp Leu Glu Thr Arg Val Ile Arg Trp Phe	
485 490 495	
gac tac ctg tgg gcc aac aaa aag act gtg gac gaa aag gaa gtg ctg	1536
Asp Tyr Leu Trp Ala Asn Lys Lys Thr Val Asp Glu Lys Glu Val Leu	
500 505 510	
aag tcg ctg ccg gat aag ctg aag gcc gaa atc gcc att aac gtg cac	1584
Lys Ser Leu Pro Asp Lys Leu Lys Ala Glu Ile Ala Ile Asn Val His	
515 520 525	
ctt gac acc ctg aag aaa gtc cgg atc ttc caa gac tgt gaa gcc ggc	1632
Leu Asp Thr Leu Lys Lys Val Arg Ile Phe Gln Asp Cys Glu Ala Gly	
530 535 540	
ctc ctg gtg gag ctc gtg ctc aag ctg cgg ccc acc gtg ttc agc ccg	1680
Leu Leu Val Glu Leu Val Leu Lys Leu Arg Pro Thr Val Phe Ser Pro	
545 550 555 560	
gga gat tac att tgc aag aag ggc gat atc ggc aaa gag atg tac atc	1728
Gly Asp Tyr Ile Cys Lys Lys Gly Asp Ile Gly Lys Glu Met Tyr Ile	
565 570 575	
atc aac gag gga aag ctg gcc gtg gtc gcg gac gac ggc gtg acc cag	1776
Ile Asn Glu Gly Lys Leu Ala Val Val Ala Asp Asp Gly Val Thr Gln	
580 585 590	
ttc gtg gtg ctg tcc gac gga tcc tac ttc ggt gaa atc tca atc ctc	1824
Phe Val Val Leu Ser Asp Gly Ser Tyr Phe Gly Glu Ile Ser Ile Leu	
595 600 605	
aac atc aag ggg tcc aag tcc ggc aac cgg aga act gcc aac att cgc	1872
Asn Ile Lys Gly Ser Lys Ser Gly Asn Arg Arg Thr Ala Asn Ile Arg	
610 615 620	
tcc atc gga tac agc gac ctg ttt tgc ctg tcc aag gat gac ctg atg	1920
Ser Ile Gly Tyr Ser Asp Leu Phe Cys Leu Ser Lys Asp Asp Leu Met	
625 630 635 640	
gag gct ctg act gag tac cct gaa gcg aag aag gct ttg gag gaa aag	1968
Glu Ala Leu Thr Glu Tyr Pro Glu Ala Lys Lys Ala Leu Glu Glu Lys	
645 650 655	
ggg cgg cag att ctg atg aag gac aat ttg atc gac gag gag ctc gca	2016
Gly Arg Gln Ile Leu Met Lys Asp Asn Leu Ile Asp Glu Glu Leu Ala	
660 665 670	
cgg gcc ggc gcc gac ccc aag gat ctc gaa gag aag gtc gaa cag ctg	2064
Arg Ala Gly Ala Asp Pro Lys Asp Leu Glu Glu Lys Val Glu Gln Leu	
675 680 685	
ggt tct tcg ctt gat acc ctg caa acc cga ttc gcg cgg ctg ctc gcc	2112
Gly Ser Ser Leu Asp Thr Leu Gln Thr Arg Phe Ala Arg Leu Leu Ala	
690 695 700	
gag tac aac gcg acc cag atg aag atg aag cag aga ctg tca cag ttg	2160
Glu Tyr Asn Ala Thr Gln Met Lys Met Lys Gln Arg Leu Ser Gln Leu	
705 710 715 720	
gaa tcc caa gtc aag ggc gga ggc gac aag ccg ctg gcg gac ggg gaa	2208
Glu Ser Gln Val Lys Gly Gly Gly Asp Lys Pro Leu Ala Asp Gly Glu	
725 730 735	
gtg ccc ggg gac gcc acc aag act gag gac aag cag cag tga	2250
Val Pro Gly Asp Ala Thr Lys Thr Glu Asp Lys Gln Gln	
740 745	

<210> SEQ ID NO 12

<211> LENGTH: 749

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 12

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Met	Ala	Lys	Ile	Asn	Thr	Gln	Tyr	Ser	His	Pro	Ser	Arg	Thr	His	Leu
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Lys	Val	Lys	Thr	Ser	Asp	Arg	Asp	Leu	Asn	Arg	Ala	Glu	Asn	Gly	Leu
			20					25					30		
Ser	Arg	Ala	His	Ser	Ser	Ser	Glu	Glu	Thr	Ser	Ser	Val	Leu	Gln	Pro
		35					40					45			
Gly	Ile	Ala	Met	Glu	Thr	Arg	Gly	Leu	Ala	Asp	Ser	Gly	Gln	Gly	Ser
	50					55					60				
Phe	Thr	Gly	Gln	Gly	Ile	Ala	Arg	Phe	Gly	Arg	Ile	Gln	Lys	Lys	Ser
65					70					75					80
Gln	Pro	Glu	Lys	Val	Val	Arg	Ala	Ala	Ser	Arg	Gly	Arg	Pro	Leu	Ile
			85						90					95	
Gly	Trp	Thr	Gln	Trp	Cys	Ala	Glu	Asp	Gly	Gly	Asp	Glu	Ser	Glu	Met
			100					105					110		
Ala	Leu	Ala	Gly	Ser	Pro	Gly	Cys	Ser	Ser	Gly	Pro	Gln	Gly	Arg	Leu
		115					120					125			
Ser	Arg	Leu	Ile	Phe	Leu	Leu	Arg	Arg	Trp	Ala	Ala	Arg	His	Val	His
	130					135						140			
His	Gln	Asp	Gln	Gly	Pro	Asp	Ser	Phe	Pro	Asp	Arg	Phe	Arg	Gly	Ala
145					150					155					160
Glu	Leu	Lys	Glu	Val	Ser	Ser	Gln	Glu	Ser	Asn	Ala	Gln	Ala	Asn	Val
			165						170					175	
Gly	Ser	Gln	Glu	Pro	Ala	Asp	Arg	Gly	Arg	Ser	Ala	Trp	Pro	Leu	Ala
			180					185					190		
Lys	Cys	Asn	Thr	Asn	Thr	Ser	Asn	Asn	Thr	Glu	Glu	Glu	Lys	Lys	Thr
		195					200					205			
Lys	Lys	Lys	Asp	Ala	Ile	Val	Val	Asp	Pro	Ser	Ser	Asn	Leu	Tyr	Tyr
	210					215					220				
Arg	Trp	Leu	Thr	Ala	Ile	Ala	Leu	Pro	Val	Phe	Tyr	Asn	Trp	Tyr	Leu
225					230					235					240
Leu	Ile	Cys	Arg	Ala	Cys	Phe	Asp	Glu	Leu	Gln	Ser	Glu	Tyr	Leu	Met
			245					250						255	
Leu	Trp	Leu	Val	Leu	Asp	Tyr	Ser	Ala	Asp	Val	Leu	Tyr	Val	Leu	Asp
			260					265					270		
Val	Leu	Val	Arg	Ala	Arg	Thr	Gly	Phe	Leu	Glu	Gln	Gly	Leu	Met	Val
		275					280					285			
Ser	Asp	Thr	Asn	Arg	Leu	Trp	Gln	His	Tyr	Lys	Thr	Thr	Thr	Gln	Phe
	290					295					300				
Lys	Leu	Asp	Val	Leu	Ser	Leu	Val	Pro	Thr	Asp	Leu	Ala	Tyr	Leu	Lys
305					310					315					320
Val	Gly	Thr	Asn	Tyr	Pro	Glu	Val	Arg	Phe	Asn	Arg	Leu	Leu	Lys	Phe
			325						330					335	
Ser	Arg	Leu	Phe	Glu	Phe	Phe	Asp	Arg	Thr	Glu	Thr	Arg	Thr	Asn	Tyr
			340					345					350		
Pro	Asn	Met	Phe	Arg	Ile	Gly	Asn	Leu	Val	Leu	Tyr	Ile	Leu	Ile	Ile
		355					360					365			
Ile	His	Trp	Asn	Ala	Cys	Ile	Tyr	Phe	Ala	Ile	Ser	Lys	Phe	Ile	Gly
	370					375					380				
Phe	Gly	Thr	Asp	Ser	Trp	Val	Tyr	Pro	Asn	Ile	Ser	Ile	Pro	Glu	His
385					390					395					400
Gly	Arg	Leu	Ser	Arg	Lys	Tyr	Ile	Tyr	Ser	Leu	Tyr	Trp	Ser	Thr	Leu
			405						410					415	
Thr	Leu	Thr	Thr	Ile	Gly	Glu	Thr	Pro	Pro	Pro	Val	Lys	Asp	Glu	Glu

	420						425				430							
Tyr	Leu	Phe	Val	Val	Val	Asp	Phe	Leu	Val	Gly	Val	Leu	Ile	Phe	Ala			
	435						440				445							
Thr	Ile	Val	Gly	Asn	Val	Gly	Ser	Met	Ile	Ser	Asn	Met	Asn	Ala	Ser			
	450						455				460							
Arg	Ala	Glu	Phe	Gln	Ala	Lys	Ile	Asp	Ser	Ile	Lys	Gln	Tyr	Met	Gln			
465	470						475						480					
Phe	Arg	Lys	Val	Thr	Lys	Asp	Leu	Glu	Thr	Arg	Val	Ile	Arg	Trp	Phe			
	485						490						495					
Asp	Tyr	Leu	Trp	Ala	Asn	Lys	Lys	Thr	Val	Asp	Glu	Lys	Glu	Val	Leu			
	500						505						510					
Lys	Ser	Leu	Pro	Asp	Lys	Leu	Lys	Ala	Glu	Ile	Ala	Ile	Asn	Val	His			
	515						520						525					
Leu	Asp	Thr	Leu	Lys	Lys	Val	Arg	Ile	Phe	Gln	Asp	Cys	Glu	Ala	Gly			
	530						535						540					
Leu	Leu	Val	Glu	Leu	Val	Leu	Lys	Leu	Arg	Pro	Thr	Val	Phe	Ser	Pro			
545	550						555						560					
Gly	Asp	Tyr	Ile	Cys	Lys	Lys	Gly	Asp	Ile	Gly	Lys	Glu	Met	Tyr	Ile			
	565						570						575					
Ile	Asn	Glu	Gly	Lys	Leu	Ala	Val	Val	Ala	Asp	Asp	Gly	Val	Thr	Gln			
	580						585						590					
Phe	Val	Val	Leu	Ser	Asp	Gly	Ser	Tyr	Phe	Gly	Glu	Ile	Ser	Ile	Leu			
	595						600						605					
Asn	Ile	Lys	Gly	Ser	Lys	Ser	Gly	Asn	Arg	Arg	Thr	Ala	Asn	Ile	Arg			
	610						615						620					
Ser	Ile	Gly	Tyr	Ser	Asp	Leu	Phe	Cys	Leu	Ser	Lys	Asp	Asp	Leu	Met			
625	630						635						640					
Glu	Ala	Leu	Thr	Glu	Tyr	Pro	Glu	Ala	Lys	Lys	Ala	Leu	Glu	Glu	Lys			
	645						650						655					
Gly	Arg	Gln	Ile	Leu	Met	Lys	Asp	Asn	Leu	Ile	Asp	Glu	Glu	Leu	Ala			
	660						665						670					
Arg	Ala	Gly	Ala	Asp	Pro	Lys	Asp	Leu	Glu	Glu	Lys	Val	Glu	Gln	Leu			
	675						680						685					
Gly	Ser	Ser	Leu	Asp	Thr	Leu	Gln	Thr	Arg	Phe	Ala	Arg	Leu	Leu	Ala			
	690						695						700					
Glu	Tyr	Asn	Ala	Thr	Gln	Met	Lys	Met	Lys	Gln	Arg	Leu	Ser	Gln	Leu			
705	710						715						720					
Glu	Ser	Gln	Val	Lys	Gly	Gly	Gly	Asp	Lys	Pro	Leu	Ala	Asp	Gly	Glu			
	725						730						735					
Val	Pro	Gly	Asp	Ala	Thr	Lys	Thr	Glu	Asp	Lys	Gln	Gln						
	740						745											

<400> SEQUENCE: 13

aag gta aag acc tca gac cgg gat ctc aat cgc gct gaa aat ggc ctc 96

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Lys	Val	Lys	Thr	Ser	Asp	Arg	Asp	Leu	Asn	Arg	Ala	Glu	Asn	Gly	Leu	
			20					25					30			
agc	aga	gcc	cac	tcg	tca	agt	gag	gag	aca	tcg	tca	gtg	ctg	cag	ccg	144
Ser	Arg	Ala	His	Ser	Ser	Ser	Glu	Glu	Thr	Ser	Ser	Val	Leu	Gln	Pro	
		35					40					45				
ggg	atc	gcc	atg	gag	acc	aga	gga	ctg	gct	gac	tcc	ggg	cag	ggc	tcc	192
Gly	Ile	Ala	Met	Glu	Thr	Arg	Gly	Leu	Ala	Asp	Ser	Gly	Gln	Gly	Ser	
	50					55					60					
ttc	acc	ggc	cag	ggg	atc	gcc	agg	ctg	tcg	cgc	ctc	atc	ttc	ttg	ctg	240
Phe	Thr	Gly	Gln	Gly	Ile	Ala	Arg	Leu	Ser	Arg	Leu	Ile	Phe	Leu	Leu	
65					70					75				80		
cgc	agg	tgg	gct	gcc	agg	cat	gtg	cac	cac	cag	gac	cag	gga	ccg	gac	288
Arg	Arg	Trp	Ala	Ala	Arg	His	Val	His	His	Gln	Asp	Gln	Gly	Pro	Asp	
			85						90					95		
tct	ttt	cct	gat	cgt	ttc	cgt	gga	gcc	gag	ctt	aag	gag	gtg	tcc	agc	336
Ser	Phe	Pro	Asp	Arg	Phe	Arg	Gly	Ala	Glu	Leu	Lys	Glu	Val	Ser	Ser	
		100					105						110			
caa	gaa	agc	aat	gcc	cag	gca	aat	gtg	ggc	agc	cag	gag	cca	gca	gac	384
Gln	Glu	Ser	Asn	Ala	Gln	Ala	Asn	Val	Gly	Ser	Gln	Glu	Pro	Ala	Asp	
	115					120					125					
aga	ggg	aga	agc	gcc	tgg	ccc	ctg	gcc	aaa	tgc	aac	act	aac	acc	agc	432
Arg	Gly	Arg	Ser	Ala	Trp	Pro	Leu	Ala	Lys	Cys	Asn	Thr	Asn	Thr	Ser	
	130					135					140					
aac	aac	acg	gag	gag	gag	aag	aag	acg	aaa	aag	aag	gat	gcg	atc	gtg	480
Asn	Asn	Thr	Glu	Glu	Glu	Lys	Lys	Thr	Lys	Lys	Lys	Asp	Ala	Ile	Val	
145					150					155				160		
gtg	gac	ccg	tcc	agc	aac	ctg	tac	tac	cgc	tgg	ctg	acc	gcc	atc	gcc	528
Val	Asp	Pro	Ser	Ser	Asn	Leu	Tyr	Tyr	Arg	Trp	Leu	Thr	Ala	Ile	Ala	
			165						170					175		
ctg	cct	gtc	ttc	tat	aac	tgg	tat	ctg	ctt	att	tgc	agg	gcc	tgt	ttc	576
Leu	Pro	Val	Phe	Tyr	Asn	Trp	Tyr	Leu	Leu	Ile	Cys	Arg	Ala	Cys	Phe	
		180						185					190			
gat	gag	ctg	cag	tcc	gag	tac	ctg	atg	ctg	tgg	ctg	gtc	ctg	gac	tac	624
Asp	Glu	Leu	Gln	Ser	Glu	Tyr	Leu	Met	Leu	Trp	Leu	Val	Leu	Asp	Tyr	
		195				200						205				
tcg	gca	gat	gtc	ctg	tat	gtc	ttg	gat	gtg	ctt	gta	cga	gct	cgg	aca	672
Ser	Ala	Asp	Val	Leu	Tyr	Val	Leu	Asp	Val	Leu	Val	Arg	Ala	Arg	Thr	
	210					215					220					
ggg	ttt	ctt	gag	caa	ggc	tta	atg	gtc	agt	gat	acc	aac	agg	ctg	tgg	720
Gly	Phe	Leu	Glu	Gln	Gly	Leu	Met	Val	Ser	Asp	Thr	Asn	Arg	Leu	Trp	
225					230					235				240		
cag	cat	tac	aag	acg	acc	acg	cag	ttc	aag	ctg	gat	gtg	ttg	tcc	ctg	768
Gln	His	Tyr	Lys	Thr	Thr	Thr	Gln	Phe	Lys	Leu	Asp	Val	Leu	Ser	Leu	
			245						250					255		
gtc	ccc	acc	gac	ctg	gct	tac	tta	aag	gtg	ggc	aca	aac	tac	cca	gaa	816
Val	Pro	Thr	Asp	Leu	Ala	Tyr	Leu	Lys	Val	Gly	Thr	Asn	Tyr	Pro	Glu	
		260						265					270			
gtg	agg	ttc	aac	cgc	cta	ctg	aag	ttt	tcc	cgg	ctc	ttt	gaa	ttc	ttt	864
Val	Arg	Phe	Asn	Arg	Leu	Leu	Lys	Phe	Ser	Arg	Leu	Phe	Glu	Phe	Phe	
		275					280					285				
gac	cgc	aca	gag	aca	agg	acc	aac	tac	ccc	aat	atg	ttc	agg	att	ggg	912
Asp	Arg	Thr	Glu	Thr	Arg	Thr	Asn	Tyr	Pro	Asn	Met	Phe	Arg	Ile	Gly	
		290				295					300					
aac	ttg	gtc	ttg	tac	att	ctc	atc	atc	atc	cac	tgg	aat	gcc	tgc	atc	960
Asn	Leu	Val	Leu	Tyr	Ile	Leu	Ile	Ile	Ile	His	Trp	Asn	Ala	Cys	Ile	
305					310					315				320		
tac	ttt	gcc	att	tcc	aag	ttc	att	ggg	ttt	ggg	aca	gac	tcc	tgg	gtc	1008
Tyr	Phe	Ala	Ile	Ser	Lys	Phe	Ile	Gly	Phe	Gly	Thr	Asp	Ser	Trp	Val	
			325						330					335		

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tac cca aac atc tca atc cca gag cat ggg cgc ctc tcc agg aag tac	1056
Tyr Pro Asn Ile Ser Ile Pro Glu His Gly Arg Leu Ser Arg Lys Tyr	
340 345 350	
att tac agt ctc tac tgg tcc acc ttg acc ctt acc acc att ggt gag	1104
Ile Tyr Ser Leu Tyr Trp Ser Thr Leu Thr Leu Thr Thr Ile Gly Glu	
355 360 365	
acc cca ccc ccc gtg aaa gat gag gag tat ctc ttt gtg gtc gta gac	1152
Thr Pro Pro Pro Val Lys Asp Glu Glu Tyr Leu Phe Val Val Val Asp	
370 375 380	
ttc ttg gtg ggt gtt ctg att ttt gcc acc att gtg ggc aat gtg ggc	1200
Phe Leu Val Gly Val Leu Ile Phe Ala Thr Ile Val Gly Asn Val Gly	
385 390 395 400	
tcc atg atc tcg aat atg aat gcc tca cgg gca gag ttc cag gcc aag	1248
Ser Met Ile Ser Asn Met Asn Ala Ser Arg Ala Glu Phe Gln Ala Lys	
405 410 415	
att gat tcc atc aag cag tac atg cag ttc cgc aag gtc acc aag gac	1296
Ile Asp Ser Ile Lys Gln Tyr Met Gln Phe Arg Lys Val Thr Lys Asp	
420 425 430	
ttg gag acg cgg gtt atc cgg tgg ttt gac tac ctg tgg gcc aac aag	1344
Leu Glu Thr Arg Val Ile Arg Trp Phe Asp Tyr Leu Trp Ala Asn Lys	
435 440 445	
aag acg gtg gat gag aag gag gtg ctc aag agc ctc cca gac aag ctg	1392
Lys Thr Val Asp Glu Lys Glu Val Leu Lys Ser Leu Pro Asp Lys Leu	
450 455 460	
aag gct gag atc gcc atc aac gtg cac ctg gac acg ctg aag aag gtt	1440
Lys Ala Glu Ile Ala Ile Asn Val His Leu Asp Thr Leu Lys Lys Val	
465 470 475 480	
cgc atc ttc cag gac tgt gag gca ggg ctg ctg gtg gag ctg gtg ctg	1488
Arg Ile Phe Gln Asp Cys Glu Ala Gly Leu Leu Val Glu Leu Val Leu	
485 490 495	
aag ctg cga ccc act gtg ttc agc cct ggg gat tat atc tgc aag aag	1536
Lys Leu Arg Pro Thr Val Phe Ser Pro Gly Asp Tyr Ile Cys Lys Lys	
500 505 510	
gga gat att ggg aag gag atg tac atc atc aac gag ggc aag ctg gcc	1584
Gly Asp Ile Gly Lys Glu Met Tyr Ile Ile Asn Glu Gly Lys Leu Ala	
515 520 525	
gtg gtg gct gat gat ggg gtc acc cag ttc gtg gtc ctc agc gat ggc	1632
Val Val Ala Asp Asp Gly Val Thr Gln Phe Val Val Leu Ser Asp Gly	
530 535 540	
agc tac ttc ggg gag atc agc att ctg aac atc aag ggg agc aag tcg	1680
Ser Tyr Phe Gly Glu Ile Ser Ile Leu Asn Ile Lys Gly Ser Lys Ser	
545 550 555 560	
ggg aac cgc agg acg gcc aac atc cgc agc att ggc tac tca gac ctg	1728
Gly Asn Arg Arg Thr Ala Asn Ile Arg Ser Ile Gly Tyr Ser Asp Leu	
565 570 575	
ttc tgc ctc tca aag gac gat ctc atg gag gcc ctc acc gag tac ccc	1776
Phe Cys Leu Ser Lys Asp Asp Leu Met Glu Ala Leu Thr Glu Tyr Pro	
580 585 590	
gaa gcc aag aag gcc ctg gag gag aaa gga cgg cag atc ctg atg aaa	1824
Glu Ala Lys Lys Ala Leu Glu Glu Lys Gly Arg Gln Ile Leu Met Lys	
595 600 605	
gac aac ctg atc gat gag gag ctg gcc agg gcg ggc gcg gac ccc aag	1872
Asp Asn Leu Ile Asp Glu Glu Leu Ala Arg Ala Gly Ala Asp Pro Lys	
610 615 620	
gac ctt gag gag aaa gtg gag cag ctg ggg tcc tcc ctg gac acc ctg	1920
Asp Leu Glu Glu Lys Val Glu Gln Leu Gly Ser Ser Leu Asp Thr Leu	
625 630 635 640	
cag acc agg ttt gca cgc ctc ctg gct gag tac aac gcc acc cag atg	1968
Gln Thr Arg Phe Ala Arg Leu Leu Ala Glu Tyr Asn Ala Thr Gln Met	
645 650 655	

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aag atg aag cag cgt ctc agc caa ctg gaa agc cag gtg aag ggt ggt      2016
Lys Met Lys Gln Arg Leu Ser Gln Leu Glu Ser Gln Val Lys Gly Gly
      660              665              670

ggg gac aag ccc ctg gct gat ggg gaa gtt ccc ggg gat gct aca aaa      2064
Gly Asp Lys Pro Leu Ala Asp Gly Glu Val Pro Gly Asp Ala Thr Lys
      675              680              685

aca gag gac aaa caa cag tga      2085
Thr Glu Asp Lys Gln Gln
      690

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<210> SEQ ID NO 14
<211> LENGTH: 694
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

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<400> SEQUENCE: 14

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Met Ala Lys Ile Asn Thr Gln Tyr Ser His Pro Ser Arg Thr His Leu
1              5              10              15

Lys Val Lys Thr Ser Asp Arg Asp Leu Asn Arg Ala Glu Asn Gly Leu
      20              25              30

Ser Arg Ala His Ser Ser Ser Glu Glu Thr Ser Ser Val Leu Gln Pro
      35              40              45

Gly Ile Ala Met Glu Thr Arg Gly Leu Ala Asp Ser Gly Gln Gly Ser
      50              55              60

Phe Thr Gly Gln Gly Ile Ala Arg Leu Ser Arg Leu Ile Phe Leu Leu
      65              70              75              80

Arg Arg Trp Ala Ala Arg His Val His His Gln Asp Gln Gly Pro Asp
      85              90              95

Ser Phe Pro Asp Arg Phe Arg Gly Ala Glu Leu Lys Glu Val Ser Ser
      100             105             110

Gln Glu Ser Asn Ala Gln Ala Asn Val Gly Ser Gln Glu Pro Ala Asp
      115             120             125

Arg Gly Arg Ser Ala Trp Pro Leu Ala Lys Cys Asn Thr Asn Thr Ser
      130             135             140

Asn Asn Thr Glu Glu Glu Lys Lys Thr Lys Lys Lys Asp Ala Ile Val
      145             150             155             160

Val Asp Pro Ser Ser Asn Leu Tyr Tyr Arg Trp Leu Thr Ala Ile Ala
      165             170             175

Leu Pro Val Phe Tyr Asn Trp Tyr Leu Leu Ile Cys Arg Ala Cys Phe
      180             185             190

Asp Glu Leu Gln Ser Glu Tyr Leu Met Leu Trp Leu Val Leu Asp Tyr
      195             200             205

Ser Ala Asp Val Leu Tyr Val Leu Asp Val Leu Val Arg Ala Arg Thr
      210             215             220

Gly Phe Leu Glu Gln Gly Leu Met Val Ser Asp Thr Asn Arg Leu Trp
      225             230             235             240

Gln His Tyr Lys Thr Thr Thr Gln Phe Lys Leu Asp Val Leu Ser Leu
      245             250             255

Val Pro Thr Asp Leu Ala Tyr Leu Lys Val Gly Thr Asn Tyr Pro Glu
      260             265             270

Val Arg Phe Asn Arg Leu Leu Lys Phe Ser Arg Leu Phe Glu Phe Phe
      275             280             285

Asp Arg Thr Glu Thr Arg Thr Asn Tyr Pro Asn Met Phe Arg Ile Gly
      290             295             300

Asn Leu Val Leu Tyr Ile Leu Ile Ile Ile His Trp Asn Ala Cys Ile

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305	310	315	320
Tyr Phe Ala Ile Ser Lys Phe Ile Gly Phe Gly Thr Asp Ser Trp Val			
	325	330	335
Tyr Pro Asn Ile Ser Ile Pro Glu His Gly Arg Leu Ser Arg Lys Tyr			
	340	345	350
Ile Tyr Ser Leu Tyr Trp Ser Thr Leu Thr Leu Thr Thr Ile Gly Glu			
	355	360	365
Thr Pro Pro Pro Val Lys Asp Glu Glu Tyr Leu Phe Val Val Val Asp			
	370	375	380
Phe Leu Val Gly Val Leu Ile Phe Ala Thr Ile Val Gly Asn Val Gly			
385	390	395	400
Ser Met Ile Ser Asn Met Asn Ala Ser Arg Ala Glu Phe Gln Ala Lys			
	405	410	415
Ile Asp Ser Ile Lys Gln Tyr Met Gln Phe Arg Lys Val Thr Lys Asp			
	420	425	430
Leu Glu Thr Arg Val Ile Arg Trp Phe Asp Tyr Leu Trp Ala Asn Lys			
	435	440	445
Lys Thr Val Asp Glu Lys Glu Val Leu Lys Ser Leu Pro Asp Lys Leu			
	450	455	460
Lys Ala Glu Ile Ala Ile Asn Val His Leu Asp Thr Leu Lys Lys Val			
465	470	475	480
Arg Ile Phe Gln Asp Cys Glu Ala Gly Leu Leu Val Glu Leu Val Leu			
	485	490	495
Lys Leu Arg Pro Thr Val Phe Ser Pro Gly Asp Tyr Ile Cys Lys Lys			
	500	505	510
Gly Asp Ile Gly Lys Glu Met Tyr Ile Ile Asn Glu Gly Lys Leu Ala			
	515	520	525
Val Val Ala Asp Asp Gly Val Thr Gln Phe Val Val Leu Ser Asp Gly			
	530	535	540
Ser Tyr Phe Gly Glu Ile Ser Ile Leu Asn Ile Lys Gly Ser Lys Ser			
545	550	555	560
Gly Asn Arg Arg Thr Ala Asn Ile Arg Ser Ile Gly Tyr Ser Asp Leu			
	565	570	575
Phe Cys Leu Ser Lys Asp Asp Leu Met Glu Ala Leu Thr Glu Tyr Pro			
	580	585	590
Glu Ala Lys Lys Ala Leu Glu Glu Lys Gly Arg Gln Ile Leu Met Lys			
	595	600	605
Asp Asn Leu Ile Asp Glu Glu Leu Ala Arg Ala Gly Ala Asp Pro Lys			
	610	615	620
Asp Leu Glu Glu Lys Val Glu Gln Leu Gly Ser Ser Leu Asp Thr Leu			
625	630	635	640
Gln Thr Arg Phe Ala Arg Leu Leu Ala Glu Tyr Asn Ala Thr Gln Met			
	645	650	655
Lys Met Lys Gln Arg Leu Ser Gln Leu Glu Ser Gln Val Lys Gly Gly			
	660	665	670
Gly Asp Lys Pro Leu Ala Asp Gly Glu Val Pro Gly Asp Ala Thr Lys			
	675	680	685
Thr Glu Asp Lys Gln Gln			
690			

<210> SEQ ID NO 15

<211> LENGTH: 2085

<212> TYPE: DNA

<213> ORGANISM: Homo sapiens

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<400> SEQUENCE: 15

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tcagaccgag atctcaatcg cgctgaaaat ggccctcagca gagcccactc gtcaagtga    120
gagacatcgt cagtgtctga gccggggatc gccatggaga ccagaggact ggctgactcc    180
gggcaggggt ccttcaccgg ccaggggatc gccaggtgt cgcgccctcat cttcttgctg    240
cgcaggtggg ctgccaggca tgtgcaccac caggaccagg gaccggactc ttttcctgat    300
cgtttcctg gagcggagct taaggaggtg tccagccaag aaagcaatgc ccaggcaaat    360
gtgggcagcc aggagccagc agacagaggg agaagcgct ggcccctggc caaatgcaac    420
actaacacca gcaacaacac ggaggaggag aagaagacga aaaagaagga tgcgatcgtg    480
gtggaccctg ccagcaacct gtactaccgc tggctgaccg ccacgcacct gcctgtcttc    540
tataactggt atctgcttat ttgcagggcc tgtttcgatg agctgcagtc cgagtacctg    600
atgtgtggc tggctcctga ctactcggca gatgtcctgt atgtcttga tgtgcttgta    660
cgagctcgga caggttttct cgagcaaggc ttaatggta gtgataccaa caggctgtgg    720
cagcattaca agacgaccac gcagttcaag ctggatgtgt tgtccctggt cccaccgac    780
ctggcttact taaaggtggg cacaaactac ccagaagtga gggtcaaccg cctactgaag    840
ttttccggc tctttgaatt ctttgaccgc acagagacaa ggaccaacta ccccaatatg    900
ttcaggattg ggaacttggc cttgtacatt ctcacatca tccactggaa tgcctgcac    960
tactttgcca tttccaagtt catttggttt gggacagact cctgggtcta cccaaacatc   1020
tcaatcccag agcatggggc cctctccagg aagtacattt acagtctcta ctggtccacc   1080
ttgacctta ccaccattgg tgagaccca cccccgtga aagatgagga gtatctcttt   1140
gtggctcgtg actcttggg ggggtgtctg atttttgcca ccattgtggg caatgtgggc   1200
tccatgatct cgaatatgaa tgccctcagg gcagagttcc aggccaaagt tgattccac   1260
aagcagtaca tgcagttccg caaggtcacc aaggacttgg agacgcgggt tatccggtgg   1320
tttgactacc tgtgggcca caagaagacg gtggatgaga aggaggtgct caagagcctc   1380
ccagacaagc tgaaggctga gatcgccac aacgtgcacc tggacacgct gaagaaggtt   1440
cgcatcttcc aggaactgtg ggcagggtg ctggtggagc tgggtgctgaa gctgcgaccc   1500
actgtgttca gccctgggga ttatatctgc aagaaggag atattgggaa ggagatgtac   1560
atcatcaacg agggcaagct ggcctggtg gctgatgatg gggtcaccca gtctgtggtc   1620
ctcagcgatg gcagctactt cggggagatc agcattctga acatcaaggg gagcaagtcg   1680
gggaaccgca ggacggccaa catccgcagc attggctact cagacctgtt ctgcctctca   1740
aaggacgac tcattggaggc cctcaccgag taccocgaag ccaagaaggc cctggaggag   1800
aaaggacggc agatcctgat gaaagacaac ctgatcgatg aggagctggc cagggcgggc   1860
gcggaaccca aggacctga ggagaaagt gagcagctgg ggtcctccct ggacaccctg   1920
cagaccaggt ttgcacgct cctggctgag tacaacgcca ccagatgaa gatgaagcag   1980
cgtctcagcc aactggaaag ccaggtgaag ggtggtgggg acaagccct ggctgatggg   2040
gaagttcccg gggatgctac aaaaacagag gacaaacaac agtga                    2085

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<210> SEQ ID NO 16

<211> LENGTH: 2107

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

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<223> OTHER INFORMATION: constructed sequence

<400> SEQUENCE: 16

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gcggcgcgcca ccatggctaa gattaacacc cagtactcac atccatcccg cactcacctc      60
aaagtcaaga cctccgatcg ggatctgaac cgggctgaga atgggctgtc gcgcgcccac      120
tcgtcgtccg aggaaccag cagcgtgctc cagccgggca tcgccatgga aactaggggg      180
ctggcggact cggacaggg atccttcaact ggacagggtg ttgccgggt gagcagactg      240
atcttctctg ttcgccgctg ggcggccaga cacgtgcacc atcaggacca gggacctgat      300
agcttccccc accgctttag gggagccgag ctgaaagaag tgtcaagcca ggagtcaaac      360
gcgcaggcca acgtcggcag ccaagagcct gcagaccggg gacgctcggc atggcggctc      420
gcaaagtgca acactaacac ttccaacaac accgaagagg aaaagaaaac caagaagaag      480
gatgcaattg tggtggaccc ttctccaac ctgtactacc gctggttgac cgccatcgcc      540
ctcccggtct ttacaattg gtatctcctt atctgccggg cctgcttcga cgaactgcaa      600
tcagagtacc tgatgctgtg gctggtgctg gactatagcg ccgatgtgct ctacgtctg      660
gatgtgctcg tgcgcgccc gaccggattc ttggaacaag gcctgatggt gtccgacacg      720
aatagactgt ggcagcacta taagaccaca acccagttca agcttgacgt gctcagcctt      780
gtgcgactg acctggccta cctgaaagtc ggaactaact acccggaagt cagattcaac      840
cgactcctga agttcagcag gctgttcgag ttctttgacc gcaccgagac tcggaccaac      900
taccctaaca tgttcgggat cggaaatctg gtgtctctaca tactgattat catccattgg      960
aacgcctgta tctatttcgc catttcgaag ttcacggtt tcggaaccga ttctgggtg      1020
taccccaaca tctcgatccc cgaacacggg cgcctgtccc ggaagtacat ctactccctg      1080
tactggtcca ctctgactct gaccacgac ggggaaaccc ctccaccgt gaaggacgaa      1140
gagtaacctg tcgtggtggt ggacttctg gtcggagtgt tgattttcgc caccattgtg      1200
ggaaacgtgg gctccatgat ctccaacatg aacgcgtcga gagctgagtt ccaagccaag      1260
atcgactcca ttaagcagta catgcagttc agaaaggcca ccaaggacct ggaaaccagg      1320
gtcatccgct ggttcgacta cctgtgggcc aacaaaaaga ctgtggacga aaaggaagtg      1380
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ctgaagaaaag tccgatctt ccaagactgt gaagccggcc tccgtgtgga gctcgtgctc      1500
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cagttcgtgg tgtgtccga cggatcctac ttcggtgaaa tctcaatcct caacatcaag      1680
gggtccaagt ccgcaaccg gagaactgcc aacattcgct ccatcgata cagcgacctg      1740
ttttgcctgt ccaaggatga cctgatggag gctctgactg agtaccctga agcgaagaag      1800
gctttggagg aaaaggggcg gcagattctg atgaaggaca atttgatcga cgaggagctc      1860
gcacgggccc gcgcgaccc caaggatctc gaagagaagg tcgaacagct gggttcttcg      1920
cttgataccc tgcaaacccg attcgcgcgg ctgctcgccg agtacaacgc gaccagatg      1980
aagatgaagc agagactgtc acagttgaa tcccaagtca agggcggagg cgacaagccg      2040
ctggcggacg gggaaagtgc cggggacgcc accaagactg aggacaagca gcagtgatca      2100
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<210> SEQ ID NO 17

<211> LENGTH: 2272

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<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: constructed sequence

<400> SEQUENCE: 17

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aaagtcaaga cctccgatcg ggatctgaac cgggctgaga atgggctgtc gcgcgcccac      120
tcgtcgtccg aggaaccag cagcgtgtc cagccgggca tcgccatgga aactaggggg      180
ctggcggact cgggacagg atccttcact ggacagggtg ttgcccggtt cgggcggatt      240
cagaagaagt ccagccgga gaaggctgtg cgggctgcca gcaggggcag gccactcatt      300
ggttgacac agtggtgcgc tgaggatggt ggagatgaat cggaatggc actggccggc      360
tctccggat gcagctcggg cccccaagg agactgagca gactgatctt cctgcttcgc      420
cgctgggcgg ccagacacgt gcaccatcag gaccagggac ctgatagctt ccccgaccgc      480
tttaggggag ccgagctgaa agaagtgtca agccaggagt caaacgcgca ggccaacgtc      540
ggcagccaag agcctgcaga cgggggacgc tcggcatggc cgctcgcaa gtgcaacact      600
aacacttcca acaacaccga agaggaaaag aaaaccaaga agaaggatgc aattgtggtg      660
gacccttctt ccaacctgta ctaccgctgg ttgaccgcca tcgcctccc ggtcttttac      720
aattggtatc tccttatctg cggggcctgc ttgcacgaac tgcaatcaga gtacctgatg      780
ctgtggctgg tgctggacta tagcgccgat gtgctctacg tcctggatgt gctcgtgcgc      840
gcccggaccg gattcttggg acaaggcctg atgggtgccg acacgaatag actgtggcag      900
cactataaga ccacaaccca gttcaagctt gacgtgtcga gccttgtgcc gactgacctg      960
gcctacctga aagtcggaac taactaccg gaagtcagat tcaaccgact cctgaagttc     1020
agcaggctgt tcgagttctt tgaccgcacc gagactcgga ccaactacc taacatgttc     1080
cggatcggaa atctggtgct ctacatactg attatcatcc attggaacgc ctgtatctat     1140
ttcgccattt cgaagtctat cggtttcgga accgattcct ggggtgaccc caacatctcg     1200
atcccgaac acggtcgctt gtcccggaag tacatctact ccctgtactg gtccactctg     1260
actctgacca cgatcgggga aacccctcca cccgtgaagg acgaagagta cctgttcgtg     1320
gtggtggact tcctggctcg agtggtgatt ttgccacca ttgtgggaaa cgtgggctcc     1380
atgatctcca acatgaacgc gtcgagagct gagttccaag ccaagatcga ctccattaag     1440
cagtacatgc agttcagaaa ggtcaccaag gacctggaaa ccagggtcat ccgctggttc     1500
gactacctgt gggccaacaa aaagactgtg gacgaaaagg aagtgctgaa gtcgctgccg     1560
gataagctga aggcggaat cgccattaac gtgcaccttg acaccctgaa gaaagtccgg     1620
atcttccaag actgtgaagc cggcctcctg gtggagctcg tgctcaagct gcggcccacc     1680
gtgttcagcc cgggagatta catttgcaag aaggcgata tcggcaaaga gatgtacatc     1740
atcaacgagg gaaagctggc cgtggctgcg gacgacggcg tgaccagtt cgtggtgctg     1800
tccgacggat cctacttcgg tgaatatcga atcctcaaca tcaaggggtc caagtccggc     1860
aaccggagaa ctgccaacat tcgctccatc ggatacagcg acctgttttg cctgtccaag     1920
gatgacctga tggaggctct gactgagtac cctgaagcga agaaggcttt ggaggaaaag     1980
gggcggcaga ttctgatgaa ggacaatttg atcgacgagg agctcgcacg ggccggcgcc     2040
gaccccaagg atctogaaga gaaggctgaa cagctggggtt ctctgcttga taccctgcaa     2100
acccgattcg cgcggtgct cgccgagtag aacgcgaccc agatgaagat gaagcagaga     2160

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ctgtcacagt	tggaatccca	agtcaagggc	ggagggcaca	agccgctggc	ggacggggaa	2220
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<210> SEQ ID NO 18
 <211> LENGTH: 2107
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: constructed sequence

<400> SEQUENCE: 18

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tcgtcaagtg	aggagacatc	gtcagtgtg	cagccgggga	tcgccatgga	gaccagagga	180
ctggctgact	cggggcaggg	ctccttcacc	ggccagggga	tcgccaggct	gtcgcgctc	240
atcttcttgc	tgcgcaggtg	ggctgccagg	catgtgcacc	accaggacca	gggaccggac	300
tcttttctg	atcgtttccg	tggagccgag	cttaaggagg	tgtccagcca	agaaagcaat	360
gcccaggcaa	atgtgggcag	ccaggagcca	gcagacagag	ggagaagcgc	ctggcccctg	420
gccc aaatgca	acactaacac	cagcaacaac	acggaggagg	agaagaagac	gaaaaagaag	480
gatgcatcg	tggtggaccc	gtccagcaac	ctgtactacc	gctggctgac	cgccatcgcc	540
ctgctgtct	tctataactg	gtatctgctt	atctgcaggg	cctgtttcga	tgagctgcag	600
tccgagtacc	tgatgtgtg	gctggctctg	gactactcgg	cagatgtcct	gtatgtcttg	660
gatgtgcttg	tacgagctcg	gacaggtttt	cttgagcaag	gcttaatggg	cagtgatacc	720
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cgctactga	agttttcccg	gctctttgaa	ttctttgacc	gcacagagac	aaggaccaac	900
tacccaata	tgttcaggat	tgggaacttg	gtctgtgaca	ttctcatcat	catccactgg	960
aatgcctgca	tctactttgc	catttccaag	ttcattgggt	ttgggacaga	ctcctgggtc	1020
taccca aaca	tctcaatccc	agagcatggg	cgctctccca	ggaagtacat	ttacagtctc	1080
tactgggtcca	ccttgaccct	taccaccatt	ggtgagaccc	cacccccctg	gaaagatgag	1140
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ggcaatgtgg	gctccatgat	ctcgaatatg	aatgcctcac	gggcagagtt	ccaggccaag	1260
attgattcca	tcaagcagta	catgcagttc	cgcaaggcca	ccaaggactt	ggagacgcgg	1320
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aaggagatgt	acatcatcaa	cgagggcaag	ctggccgtgg	tggtgatga	tggggtcacc	1620
cagttcgtgg	tcctcagcga	tggcagctac	ttcggggaga	tcagcattct	gaacatcaag	1680
gggagcaagt	cggggaaccg	caggacggcc	aacatccgca	gcattggcta	ctcagacctg	1740
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gccctggagg	agaaaggacg	gcagatcctg	atgaaagaca	acctgatcga	tgaggagctg	1860
gccagggcgg	gcgcggaccc	caaggacctt	gaggagaaa	tgagcagct	ggggtcctcc	1920
ctggacaccc	tcagaccag	gtttgcacgc	ctcctggctg	agtacaacgc	caccagatg	1980

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aagatgaagc agcgtctcag ccaactggaa agccaggtga aggggtggtg ggacaagccc 2040
ctggctgatg gggaagttcc cggggatgct acaaaaacag aggacaaaca acagtgatca 2100
tagatct 2107

<210> SEQ ID NO 19
<211> LENGTH: 2430
<212> TYPE: DNA
<213> ORGANISM: Homo sapiens
<220> FEATURE:
<221> NAME/KEY: CDS
<222> LOCATION: (1) .. (2430)

<400> SEQUENCE: 19

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Met Phe Lys Ser Leu Thr Lys Val Asn Lys Val Lys Pro Ile Gly Glu
1 5 10 15

aac aat gag aat gaa caa agt tct cgt cgg aat gaa gaa ggc tct cac 96
Asn Asn Glu Asn Glu Gln Ser Ser Arg Arg Asn Glu Glu Gly Ser His
20 25 30

cca agt aat cag tct cag caa acc aca gca cag gaa gaa aac aaa ggt 144
Pro Ser Asn Gln Ser Gln Gln Thr Thr Ala Gln Glu Glu Asn Lys Gly
35 40 45

gaa gag aaa tct ctc aaa acc aag tca act cca gtc acg tct gaa gag 192
Glu Glu Lys Ser Leu Lys Thr Lys Ser Thr Pro Val Thr Ser Glu Glu
50 55 60

cca cac acc aac ata caa gac aaa ctc tcc aag aaa aat tcc tct gga 240
Pro His Thr Asn Ile Gln Asp Lys Leu Ser Lys Lys Asn Ser Ser Gly
65 70 75 80

gat ctg acc aca aac cct gac cct caa aat gca gca gaa cca act gga 288
Asp Leu Thr Thr Asn Pro Asp Pro Gln Asn Ala Ala Glu Pro Thr Gly
85 90 95

aca gtg cca gag cag aag gaa atg gac ccc ggg aaa gaa ggt cca aac 336
Thr Val Pro Glu Gln Lys Glu Met Asp Pro Gly Lys Glu Gly Pro Asn
100 105 110

agc cca caa aac aaa ccg cct gca gct cct gtt ata aat gag tat gcc 384
Ser Pro Gln Asn Lys Pro Pro Ala Ala Pro Val Ile Asn Glu Tyr Ala
115 120 125

gat gcc cag cta cac aac ctg gtg aaa aga atg cgt caa aga aca gcc 432
Asp Ala Gln Leu His Asn Leu Val Lys Arg Met Arg Gln Arg Thr Ala
130 135 140

ctc tac aag aaa aag ttg gta gag gga gat ctc tcc tca ccc gaa gcc 480
Leu Tyr Lys Lys Lys Leu Val Glu Gly Asp Leu Ser Ser Pro Glu Ala
145 150 155 160

agc cca caa act gca aag ccc acg gct gta cca cca gta aaa gaa agc 528
Ser Pro Gln Thr Ala Lys Pro Thr Ala Val Pro Pro Val Lys Glu Ser
165 170 175

gat gat aag cca aca gaa cat tac tac agg ctg ttg tgg ttc aaa gtc 576
Asp Asp Lys Pro Thr Glu His Tyr Arg Leu Leu Trp Phe Lys Val
180 185 190

aaa aag atg cct tta aca gag tac tta aag cga att aaa ctt cca aac 624
Lys Lys Met Pro Leu Thr Glu Tyr Leu Lys Arg Ile Lys Leu Pro Asn
195 200 205

agc ata gat tca tac aca gat cga ctc tat ctc ctg tgg ctc ttg ctt 672
Ser Ile Asp Ser Tyr Thr Asp Arg Leu Tyr Leu Leu Trp Leu Leu Leu
210 215 220

gtc act ctt gcc tat aac tgg aac tgc tgt ttt ata cca ctg cgc ctc 720
Val Thr Leu Ala Tyr Asn Trp Asn Cys Cys Phe Ile Pro Leu Arg Leu
225 230 235 240

gtc ttc cca tat caa acc gca gac aac ata cac tac tgg ctt att gcg 768

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Val	Phe	Pro	Tyr	Gln	Thr	Ala	Asp	Asn	Ile	His	Tyr	Trp	Leu	Ile	Ala	
				245					250					255		
gac	atc	ata	tgt	gat	atc	atc	tac	ctt	tat	gat	atg	cta	ttt	atc	cag	816
Asp	Ile	Ile	Cys	Asp	Ile	Ile	Tyr	Leu	Tyr	Asp	Met	Leu	Phe	Ile	Gln	
			260					265					270			
ccc	aga	ctc	cag	ttt	gta	aga	gga	gga	gac	ata	ata	gtg	gat	tca	aat	864
Pro	Arg	Leu	Gln	Phe	Val	Arg	Gly	Gly	Asp	Ile	Ile	Val	Asp	Ser	Asn	
			275				280					285				
gag	cta	agg	aaa	cac	tac	agg	act	tct	aca	aaa	ttt	cag	ttg	gat	gtc	912
Glu	Leu	Arg	Lys	His	Tyr	Arg	Thr	Ser	Thr	Lys	Phe	Gln	Leu	Asp	Val	
	290					295				300						
gca	tca	ata	ata	cca	ttt	gat	att	tgc	tac	ctc	ttc	ttt	ggg	ttt	aat	960
Ala	Ser	Ile	Ile	Pro	Phe	Asp	Ile	Cys	Tyr	Leu	Phe	Phe	Gly	Phe	Asn	
	305				310					315					320	
cca	atg	ttt	aga	gca	aat	agg	atg	tta	aag	tac	act	tca	ttt	ttt	gaa	1008
Pro	Met	Phe	Arg	Ala	Asn	Arg	Met	Leu	Lys	Tyr	Thr	Ser	Phe	Phe	Glu	
				325				330						335		
ttt	aat	cat	cac	cta	gag	tct	ata	atg	gac	aaa	gca	tat	atc	tac	aga	1056
Phe	Asn	His	His	Leu	Glu	Ser	Ile	Met	Asp	Lys	Ala	Tyr	Ile	Tyr	Arg	
			340				345						350			
gtt	att	cga	aca	act	gga	tac	ttg	ctg	ttt	att	ctg	cac	att	aat	gcc	1104
Val	Ile	Arg	Thr	Thr	Gly	Tyr	Leu	Leu	Phe	Ile	Leu	His	Ile	Asn	Ala	
		355				360						365				
tgt	gtt	tat	tac	tgg	gct	tca	aac	tat	gaa	gga	att	ggc	act	act	aga	1152
Cys	Val	Tyr	Tyr	Trp	Ala	Ser	Asn	Tyr	Glu	Gly	Ile	Gly	Thr	Thr	Arg	
	370				375					380						
tgg	gtg	tat	gat	ggg	gaa	gga	aac	gag	tat	ctg	aga	tgt	tat	tat	tgg	1200
Trp	Val	Tyr	Asp	Gly	Glu	Gly	Asn	Glu	Tyr	Leu	Arg	Cys	Tyr	Tyr	Trp	
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gca	gtt	cga	act	tta	att	acc	att	ggg	ggc	ctt	cca	gaa	cca	caa	act	1248
Ala	Val	Arg	Thr	Leu	Ile	Thr	Ile	Gly	Gly	Leu	Pro	Glu	Pro	Gln	Thr	
			405					410						415		
tta	ttt	gaa	att	gtt	ttt	caa	ctc	ttg	aat	ttt	ttt	tct	gga	gtt	ttt	1296
Leu	Phe	Glu	Ile	Val	Phe	Gln	Leu	Leu	Asn	Phe	Phe	Ser	Gly	Val	Phe	
		420					425						430			
gtg	ttc	tcc	agt	tta	att	ggg	cag	atg	aga	gat	gtg	att	gga	gca	gct	1344
Val	Phe	Ser	Ser	Leu	Ile	Gly	Gln	Met	Arg	Asp	Val	Ile	Gly	Ala	Ala	
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aca	gcc	aat	cag	aac	tac	ttc	cgc	gcc	tgc	atg	gat	gac	acc	att	gcc	1392
Thr	Ala	Asn	Gln	Asn	Tyr	Phe	Arg	Ala	Cys	Met	Asp	Asp	Thr	Ile	Ala	
	450					455					460					
tac	atg	aac	aat	tac	tcc	att	cct	aaa	ctt	gtg	caa	aag	cga	gtt	cgg	1440
Tyr	Met	Asn	Asn	Tyr	Ser	Ile	Pro	Lys	Leu	Val	Gln	Lys	Arg	Val	Arg	
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act	tgg	tat	gaa	tat	aca	tgg	gac	tct	caa	aga	atg	cta	gat	gag	tct	1488
Thr	Trp	Tyr	Glu	Tyr	Thr	Trp	Asp	Ser	Gln	Arg	Met	Leu	Asp	Glu	Ser	
			485				490							495		
gat	ttg	ctt	aag	acc	cta	cca	act	acg	gtc	cag	tta	gcc	ctc	gcc	att	1536
Asp	Leu	Leu	Lys	Thr	Leu	Pro	Thr	Thr	Val	Gln	Leu	Ala	Leu	Ala	Ile	
			500				505					510				
gat	gtg	aac	ttc	agc	atc	atc	agc	aaa	gtc	gac	ttg	ttc	aag	ggg	tgt	1584
Asp	Val	Asn	Phe	Ser	Ile	Ile	Ser	Lys	Val	Asp	Leu	Phe	Lys	Gly	Cys	
			515				520					525				
gat	aca	cag	atg	att	tat	gac	atg	ttg	cta	aga	ttg	aaa	tcc	gtt	ctc	1632
Asp	Thr	Gln	Met	Ile	Tyr	Asp	Met	Leu	Leu	Arg	Leu	Lys	Ser	Val	Leu	
	530					535				540						
tat	ttg	cct	ggg	gac	ttt	gtc	tgc	aaa	aag	gga	gaa	att	ggc	aag	gaa	1680
Tyr	Leu	Pro	Gly	Asp	Phe	Val	Cys	Lys	Lys	Gly	Glu	Ile	Gly	Lys	Glu	
	545					550				555					560	

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atg tat atc atc aag cat gga gaa gtc caa gtt ctt gga ggc cct gat	1728
Met Tyr Ile Ile Lys His Gly Glu Val Gln Val Leu Gly Gly Pro Asp	
565 570 575	
ggg act aaa gtt ctg gtt act ctg aaa gct ggg tcg gtg ttt gga gaa	1776
Gly Thr Lys Val Leu Val Thr Leu Lys Ala Gly Ser Val Phe Gly Glu	
580 585 590	
atc agc ctt cta gca gca gga gga gga aac cgt cga act gcc aat gtg	1824
Ile Ser Leu Leu Ala Ala Gly Gly Asn Arg Arg Thr Ala Asn Val	
595 600 605	
gtg gcc cac ggg ttt gcc aat ctt tta act cta gac aaa aag acc ctc	1872
Val Ala His Gly Phe Ala Asn Leu Leu Thr Leu Asp Lys Lys Thr Leu	
610 615 620	
caa gaa att cta gtg cat tat cca gat tct gaa agg atc ctc atg aag	1920
Gln Glu Ile Leu Val His Tyr Pro Asp Ser Glu Arg Ile Leu Met Lys	
625 630 635 640	
aaa gcc aga gtg ctt tta aag cag aag gct aag acc gca gaa gca acc	1968
Lys Ala Arg Val Leu Lys Gln Lys Ala Lys Thr Ala Glu Ala Thr	
645 650 655	
cct cca aga aaa gat ctt gcc ctc ctc ttc cca ccg aaa gaa gag aca	2016
Pro Pro Arg Lys Asp Leu Ala Leu Leu Phe Pro Pro Lys Glu Glu Thr	
660 665 670	
ccc aaa ctg ttt aaa act ctc cta gga ggc aca gga aaa gca agt ctt	2064
Pro Lys Leu Phe Lys Thr Leu Leu Gly Gly Thr Gly Lys Ala Ser Leu	
675 680 685	
gca aga cta ctc aaa ttg aag cga gag caa gca gct cag aag aaa gaa	2112
Ala Arg Leu Leu Lys Leu Lys Arg Glu Gln Ala Ala Gln Lys Lys Glu	
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aat tct gaa gga gga gag gaa gaa gga aaa gaa aat gaa gat aaa caa	2160
Asn Ser Glu Gly Gly Glu Glu Glu Gly Lys Glu Asn Glu Asp Lys Gln	
705 710 715 720	
aaa gaa aat gaa gat aaa caa aaa gaa aat gaa gat aaa gga aaa gaa	2208
Lys Glu Asn Glu Asp Lys Gln Lys Glu Asn Glu Asp Lys Gly Lys Glu	
725 730 735	
aat gaa gat aaa gat aaa gga aga gag cca gaa gag aag cca ctg gac	2256
Asn Glu Asp Lys Asp Lys Gly Arg Glu Pro Glu Glu Lys Pro Leu Asp	
740 745 750	
aga cct gaa tgt aca gca agt cct att gca gtg gag gaa gaa ccc cac	2304
Arg Pro Glu Cys Thr Ala Ser Pro Ile Ala Val Glu Glu Glu Pro His	
755 760 765	
tca gtt aga agg aca gtt tta ccc aga ggg act tct cgt caa tca ctc	2352
Ser Val Arg Arg Thr Val Leu Pro Arg Gly Thr Ser Arg Gln Ser Leu	
770 775 780	
att atc agc atg gct cct tct gct gag ggc gga gaa gag gtt ctt act	2400
Ile Ile Ser Met Ala Pro Ser Ala Glu Gly Gly Glu Glu Val Leu Thr	
785 790 795 800	
att gaa gtc aaa gaa aag gct aag caa taa	2430
Ile Glu Val Lys Glu Lys Ala Lys Gln	
805	

<210> SEQ ID NO 20

<211> LENGTH: 809

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 20

Met Phe Lys Ser Leu Thr Lys Val Asn Lys Val Lys Pro Ile Gly Glu
1 5 10 15

Asn Asn Glu Asn Glu Gln Ser Ser Arg Arg Asn Glu Glu Gly Ser His
20 25 30

Pro Ser Asn Gln Ser Gln Gln Thr Thr Ala Gln Glu Glu Asn Lys Gly

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35					40					45					
Glu 50	Glu	Lys	Ser	Leu	Lys 55	Thr	Lys	Ser	Thr	Pro 60	Val	Thr	Ser	Glu	Glu
Pro 65	His	Thr	Asn	Ile 70	Gln	Asp	Lys	Leu	Ser 75	Lys	Asn	Ser	Ser	Ser	Gly 80
Asp	Leu	Thr	Thr 85	Asn	Pro	Asp	Pro	Gln 90	Asn	Ala	Ala	Glu	Pro 95	Thr	Gly
Thr	Val	Pro	Glu 100	Gln	Lys	Glu	Met	Asp 105	Pro	Gly	Lys	Glu 110	Gly	Pro	Asn
Ser	Pro	Gln	Asn 115	Lys	Pro	Pro	Ala 120	Ala	Pro	Val	Ile	Asn 125	Glu	Tyr	Ala
Asp	Ala	Gln	Leu 130	His	Asn	Leu 135	Val	Lys	Arg	Met	Arg 140	Gln	Arg	Thr	Ala
Leu 145	Tyr	Lys	Lys	Lys 150	Leu	Val	Glu	Gly	Asp 155	Leu	Ser	Ser	Pro	Glu	Ala 160
Ser	Pro	Gln	Thr	Ala 165	Lys	Pro	Thr	Ala	Val 170	Pro	Pro	Val	Lys	Glu 175	Ser
Asp	Asp	Lys	Pro 180	Thr	Glu	His	Tyr	Tyr 185	Arg	Leu	Leu	Trp	Phe 190	Lys	Val
Lys	Lys	Met 195	Pro	Leu	Thr	Glu	Tyr 200	Leu	Lys	Arg	Ile	Lys 205	Leu	Pro	Asn
Ser	Ile	Asp	Ser	Tyr	Thr	Asp 215	Arg	Leu	Tyr	Leu	Leu 220	Trp	Leu	Leu	Leu
Val 225	Thr	Leu	Ala	Tyr	Asn 230	Trp	Asn	Cys	Cys	Phe 235	Ile	Pro	Leu	Arg	Leu 240
Val	Phe	Pro	Tyr	Gln 245	Thr	Ala	Asp	Asn	Ile 250	His	Tyr	Trp	Leu	Ile 255	Ala
Asp	Ile	Ile	Cys 260	Asp	Ile	Ile	Tyr	Leu 265	Tyr	Asp	Met	Leu	Phe 270	Ile	Gln
Pro	Arg	Leu	Gln 275	Phe	Val	Arg	Gly 280	Gly	Asp	Ile	Ile	Val 285	Asp	Ser	Asn
Glu 290	Leu	Arg	Lys	His	Tyr	Arg 295	Thr	Ser	Thr	Lys	Phe 300	Gln	Leu	Asp	Val
Ala 305	Ser	Ile	Ile	Pro	Phe 310	Asp	Ile	Cys	Tyr	Leu	Phe 315	Phe	Gly	Phe	Asn 320
Pro	Met	Phe	Arg	Ala 325	Asn	Arg	Met	Leu	Lys 330	Tyr	Thr	Ser	Phe	Phe 335	Glu
Phe	Asn	His	His 340	Leu	Glu	Ser	Ile	Met	Asp 345	Lys	Ala	Tyr	Ile	Tyr	Arg
Val	Ile	Arg	Thr	Thr	Gly	Tyr	Leu 360	Leu	Phe	Ile	Leu	His 365	Ile	Asn	Ala
Cys 370	Val	Tyr	Tyr	Trp	Ala	Ser 375	Asn	Tyr	Glu	Gly	Ile 380	Gly	Thr	Thr	Arg
Trp 385	Val	Tyr	Asp	Gly	Glu	Gly 390	Asn	Glu	Tyr	Leu	Arg 395	Cys	Tyr	Tyr	Trp 400
Ala	Val	Arg	Thr	Leu 405	Ile	Thr	Ile	Gly	Gly 410	Leu	Pro	Glu	Pro	Gln 415	Thr
Leu	Phe	Glu	Ile 420	Val	Phe	Gln	Leu	Leu 425	Asn	Phe	Phe	Ser	Gly 430	Val	Phe
Val	Phe	Ser	Ser	Leu	Ile	Gly 440	Gln	Met	Arg	Asp	Val 445	Ile	Gly	Ala	Ala
Thr	Ala	Asn	Gln	Asn	Tyr	Phe 455	Arg	Ala	Cys	Met	Asp 460	Asp	Thr	Ile	Ala

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Tyr Met Asn Asn Tyr Ser Ile Pro Lys Leu Val Gln Lys Arg Val Arg
 465 470 475 480
 Thr Trp Tyr Glu Tyr Thr Trp Asp Ser Gln Arg Met Leu Asp Glu Ser
 485 490 495
 Asp Leu Leu Lys Thr Leu Pro Thr Thr Val Gln Leu Ala Leu Ala Ile
 500 505 510
 Asp Val Asn Phe Ser Ile Ile Ser Lys Val Asp Leu Phe Lys Gly Cys
 515 520 525
 Asp Thr Gln Met Ile Tyr Asp Met Leu Leu Arg Leu Lys Ser Val Leu
 530 535 540
 Tyr Leu Pro Gly Asp Phe Val Cys Lys Lys Gly Glu Ile Gly Lys Glu
 545 550 555 560
 Met Tyr Ile Ile Lys His Gly Glu Val Gln Val Leu Gly Gly Pro Asp
 565 570 575
 Gly Thr Lys Val Leu Val Thr Leu Lys Ala Gly Ser Val Phe Gly Glu
 580 585 590
 Ile Ser Leu Leu Ala Ala Gly Gly Gly Asn Arg Arg Thr Ala Asn Val
 595 600 605
 Val Ala His Gly Phe Ala Asn Leu Leu Thr Leu Asp Lys Lys Thr Leu
 610 615 620
 Gln Glu Ile Leu Val His Tyr Pro Asp Ser Glu Arg Ile Leu Met Lys
 625 630 635 640
 Lys Ala Arg Val Leu Leu Lys Gln Lys Ala Lys Thr Ala Glu Ala Thr
 645 650 655
 Pro Pro Arg Lys Asp Leu Ala Leu Leu Phe Pro Pro Lys Glu Glu Thr
 660 665 670
 Pro Lys Leu Phe Lys Thr Leu Leu Gly Gly Thr Gly Lys Ala Ser Leu
 675 680 685
 Ala Arg Leu Leu Lys Leu Lys Arg Glu Gln Ala Ala Gln Lys Lys Glu
 690 695 700
 Asn Ser Glu Gly Gly Glu Glu Glu Gly Lys Glu Asn Glu Asp Lys Gln
 705 710 715 720
 Lys Glu Asn Glu Asp Lys Gln Lys Glu Asn Glu Asp Lys Gly Lys Glu
 725 730 735
 Asn Glu Asp Lys Asp Lys Gly Arg Glu Pro Glu Glu Lys Pro Leu Asp
 740 745 750
 Arg Pro Glu Cys Thr Ala Ser Pro Ile Ala Val Glu Glu Glu Pro His
 755 760 765
 Ser Val Arg Arg Thr Val Leu Pro Arg Gly Thr Ser Arg Gln Ser Leu
 770 775 780
 Ile Ile Ser Met Ala Pro Ser Ala Glu Gly Gly Glu Glu Val Leu Thr
 785 790 795 800
 Ile Glu Val Lys Glu Lys Ala Lys Gln
 805

<210> SEQ ID NO 21
 <211> LENGTH: 2430
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: constructed sequence
 <220> FEATURE:
 <221> NAME/KEY: CDS
 <222> LOCATION: (1)..(2430)
 <400> SEQUENCE: 21

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atg ttt aaa tcg ctg aca aaa gtc aac aag gtg aag cct ata gga gag	48
Met Phe Lys Ser Leu Thr Lys Val Asn Lys Val Lys Pro Ile Gly Glu	
1 5 10 15	
aac aat gag aat gaa caa agt tct cgt cgg aat gaa gaa ggc tct cac	96
Asn Asn Glu Asn Glu Gln Ser Ser Arg Arg Asn Glu Glu Gly Ser His	
20 25 30	
cca agt aat cag tct cag caa acc aca gca cag gaa gaa aac aaa ggt	144
Pro Ser Asn Gln Ser Gln Gln Thr Thr Ala Gln Glu Glu Asn Lys Gly	
35 40 45	
gaa gag aaa tct ctc aaa acc aag tca act cca gtc acg tct gaa gag	192
Glu Glu Lys Ser Leu Lys Thr Lys Ser Thr Pro Val Thr Ser Glu Glu	
50 55 60	
cca cac acc aac ata caa gac aaa ctc tcc aag aaa aat tcc tct gga	240
Pro His Thr Asn Ile Gln Asp Lys Leu Ser Lys Lys Asn Ser Ser Gly	
65 70 75 80	
gat ctg acc aca aac cct gac cct caa aat gca gca gaa cca act gga	288
Asp Leu Thr Thr Asn Pro Asp Pro Gln Asn Ala Ala Glu Pro Thr Gly	
85 90 95	
aca gtg cca gag cag aag gaa atg gac ccc ggg aaa gaa ggt cca aac	336
Thr Val Pro Glu Gln Lys Glu Met Asp Pro Gly Lys Glu Gly Pro Asn	
100 105 110	
agc cca caa aac aaa ccg cca gca gct cct gtt ata aat gag tat gcc	384
Ser Pro Gln Asn Lys Pro Pro Ala Ala Pro Val Ile Asn Glu Tyr Ala	
115 120 125	
gat gcc cag cta cac aac ctg gtg aaa aga atg cgt caa aga aca gcc	432
Asp Ala Gln Leu His Asn Leu Val Lys Arg Met Arg Gln Arg Thr Ala	
130 135 140	
ctc tac aag aaa aag ttg gta gag gga gat ctc tcc tca ccc gaa gcc	480
Leu Tyr Lys Lys Lys Leu Val Glu Gly Asp Leu Ser Ser Pro Glu Ala	
145 150 155 160	
agc cca caa act gca aag ccc acg gct gta cca cca gta aaa gaa agc	528
Ser Pro Gln Thr Ala Lys Pro Thr Ala Val Pro Pro Val Lys Glu Ser	
165 170 175	
gat gat aag cca aca gaa cat tac tac agg ctg ttg tgg ttc aaa gtc	576
Asp Asp Lys Pro Thr Glu His Tyr Tyr Arg Leu Leu Trp Phe Lys Val	
180 185 190	
aaa aag atg cct tta aca gag tac tta aag cga att aaa ctt cca aac	624
Lys Lys Met Pro Leu Thr Glu Tyr Leu Lys Arg Ile Lys Leu Pro Asn	
195 200 205	
agc ata gat tca tac aca gat cga ctc tat ctc ctg tgg ctc ttg ctt	672
Ser Ile Asp Ser Tyr Thr Asp Arg Leu Tyr Leu Leu Trp Leu Leu Leu	
210 215 220	
gtc act ctt gcc tat aac tgg aac tgc tgt ttt ata cca ctg cgc ctc	720
Val Thr Leu Ala Tyr Asn Trp Asn Cys Cys Phe Ile Pro Leu Arg Leu	
225 230 235 240	
gtc ttc cca tat caa acc gca gac aac ata cac tac tgg ctt att gcg	768
Val Phe Pro Tyr Gln Thr Ala Asp Asn Ile His Tyr Trp Leu Ile Ala	
245 250 255	
gac atc atc tgt gat atc atc tac ctt tat gat atg cta ttt atc cag	816
Asp Ile Ile Cys Asp Ile Ile Tyr Leu Tyr Asp Met Leu Phe Ile Gln	
260 265 270	
ccc aga ctc cag ttt gta aga gga gga gac ata ata gtg gat tca aat	864
Pro Arg Leu Glu Phe Val Arg Gly Gly Asp Ile Ile Val Asp Ser Asn	
275 280 285	
gag cta agg aaa cac tac agg act tct aca aaa ttt cag ttg gat gtc	912
Glu Leu Arg Lys His Tyr Arg Thr Ser Thr Lys Phe Gln Leu Asp Val	
290 295 300	
gca tca ata ata cca ttt gat att tgc tac ctc ttc ttt ggg ttt aat	960
Ala Ser Ile Ile Pro Phe Asp Ile Cys Tyr Leu Phe Phe Gly Phe Asn	

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305	310	315	320	
cca atg ttt aga gca aat agg atg tta aag tac act tca ttt ttt gaa				1008
Pro Met Phe Arg Ala Asn Arg Met Leu Lys Tyr Thr Ser Phe Phe Glu				
	325	330	335	
ttt aat cat cac cta gag tct ata atg gac aaa gca tat atc tac aga				1056
Phe Asn His His Leu Glu Ser Ile Met Asp Lys Ala Tyr Ile Tyr Arg				
	340	345	350	
gtt att cga aca act gga tac ttg ctg ttt att ctg cac att aat gcc				1104
Val Ile Arg Thr Thr Gly Tyr Leu Leu Phe Ile Leu His Ile Asn Ala				
	355	360	365	
tgt gtt tat tac tgg gct tca aac tat gaa gga att ggc act act aga				1152
Cys Val Tyr Tyr Trp Ala Ser Asn Tyr Glu Gly Ile Gly Thr Thr Arg				
	370	375	380	
tgg gtg tat gat ggg gaa gga aac gag tat ctg aga tgt tat tat tgg				1200
Trp Val Tyr Asp Gly Glu Gly Asn Glu Tyr Leu Arg Cys Tyr Tyr Trp				
	385	390	395	400
gca gtt cga act tta att acc att ggt ggc ctt cca gaa cca caa act				1248
Ala Val Arg Thr Leu Ile Thr Ile Gly Gly Leu Pro Glu Pro Gln Thr				
	405	410	415	
tta ttt gaa att gtt ttt caa ctc ttg aat ttt ttt tct gga gtt ttt				1296
Leu Phe Glu Ile Val Phe Gln Leu Leu Asn Phe Phe Ser Gly Val Phe				
	420	425	430	
gtg ttc tcc agt tta att ggt cag atg aga gat gtg att gga gca gct				1344
Val Phe Ser Ser Leu Ile Gly Gln Met Arg Asp Val Ile Gly Ala Ala				
	435	440	445	
aca gcc aat cag aac tac ttc cgc gcc tgc atg gat gac acc att gcc				1392
Thr Ala Asn Gln Asn Tyr Phe Arg Ala Cys Met Asp Asp Thr Ile Ala				
	450	455	460	
tac atg aac aat tac tcc att cct aaa ctt gtg caa aag cga gtt cgg				1440
Tyr Met Asn Asn Tyr Ser Ile Pro Lys Leu Val Gln Lys Arg Val Arg				
	465	470	475	480
act tgg tat gaa tat aca tgg gac tct caa aga atg cta gat gag tct				1488
Thr Trp Tyr Glu Tyr Thr Trp Asp Ser Gln Arg Met Leu Asp Glu Ser				
	485	490	495	
gat ttg ctt aag acc cta cca act acg gtc cag tta gcc ctc gcc att				1536
Asp Leu Leu Lys Thr Leu Pro Thr Thr Val Gln Leu Ala Leu Ala Ile				
	500	505	510	
gat gtg aac ttc agc atc atc agc aaa gtt gac ttg ttc aag ggt tgt				1584
Asp Val Asn Phe Ser Ile Ile Ser Lys Val Asp Leu Phe Lys Gly Cys				
	515	520	525	
gat aca cag atg att tat gac atg ttg cta aga ttg aaa tcc gtt ctc				1632
Asp Thr Gln Met Ile Tyr Asp Met Leu Leu Arg Leu Lys Ser Val Leu				
	530	535	540	
tat ttg cct ggt gac ttt gtc tgc aaa aag gga gaa att ggc aag gaa				1680
Tyr Leu Pro Gly Asp Phe Val Cys Lys Lys Gly Glu Ile Gly Lys Glu				
	545	550	555	560
atg tat atc atc aag cat gga gaa gtc caa gtt ctt gga ggc cct gat				1728
Met Tyr Ile Ile Lys His Gly Glu Val Gln Val Leu Gly Gly Pro Asp				
	565	570	575	
ggg act aaa gtt ctg gtt act ctg aaa gct ggg tcg gtg ttt gga gaa				1776
Gly Thr Lys Val Leu Val Thr Leu Lys Ala Gly Ser Val Phe Gly Glu				
	580	585	590	
atc agc ctt cta gca gca gga gga gga aac cgt cga act gcc aat gtg				1824
Ile Ser Leu Leu Ala Ala Gly Gly Gly Asn Arg Arg Thr Ala Asn Val				
	595	600	605	
gtg gcc cac ggg ttt gcc aat ctt tta act cta gac aaa aag acc ctc				1872
Val Ala His Gly Phe Ala Asn Leu Leu Thr Leu Asp Lys Lys Thr Leu				
	610	615	620	
caa gaa att cta gtg cat tat cca gat tct gaa aga atc ctc atg aag				1920

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Gln	Glu	Ile	Leu	Val	His	Tyr	Pro	Asp	Ser	Glu	Arg	Ile	Leu	Met	Lys	
625					630					635					640	
aaa	gcc	aga	gtg	ctt	tta	aag	cag	aag	gct	aag	acc	gca	gaa	gca	acc	1968
Lys	Ala	Arg	Val	Leu	Leu	Lys	Gln	Lys	Ala	Lys	Thr	Ala	Glu	Ala	Thr	
			645					650					655			
cct	cca	aga	aaa	gat	ctt	gcc	ctc	ctc	ttc	cca	ccg	aaa	gaa	gag	aca	2016
Pro	Pro	Arg	Lys	Asp	Leu	Ala	Leu	Leu	Phe	Pro	Pro	Lys	Glu	Glu	Thr	
			660				665					670				
ccc	aaa	ctg	ttt	aaa	act	ctc	cta	gga	ggc	aca	gga	aaa	gca	agt	ctt	2064
Pro	Lys	Leu	Phe	Lys	Thr	Leu	Leu	Gly	Gly	Thr	Gly	Lys	Ala	Ser	Leu	
		675				680					685					
gca	aga	cta	ctc	aaa	ttg	aag	cga	gag	caa	gca	gct	cag	aag	aaa	gaa	2112
Ala	Arg	Leu	Leu	Lys	Leu	Lys	Arg	Glu	Gln	Ala	Ala	Gln	Lys	Lys	Glu	
		690			695					700						
aat	tct	gaa	gga	gga	gag	gaa	gaa	gga	aaa	gaa	aat	gaa	gat	aaa	caa	2160
Asn	Ser	Glu	Gly	Gly	Glu	Glu	Glu	Gly	Lys	Glu	Asn	Glu	Asp	Lys	Gln	
705					710				715						720	
aaa	gaa	aat	gaa	gat	aaa	caa	aaa	gaa	aat	gaa	gat	aaa	gga	aaa	gaa	2208
Lys	Glu	Asn	Glu	Asp	Lys	Gln	Lys	Glu	Asn	Glu	Asp	Lys	Gly	Lys	Glu	
			725					730					735			
aat	gaa	gat	aaa	gat	aaa	gga	aga	gag	cca	gaa	gag	aag	cca	ctg	gac	2256
Asn	Glu	Asp	Lys	Asp	Lys	Gly	Arg	Glu	Pro	Glu	Glu	Lys	Pro	Leu	Asp	
			740				745					750				
aga	cct	gaa	tgt	aca	gca	agt	cct	att	gca	gtg	gag	gaa	gaa	ccc	cac	2304
Arg	Pro	Glu	Cys	Thr	Ala	Ser	Pro	Ile	Ala	Val	Glu	Glu	Glu	Pro	His	
		755				760					765					
tca	gtt	aga	agg	aca	gtt	tta	ccc	aga	ggg	act	tct	cgt	caa	tca	ctc	2352
Ser	Val	Arg	Arg	Thr	Val	Leu	Pro	Arg	Gly	Thr	Ser	Arg	Gln	Ser	Leu	
	770				775					780						
att	atc	agc	atg	gct	cct	tct	gct	gag	ggc	gga	gaa	gag	gtt	ctt	act	2400
Ile	Ile	Ser	Met	Ala	Pro	Ser	Ala	Glu	Gly	Gly	Glu	Glu	Val	Leu	Thr	
785				790				795					800			
att	gaa	gtc	aaa	gaa	aag	gct	aag	caa	tga							2430
Ile	Glu	Val	Lys	Glu	Lys	Ala	Lys	Gln								
			805													

<210> SEQ ID NO 22

<211> LENGTH: 809

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 22

Met	Phe	Lys	Ser	Leu	Thr	Lys	Val	Asn	Lys	Val	Lys	Pro	Ile	Gly	Glu	
1				5				10					15			
Asn	Asn	Glu	Asn	Glu	Gln	Ser	Ser	Arg	Arg	Asn	Glu	Glu	Gly	Ser	His	
		20					25					30				
Pro	Ser	Asn	Gln	Ser	Gln	Gln	Thr	Thr	Ala	Gln	Glu	Glu	Asn	Lys	Gly	
		35				40					45					
Glu	Glu	Lys	Ser	Leu	Lys	Thr	Lys	Ser	Thr	Pro	Val	Thr	Ser	Glu	Glu	
	50				55					60						
Pro	His	Thr	Asn	Ile	Gln	Asp	Lys	Leu	Ser	Lys	Lys	Asn	Ser	Ser	Gly	
	65			70					75					80		
Asp	Leu	Thr	Thr	Asn	Pro	Asp	Pro	Gln	Asn	Ala	Ala	Glu	Pro	Thr	Gly	
		85					90						95			
Thr	Val	Pro	Glu	Gln	Lys	Glu	Met	Asp	Pro	Gly	Lys	Glu	Gly	Pro	Asn	
		100					105						110			
Ser	Pro	Gln	Asn	Lys	Pro	Pro	Ala	Ala	Pro	Val	Ile	Asn	Glu	Tyr	Ala	

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115	120	125
Asp Ala Gln Leu His Asn Leu Val Lys Arg Met Arg Gln Arg Thr Ala 130 135 140		
Leu Tyr Lys Lys Lys Leu Val Glu Gly Asp Leu Ser Ser Pro Glu Ala 145 150 155 160		
Ser Pro Gln Thr Ala Lys Pro Thr Ala Val Pro Pro Val Lys Glu Ser 165 170 175		
Asp Asp Lys Pro Thr Glu His Tyr Tyr Arg Leu Leu Trp Phe Lys Val 180 185 190		
Lys Lys Met Pro Leu Thr Glu Tyr Leu Lys Arg Ile Lys Leu Pro Asn 195 200 205		
Ser Ile Asp Ser Tyr Thr Asp Arg Leu Tyr Leu Leu Trp Leu Leu Leu 210 215 220		
Val Thr Leu Ala Tyr Asn Trp Asn Cys Cys Phe Ile Pro Leu Arg Leu 225 230 235 240		
Val Phe Pro Tyr Gln Thr Ala Asp Asn Ile His Tyr Trp Leu Ile Ala 245 250 255		
Asp Ile Ile Cys Asp Ile Ile Tyr Leu Tyr Asp Met Leu Phe Ile Gln 260 265 270		
Pro Arg Leu Gln Phe Val Arg Gly Gly Asp Ile Ile Val Asp Ser Asn 275 280 285		
Glu Leu Arg Lys His Tyr Arg Thr Ser Thr Lys Phe Gln Leu Asp Val 290 295 300		
Ala Ser Ile Ile Pro Phe Asp Ile Cys Tyr Leu Phe Phe Gly Phe Asn 305 310 315 320		
Pro Met Phe Arg Ala Asn Arg Met Leu Lys Tyr Thr Ser Phe Phe Glu 325 330 335		
Phe Asn His His Leu Glu Ser Ile Met Asp Lys Ala Tyr Ile Tyr Arg 340 345 350		
Val Ile Arg Thr Thr Gly Tyr Leu Leu Phe Ile Leu His Ile Asn Ala 355 360 365		
Cys Val Tyr Tyr Trp Ala Ser Asn Tyr Glu Gly Ile Gly Thr Thr Arg 370 375 380		
Trp Val Tyr Asp Gly Glu Gly Asn Glu Tyr Leu Arg Cys Tyr Tyr Trp 385 390 395 400		
Ala Val Arg Thr Leu Ile Thr Ile Gly Gly Leu Pro Glu Pro Gln Thr 405 410 415		
Leu Phe Glu Ile Val Phe Gln Leu Leu Asn Phe Phe Ser Gly Val Phe 420 425 430		
Val Phe Ser Ser Leu Ile Gly Gln Met Arg Asp Val Ile Gly Ala Ala 435 440 445		
Thr Ala Asn Gln Asn Tyr Phe Arg Ala Cys Met Asp Asp Thr Ile Ala 450 455 460		
Tyr Met Asn Asn Tyr Ser Ile Pro Lys Leu Val Gln Lys Arg Val Arg 465 470 475 480		
Thr Trp Tyr Glu Tyr Thr Trp Asp Ser Gln Arg Met Leu Asp Glu Ser 485 490 495		
Asp Leu Leu Lys Thr Leu Pro Thr Thr Val Gln Leu Ala Leu Ala Ile 500 505 510		
Asp Val Asn Phe Ser Ile Ile Ser Lys Val Asp Leu Phe Lys Gly Cys 515 520 525		
Asp Thr Gln Met Ile Tyr Asp Met Leu Leu Arg Leu Lys Ser Val Leu 530 535 540		

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Tyr Leu Pro Gly Asp Phe Val Cys Lys Lys Gly Glu Ile Gly Lys Glu
 545 550 555 560
 Met Tyr Ile Ile Lys His Gly Glu Val Gln Val Leu Gly Gly Pro Asp
 565 570 575
 Gly Thr Lys Val Leu Val Thr Leu Lys Ala Gly Ser Val Phe Gly Glu
 580 585 590
 Ile Ser Leu Leu Ala Ala Gly Gly Gly Asn Arg Arg Thr Ala Asn Val
 595 600 605
 Val Ala His Gly Phe Ala Asn Leu Leu Thr Leu Asp Lys Lys Thr Leu
 610 615 620
 Gln Glu Ile Leu Val His Tyr Pro Asp Ser Glu Arg Ile Leu Met Lys
 625 630 635 640
 Lys Ala Arg Val Leu Leu Lys Gln Lys Ala Lys Thr Ala Glu Ala Thr
 645 650 655
 Pro Pro Arg Lys Asp Leu Ala Leu Leu Phe Pro Pro Lys Glu Glu Thr
 660 665 670
 Pro Lys Leu Phe Lys Thr Leu Leu Gly Gly Thr Gly Lys Ala Ser Leu
 675 680 685
 Ala Arg Leu Leu Lys Leu Lys Arg Glu Gln Ala Ala Gln Lys Lys Glu
 690 695 700
 Asn Ser Glu Gly Gly Glu Glu Glu Gly Lys Glu Asn Glu Asp Lys Gln
 705 710 715 720
 Lys Glu Asn Glu Asp Lys Gln Lys Glu Asn Glu Asp Lys Gly Lys Glu
 725 730 735
 Asn Glu Asp Lys Asp Lys Gly Arg Glu Pro Glu Glu Lys Pro Leu Asp
 740 745 750
 Arg Pro Glu Cys Thr Ala Ser Pro Ile Ala Val Glu Glu Glu Pro His
 755 760 765
 Ser Val Arg Arg Thr Val Leu Pro Arg Gly Thr Ser Arg Gln Ser Leu
 770 775 780
 Ile Ile Ser Met Ala Pro Ser Ala Glu Gly Gly Glu Glu Val Leu Thr
 785 790 795 800
 Ile Glu Val Lys Glu Lys Ala Lys Gln
 805

<210> SEQ ID NO 23
 <211> LENGTH: 2454
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: constructed sequence
 <220> FEATURE:
 <221> NAME/KEY: misc_feature
 <222> LOCATION: (1)..(12)
 <223> OTHER INFORMATION: modified end with NotI site and Kozak
 <220> FEATURE:
 <221> NAME/KEY: misc_feature
 <222> LOCATION: (1)..(8)
 <223> OTHER INFORMATION: NotI site for subcloning
 <220> FEATURE:
 <221> NAME/KEY: CDS
 <222> LOCATION: (13)..(2448)
 <223> OTHER INFORMATION: ORF with silent mutations (stop codon and
 restriction sites BamHI, PstI, SalI, and NdeI)
 <220> FEATURE:
 <221> NAME/KEY: misc_feature
 <222> LOCATION: (2440)..(2442)
 <223> OTHER INFORMATION: modified stop codon
 <220> FEATURE:
 <221> NAME/KEY: misc_feature
 <222> LOCATION: (2440)..(2445)

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<223> OTHER INFORMATION: BclI site to facilitate addition of epitope tag
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (2446)..(2448)
<223> OTHER INFORMATION: additional stop codon
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (2449)..(2454)
<223> OTHER INFORMATION: PstI site for subcloning

<400> SEQUENCE: 23

gcggccgccca cc atg ttt aaa tcg ctg aca aaa gtc aac aag gtg aag cct      51
Met Phe Lys Ser Leu Thr Lys Val Asn Lys Val Lys Pro
      1              5              10

ata gga gag aac aat gag aat gaa caa agt tct cgt cgg aat gaa gaa      99
Ile Gly Glu Asn Asn Glu Asn Glu Gln Ser Ser Arg Arg Asn Glu Glu
      15              20              25

ggc tct cac cca agt aat cag tct cag caa acc aca gca cag gaa gaa      147
Gly Ser His Pro Ser Asn Gln Ser Gln Gln Thr Thr Ala Gln Glu Glu
      30              35              40              45

aac aaa ggt gaa gag aaa tct ctc aaa acc aag tca act cca gtc acg      195
Asn Lys Gly Glu Glu Lys Ser Leu Lys Thr Lys Ser Thr Pro Val Thr
              50              55              60

tct gaa gag cca cac acc aac ata caa gac aaa ctc tcc aag aaa aat      243
Ser Glu Glu Pro His Thr Asn Ile Gln Asp Lys Leu Ser Lys Lys Asn
      65              70              75

tcc tct gga gat ctg acc aca aac cct gac cct caa aat gca gca gaa      291
Ser Ser Gly Asp Leu Thr Thr Asn Pro Asp Pro Gln Asn Ala Ala Glu
      80              85              90

cca act gga aca gtg cca gag cag aag gaa atg gac ccc ggg aaa gaa      339
Pro Thr Gly Thr Val Pro Glu Gln Lys Glu Met Asp Pro Gly Lys Glu
      95              100             105

ggg cca aac agc cca caa aac aaa ccg cca gca gct cct gtt ata aat      387
Gly Pro Asn Ser Pro Gln Asn Lys Pro Pro Ala Ala Pro Val Ile Asn
      110             115             120             125

gag tat gcc gat gcc cag cta cac aac ctg gtg aaa aga atg cgt caa      435
Glu Tyr Ala Asp Ala Gln Leu His Asn Leu Val Lys Arg Met Arg Gln
      130             135             140

aga aca gcc ctc tac aag aaa aag ttg gta gag gga gat ctc tcc tca      483
Arg Thr Ala Leu Tyr Lys Lys Lys Leu Val Glu Gly Asp Leu Ser Ser
      145             150             155

ccc gaa gcc agc cca caa act gca aag ccc acg gct gta cca cca gta      531
Pro Glu Ala Ser Pro Gln Thr Ala Lys Pro Thr Ala Val Pro Pro Val
      160             165             170

aaa gaa agc gat gat aag cca aca gaa cat tac tac agg ctg ttg tgg      579
Lys Glu Ser Asp Asp Lys Pro Thr Glu His Tyr Tyr Arg Leu Leu Trp
      175             180             185

ttc aaa gtc aaa aag atg cct tta aca gag tac tta aag cga att aaa      627
Phe Lys Val Lys Lys Met Pro Leu Thr Glu Tyr Leu Lys Arg Ile Lys
      190             195             200             205

ctt cca aac agc ata gat tca tac aca gat cga ctc tat ctc ctg tgg      675
Leu Pro Asn Ser Ile Asp Ser Tyr Thr Asp Arg Leu Tyr Leu Leu Trp
      210             215             220

ctc ttg ctt gtc act ctt gcc tat aac tgg aac tgc tgt ttt ata cca      723
Leu Leu Leu Val Thr Leu Ala Tyr Asn Trp Asn Cys Cys Phe Ile Pro
      225             230             235

ctg cgc ctc gtc ttc cca tat caa acc gca gac aac ata cac tac tgg      771
Leu Arg Leu Val Phe Pro Tyr Gln Thr Ala Asp Asn Ile His Tyr Trp
      240             245             250

ctt att gcg gac atc atc tgt gat atc atc tac ctt tat gat atg cta      819
Leu Ile Ala Asp Ile Ile Cys Asp Ile Ile Tyr Leu Tyr Asp Met Leu
      255             260             265

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ttt atc cag ccc aga ctc cag ttt gta aga gga gga gac ata ata gtg	867
Phe Ile Gln Pro Arg Leu Gln Phe Val Arg Gly Gly Asp Ile Ile Val	
270 275 280 285	
gat tca aat gag cta agg aaa cac tac agg act tct aca aaa ttt cag	915
Asp Ser Asn Glu Leu Arg Lys His Tyr Arg Thr Ser Thr Lys Phe Gln	
290 295 300	
ttg gat gtc gca tca ata ata cca ttt gat att tgc tac ctc ttc ttt	963
Leu Asp Val Ala Ser Ile Ile Pro Phe Asp Ile Cys Tyr Leu Phe Phe	
305 310 315	
ggg ttt aat cca atg ttt aga gca aat agg atg tta aag tac act tca	1011
Gly Phe Asn Pro Met Phe Arg Ala Asn Arg Met Leu Lys Tyr Thr Ser	
320 325 330	
ttt ttt gaa ttt aat cat cac cta gag tct ata atg gac aaa gca tat	1059
Phe Phe Glu Phe Asn His His Leu Glu Ser Ile Met Asp Lys Ala Tyr	
335 340 345	
atc tac aga gtt att cga aca act gga tac ttg ctg ttt att ctg cac	1107
Ile Tyr Arg Val Ile Arg Thr Thr Gly Tyr Leu Leu Phe Ile Leu His	
350 355 360 365	
att aat gcc tgt gtt tat tac tgg gct tca aac tat gaa gga att ggc	1155
Ile Asn Ala Cys Val Tyr Tyr Trp Ala Ser Asn Tyr Glu Gly Ile Gly	
370 375 380	
act act aga tgg gtg tat gat ggg gaa gga aac gag tat ctg aga tgt	1203
Thr Thr Arg Trp Val Tyr Asp Gly Glu Gly Asn Glu Tyr Leu Arg Cys	
385 390 395	
tat tat tgg gca gtt cga act tta att acc att ggt ggc ctt cca gaa	1251
Tyr Tyr Trp Ala Val Arg Thr Leu Ile Thr Ile Gly Gly Leu Pro Glu	
400 405 410	
cca caa act tta ttt gaa att gtt ttt caa ctc ttg aat ttt ttt tct	1299
Pro Gln Thr Leu Phe Glu Ile Val Phe Gln Leu Leu Asn Phe Phe Ser	
415 420 425	
gga gtt ttt gtg ttc tcc agt tta att ggt cag atg aga gat gtg att	1347
Gly Val Phe Val Phe Ser Ser Leu Ile Gly Gln Met Arg Asp Val Ile	
430 435 440 445	
gga gca gct aca gcc aat cag aac tac ttc cgc gcc tgc atg gat gac	1395
Gly Ala Ala Thr Ala Asn Gln Asn Tyr Phe Arg Ala Cys Met Asp Asp	
450 455 460	
acc att gcc tac atg aac aat tac tcc att cct aaa ctt gtg caa aag	1443
Thr Ile Ala Tyr Met Asn Asn Tyr Ser Ile Pro Lys Leu Val Gln Lys	
465 470 475	
cga gtt cgg act tgg tat gaa tat aca tgg gac tct caa aga atg cta	1491
Arg Val Arg Thr Trp Tyr Glu Tyr Thr Trp Asp Ser Gln Arg Met Leu	
480 485 490	
gat gag tct gat ttg ctt aag acc cta cca act acg gtc cag tta gcc	1539
Asp Glu Ser Asp Leu Leu Lys Thr Leu Pro Thr Thr Val Gln Leu Ala	
495 500 505	
ctc gcc att gat gtg aac ttc agc atc atc agc aaa gtt gac ttg ttc	1587
Leu Ala Ile Asp Val Asn Phe Ser Ile Ile Ser Lys Val Asp Leu Phe	
510 515 520 525	
aag ggt tgt gat aca cag atg att tat gac atg ttg cta aga ttg aaa	1635
Lys Gly Cys Asp Thr Gln Met Ile Tyr Asp Met Leu Leu Arg Leu Lys	
530 535 540	
tcc gtt ctc tat ttg cct ggt gac ttt gtc tgc aaa aag gga gaa att	1683
Ser Val Leu Tyr Leu Pro Gly Asp Phe Val Cys Lys Lys Gly Glu Ile	
545 550 555	
ggc aag gaa atg tat atc atc aag cat gga gaa gtc caa gtt ctt gga	1731
Gly Lys Glu Met Tyr Ile Ile Lys His Gly Glu Val Gln Val Leu Gly	
560 565 570	
ggc cct gat ggt act aaa gtt ctg gtt act ctg aaa gct ggg tcg gtg	1779
Gly Pro Asp Gly Thr Lys Val Leu Val Thr Leu Lys Ala Gly Ser Val	

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575	580	585	
ttt gga gaa atc agc ctt cta gca gca gga gga gga aac cgt cga act Phe Gly Glu Ile Ser Leu Leu Ala Ala Gly Gly Gly Asn Arg Arg Thr 590 595 600 605			1827
gcc aat gtg gtg gcc cac ggg ttt gcc aat ctt tta act cta gac aaa Ala Asn Val Val Ala His Gly Phe Ala Asn Leu Leu Thr Leu Asp Lys 610 615 620			1875
aag acc ctc caa gaa att cta gtg cat tat cca gat tct gaa aga atc Lys Thr Leu Gln Glu Ile Leu Val His Tyr Pro Asp Ser Glu Arg Ile 625 630 635			1923
ctc atg aag aaa gcc aga gtg ctt tta aag cag aag gct aag acc gca Leu Met Lys Lys Ala Arg Val Leu Leu Lys Gln Lys Ala Lys Thr Ala 640 645 650			1971
gaa gca acc cct cca aga aaa gat ctt gcc ctc ctc ttc cca ccg aaa Glu Ala Thr Pro Pro Arg Lys Asp Leu Ala Leu Leu Phe Pro Pro Lys 655 660 665			2019
gaa gag aca ccc aaa ctg ttt aaa act ctc cta gga ggc aca gga aaa Glu Glu Thr Pro Lys Leu Phe Lys Thr Leu Leu Gly Gly Thr Gly Lys 670 675 680 685			2067
gca agt ctt gca aga cta ctc aaa ttg aag cga gag caa gca gct cag Ala Ser Leu Ala Arg Leu Leu Lys Leu Lys Arg Glu Gln Ala Ala Gln 690 695 700			2115
aag aaa gaa aat tct gaa gga gga gag gaa gaa gga aaa gaa aat gaa Lys Lys Glu Asn Ser Glu Gly Gly Glu Glu Glu Gly Lys Glu Asn Glu 705 710 715			2163
gat aaa caa aaa gaa aat gaa gat aaa caa aaa gaa aat gaa gat aaa Asp Lys Gln Lys Glu Asn Glu Asp Lys Gln Lys Glu Asn Glu Asp Lys 720 725 730			2211
gga aaa gaa aat gaa gat aaa gat aaa gga aga gag cca gaa gag aag Gly Lys Glu Asn Glu Asp Lys Asp Lys Gly Arg Glu Pro Glu Glu Lys 735 740 745			2259
cca ctg gac aga cct gaa tgt aca gca agt cct att gca gtg gag gaa Pro Leu Asp Arg Pro Glu Cys Thr Ala Ser Pro Ile Ala Val Glu Glu 750 755 760 765			2307
gaa ccc cac tca gtt aga agg aca gtt tta ccc aga ggg act tct cgt Glu Pro His Ser Val Arg Arg Thr Val Leu Pro Arg Gly Thr Ser Arg 770 775 780			2355
caa tca ctc att atc agc atg gct cct tct gct gag ggc gga gaa gag Gln Ser Leu Ile Ile Ser Met Ala Pro Ser Ala Glu Gly Gly Glu Glu 785 790 795			2403
gtt ctt act att gaa gtc aaa gaa aag gct aag caa tga tca taa Val Leu Thr Ile Glu Val Lys Glu Lys Ala Lys Gln Ser 800 805 810			2448
ctgcag			2454

<210> SEQ ID NO 24
 <211> LENGTH: 809
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 24

Met Phe Lys Ser Leu Thr Lys Val Asn Lys Val Lys Pro Ile Gly Glu
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 Asn Asn Glu Asn Glu Gln Ser Ser Arg Arg Asn Glu Glu Gly Ser His
 20 25 30
 Pro Ser Asn Gln Ser Gln Gln Thr Thr Ala Gln Glu Glu Asn Lys Gly
 35 40 45

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Glu	Glu	Lys	Ser	Leu	Lys	Thr	Lys	Ser	Thr	Pro	Val	Thr	Ser	Glu	Glu
50						55					60				
Pro	His	Thr	Asn	Ile	Gln	Asp	Lys	Leu	Ser	Lys	Lys	Asn	Ser	Ser	Gly
65					70					75					80
Asp	Leu	Thr	Thr	Asn	Pro	Asp	Pro	Gln	Asn	Ala	Ala	Glu	Pro	Thr	Gly
				85				90						95	
Thr	Val	Pro	Glu	Gln	Lys	Glu	Met	Asp	Pro	Gly	Lys	Glu	Gly	Pro	Asn
			100					105					110		
Ser	Pro	Gln	Asn	Lys	Pro	Pro	Ala	Ala	Pro	Val	Ile	Asn	Glu	Tyr	Ala
			115				120					125			
Asp	Ala	Gln	Leu	His	Asn	Leu	Val	Lys	Arg	Met	Arg	Gln	Arg	Thr	Ala
	130					135					140				
Leu	Tyr	Lys	Lys	Lys	Leu	Val	Glu	Gly	Asp	Leu	Ser	Ser	Pro	Glu	Ala
145					150					155					160
Ser	Pro	Gln	Thr	Ala	Lys	Pro	Thr	Ala	Val	Pro	Pro	Val	Lys	Glu	Ser
				165					170					175	
Asp	Asp	Lys	Pro	Thr	Glu	His	Tyr	Tyr	Arg	Leu	Leu	Trp	Phe	Lys	Val
			180					185					190		
Lys	Lys	Met	Pro	Leu	Thr	Glu	Tyr	Leu	Lys	Arg	Ile	Lys	Leu	Pro	Asn
		195					200					205			
Ser	Ile	Asp	Ser	Tyr	Thr	Asp	Arg	Leu	Tyr	Leu	Leu	Trp	Leu	Leu	Leu
	210					215					220				
Val	Thr	Leu	Ala	Tyr	Asn	Trp	Asn	Cys	Cys	Phe	Ile	Pro	Leu	Arg	Leu
225					230					235					240
Val	Phe	Pro	Tyr	Gln	Thr	Ala	Asp	Asn	Ile	His	Tyr	Trp	Leu	Ile	Ala
				245					250					255	
Asp	Ile	Ile	Cys	Asp	Ile	Ile	Tyr	Leu	Tyr	Asp	Met	Leu	Phe	Ile	Gln
			260					265					270		
Pro	Arg	Leu	Gln	Phe	Val	Arg	Gly	Gly	Asp	Ile	Ile	Val	Asp	Ser	Asn
		275					280					285			
Glu	Leu	Arg	Lys	His	Tyr	Arg	Thr	Ser	Thr	Lys	Phe	Gln	Leu	Asp	Val
	290					295					300				
Ala	Ser	Ile	Ile	Pro	Phe	Asp	Ile	Cys	Tyr	Leu	Phe	Phe	Gly	Phe	Asn
305					310					315					320
Pro	Met	Phe	Arg	Ala	Asn	Arg	Met	Leu	Lys	Tyr	Thr	Ser	Phe	Phe	Glu
				325					330					335	
Phe	Asn	His	His	Leu	Glu	Ser	Ile	Met	Asp	Lys	Ala	Tyr	Ile	Tyr	Arg
			340					345					350		
Val	Ile	Arg	Thr	Thr	Gly	Tyr	Leu	Leu	Phe	Ile	Leu	His	Ile	Asn	Ala
		355					360					365			
Cys	Val	Tyr	Tyr	Trp	Ala	Ser	Asn	Tyr	Glu	Gly	Ile	Gly	Thr	Thr	Arg
	370					375					380				
Trp	Val	Tyr	Asp	Gly	Glu	Gly	Asn	Glu	Tyr	Leu	Arg	Cys	Tyr	Tyr	Trp
385					390					395					400
Ala	Val	Arg	Thr	Leu	Ile	Thr	Ile	Gly	Gly	Leu	Pro	Glu	Pro	Gln	Thr
				405					410					415	
Leu	Phe	Glu	Ile	Val	Phe	Gln	Leu	Leu	Asn	Phe	Phe	Ser	Gly	Val	Phe
			420					425					430		
Val	Phe	Ser	Ser	Leu	Ile	Gly	Gln	Met	Arg	Asp	Val	Ile	Gly	Ala	Ala
		435					440					445			
Thr	Ala	Asn	Gln	Asn	Tyr	Phe	Arg	Ala	Cys	Met	Asp	Asp	Thr	Ile	Ala
	450					455					460				
Tyr	Met	Asn	Asn	Tyr	Ser	Ile	Pro	Lys	Leu	Val	Gln	Lys	Arg	Val	Arg

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465	470	475	480
Thr Trp Tyr Glu Tyr Thr Trp Asp Ser Gln Arg Met Leu Asp Glu Ser			
	485	490	495
Asp Leu Leu Lys Thr Leu Pro Thr Thr Val Gln Leu Ala Leu Ala Ile			
	500	505	510
Asp Val Asn Phe Ser Ile Ile Ser Lys Val Asp Leu Phe Lys Gly Cys			
	515	520	525
Asp Thr Gln Met Ile Tyr Asp Met Leu Leu Arg Leu Lys Ser Val Leu			
	530	535	540
Tyr Leu Pro Gly Asp Phe Val Cys Lys Lys Gly Glu Ile Gly Lys Glu			
	545	550	555
Met Tyr Ile Ile Lys His Gly Glu Val Gln Val Leu Gly Gly Pro Asp			
	565	570	575
Gly Thr Lys Val Leu Val Thr Leu Lys Ala Gly Ser Val Phe Gly Glu			
	580	585	590
Ile Ser Leu Leu Ala Ala Gly Gly Gly Asn Arg Arg Thr Ala Asn Val			
	595	600	605
Val Ala His Gly Phe Ala Asn Leu Leu Thr Leu Asp Lys Lys Thr Leu			
	610	615	620
Gln Glu Ile Leu Val His Tyr Pro Asp Ser Glu Arg Ile Leu Met Lys			
	625	630	635
Lys Ala Arg Val Leu Leu Lys Gln Lys Ala Lys Thr Ala Glu Ala Thr			
	645	650	655
Pro Pro Arg Lys Asp Leu Ala Leu Leu Phe Pro Pro Lys Glu Glu Thr			
	660	665	670
Pro Lys Leu Phe Lys Thr Leu Leu Gly Gly Thr Gly Lys Ala Ser Leu			
	675	680	685
Ala Arg Leu Leu Lys Leu Lys Arg Glu Gln Ala Ala Gln Lys Lys Glu			
	690	695	700
Asn Ser Glu Gly Gly Glu Glu Glu Gly Lys Glu Asn Glu Asp Lys Gln			
	705	710	715
Lys Glu Asn Glu Asp Lys Gln Lys Glu Asn Glu Asp Lys Gly Lys Glu			
	725	730	735
Asn Glu Asp Lys Asp Lys Gly Arg Glu Pro Glu Glu Lys Pro Leu Asp			
	740	745	750
Arg Pro Glu Cys Thr Ala Ser Pro Ile Ala Val Glu Glu Glu Pro His			
	755	760	765
Ser Val Arg Arg Thr Val Leu Pro Arg Gly Thr Ser Arg Gln Ser Leu			
	770	775	780
Ile Ile Ser Met Ala Pro Ser Ala Glu Gly Gly Glu Glu Val Leu Thr			
	785	790	795
Ile Glu Val Lys Glu Lys Ala Lys Gln			
	805		

What is claimed is:

1. A method for treating achromatopsia in a human subject in need thereof, said method comprising administering to the human subject subretinally an AAV vector comprising an AAV capsid and a nucleic acid sequence packaged therein, wherein the nucleic acid sequence comprises i) AAV ITR sequences, ii) a nucleic acid sequence encoding human cyclic nucleotide gated channel alpha 3 (CNGA3), and iii) expression control sequences that direct expression of the CNGA3 in a host cell, wherein the nucleic acid sequence encoding the human CNGA3 sequence comprises SEQ ID NO: 9 or SEQ ID NO: 11.

2. The method of claim 1, wherein nucleic acid sequence encoding human CNGA3 sequence comprises SEQ ID NO: 9 or SEQ ID NO: 11.

3. The method of claim 1, wherein the expression control sequences comprise an ocular cell-specific promoter.

4. The method of claim 1, wherein the AAV vector is administered in combination with another therapy.

5. The method of claim 1, wherein the AAV vector is administered at a dosage of about 10^9 to about 10^{13} vector genomes (VG).

6. The method of claim 1, wherein the AAV vector is administered in a volume of about 100 μ L to about 500 μ L.

7. The method of claim 1, wherein the AAV vector is administered more than once.

8. The method of claim 1, wherein the expression control sequences comprise a chicken β -actin (CBA) promoter with cytomegalovirus (CMV) enhancer elements.

9. The method of claim 1, wherein the expression control sequences comprise a rhodopsin kinase promoter.

10. The method of claim 1, wherein the AAV capsid is an AAV2 capsid, an AAV5 capsid, an AAV8 capsid, an AAV9 capsid, an AAV8 bp capsid, an AAV7m8 capsid, or a variant thereof.

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